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Modeling and Analysis of Cascading Failure in Complex Power Networks



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Thesis Declaration

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Abstract

This thesis is dedicated to extensive studies on cascading failure modeling and analysis of complex power systems. Novel models and analytical approaches have been proposed to better understand the complex mechanisms during the cascading failure.

To be specific, a novel Power System Dynamic (PSD)-based Cascading Failure Simulation Platform (CFSP) is proposed, which takes into consideration PSD, load models, relay models, and automatic load shedding mechanisms. The proposed PSD-based CFSP is tested using the IEEE 68-bus power system, and validated by studying the cascading failure propagation profile, which demonstrates great agreement with historical power system cascading failure patterns. The comparison between CFSP and a Complex Network (CN)-based cascading failure model has been conducted to vigorously study the impact of PSD. The failure propagation profiles of the two models match well for the slow cascade stage when PSD do not dominate the cascading process, but substantial difference appears when large-scale failure starts, i.e., fast cascade stage. Moreover, the impact of varied transmission line capacity on blackout sizes is also investigated for both CN-based model and the proposed CFSP, with comparison presented and analyzed.

Moreover, a novel stochastic cascading failure model considering the interdependency between the cyber network and power network has been introduced. A multi-agent system and a novel protection relay model are incorporated into the proposed model. To apply the proposed analytical method, a test power system, IEEE 68-bus power system, is used to study the impacts of a range of interdependencies and cyber network topological structures on the cascading failure. Simulation results show the necessity and effects of consideration of cyber communication network when investigating power system cascading failures. The study also provides useful information on robustness and vulnerability of a particular power grid, given different communication topologies and interdependencies.

The major original contributions reported in this thesis include:

- Development of the mathematical models of PSD-based cascading failure simulator platform.
- Inverse definite minimum time-based relay models incorporated.
- Introducing the cascading failure model that is easily modified and well trade-off between the accuracy and efficiency.
- Performing a multi-step attack strategy that could lead to cascading failures on the complex power system.
- Analyzing the cascading failure through complex network theory considering interdependency between the cyber network and power network.
- Multi-agent system incorporated into the cascading failure model.
- A novel transmission line protection system is introduced based on the electric transmission line relays, where circuit breakers are placed at both ends of the line.

- Development of a stochastic model considering cascading failure propagation in CPPS, where a well trade-off between the complexity and computation speed is carefully considered.

Since the complex power systems used in this thesis are with IEEE standard realistic configurations, the simulations are able to emulate the real-world cases, which makes the research project bear both academic and industrial values. The proposed models and analytical approaches allow network operators to make more sensible and timely decisions to mitigate the adverse effects of cascading failures in power systems, and provide useful information for the robustness, vulnerability and economic aspects for complex power systems given various network conditions and topological features.

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AUTHORSHIP DECLARATION: CO-AUTHORED PUBLICATIONS

This thesis contains a part of the research articles that have been published during my PhD study since November 2016, which are rearranged and slightly modified to achieve high consistency and coherence in the content.

Details of work 1:

Hengdao Guo, Ciyan Zheng, Herbert Iu, Tyrone Fernando, "A critical review of cascading failure analysis and modeling of power system", Renewable and Sustainable Energy Reviews, vol.80, pp.9-22, 2017.

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Co-author signatures and dates: (only signatures of Professor Tyrone and Herbert, my PhD supervisors, are available upon the submission time)

[redacted]

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[redacted]

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Location in thesis: Chapter 5

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Tyrone Fernando

Date: 19-03-2020

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Chapter 1

Introduction

1.1 Background

Power system, one of the most complex networks in modern society, plays a significant role in people's daily lives. It is a huge system with interconnections between service areas, and can be regarded as the backbone of our modern cities. Power system normally consists of generating stations, transmission lines, and distribution substations/lines. Power stations generate electricity, which is transferred through high voltage transmission lines from distant resources to demand centers, and finally distributed by distribution substations to industrial, commercial, and residential customers. The power system ensures our safety and security, and is vital to the society. Recent years, rapid development of smart grid and integration of renewable energies has brought significant changes to the power system in a more efficient, effective, economical, and socially friendly fashion [3]. On the other hand, these great changes have also brought new challenges, especially the stability and reliability issues and the subsequent occurrence of blackouts [4]. Cascading failures, one of the severest power system instability problems, may cause large-scale damages and massive economic losses to society. Cascading failures are defined as "the uncontrolled loss of bulk electric system facilities triggered by an incident (or condition) at any location resulting in the interruption of electric service that cannot be restrained from spreading beyond a pre-determined area [5]." Generally, these cascading failures may initially start with any component of the power system, e.g., the failure of buses or transmission lines. The initial failure may cause power flow redistribution and affect other transmission lines or buses, which in turn may get overloaded and therefore forming the cascading effect. These large-scale cascading blackouts are infrequent but substantially risky due to their catastrophic consequences. Blackout data from North America suggest that the frequency of blackouts with scale over a certain threshold follows a power-law distribution, rather than the common Gauss distribution [6–8]. This evidence demonstrates a much higher large-scale blackouts probability than we thought. Some major blackouts are listed in Table 1.1, where the economic losses and horrible effect are shown. These severe events draw wide attention from both academia and industry, as such large blackouts are more complicated than normal electric outages caused by small disturbances. Cascading failure is found to be the key factor leading to a large blackout from the reports of those blackouts [1, 9–12]. For example, in 2003 North America blackout, the blackout was initiated by a 345kV transmission line failure that was caused by its contact to the tree. At the same time, there was an communication error in the control room. These two

Table 1.1: Major blackouts since 2003.

Blackout location	Date	People affected (Millions)	Loss of load (MW)	Estimated cost(Million dollars)	Time duration	Improvements after blackout
North America	14-15 August 2003	50	61.8	Over 10,000	5-72h	Introduce higher reliability standards for North American electricity industry
London	28 August 2003	0.5	724	N.A.	>30mins	Enhance cooperation between utility companies
Italy	28 September 2003	57	24	1200	5-9h	Implement Day-Ahead Congestion Forecast (DACF)
Pakistan	24 September 2006	160	11.16	N.A.	5-6h	N.A.
Europe	4 November 2006	45	14.5	N.A.	2h	Amendments UCTE Operation Handbook
Colombia	26 April 2007	41	6.644	130	>4h	Improvements of communication channels among the control centers
Brazil and Paraguay	10-11 November 2009	87	24,436	N.A.	4-6h	Introduce House Load Operation (HLO) and new restoration strategies
Brazil	4 February 2011	40	8.884	N.A.	>3h	Implement new islanding protection scheme
India	30-31 July 2012	670	48	6000	2-8h	New load shedding strategies
Turkey	31 March 2015	70	32.2	700	>7h	Improve overload monitoring and protection of transmission lines
Sri Lanka	13 March 2016	21	800	N.A.	>4h	Adopting 'must run units'
Kenya	7 June 2016	44	N.A.	N.A.	<3h	N.A.
South Australia	28 September 2016	1.7	483.7	367	>3h	New rules on wind generation, System restart Ancillary Services (SRAS), Rate of Change of Frequency (RoCoF)
Taiwan	15 August 2017	6.68	4000	Over 3	>3h	N.A.
Venezuela	7-14 March 2019	30	N.A.	Over 1400	>168h	N.A.

contingencies together initiated the cascading blackouts, as the communication error made the power system operator underestimate the possbile results of the transmission line failure, whose trip finally led to the overload of another 345kV transmission line. Finally, more transmission lines and generation units failed and the whole power system collapsed.

The complex mechanisms during the cascading failure make it difficult to anayze through conventional power system analysis approaches and models. The consequence of this is the emergence of many researchers and research groups, as well as a variety of novel models and analysis tools, focusing on understanding, prediction, prevention, mitigation and restoration of cascading failures [13–15].

1.2 Motivations

Large-scale cascading outages brought disastrous impact and great economic losses to our society, while the related research works are far from accomplished. Traditional research works are focusing more on the single or initial several failures, rather than the whole procedure of the cascading failure process. Many commercial power system analysis tools, such as Electromagnetic Transients Program (EMTP) and PowerWorld [16, 17], are not able to reflect the whole cascading failure propagation and procedure. These tools are not designed for cascading failure analysis, and thus having a large learning curve and are difficult to use. When considering the whole cascading failure procedure with considerable computations, such tools may fail to provide a timely solution. Last but not least, some of these commercial tools also require the purchase of parallel simulations, which would be a large budget for the majority of research groups. Some other cascading failure modeling tools provided by researchers, such as [18–20], fail to consider essential electrical properties, which might lead to erroneous and misleading results [21]. In addition, most of the analysis tools mentioned above have limited functions on PSD. Therefore, a Power System Dynamic(PSD)-based cascading failure simulation platform is introduced, which is easily modified and have a well trade-off between the accuracy and efficiency.

Furthermore, the complex power network is increasingly coupled as Cyber-Physical Power System (CPPS) nowadays. In a CPPS, cyber devices are used to collect real-time data and receive control signals, in order to assist normal operation of electrical network. Electrical network provides necessary power for cyber devices to operate normally. Previous studies provide over-simplified models that fail to catch the necessary features of the power system and the cascading failure. Further, there are many probabilistic failures during the cascading failure, such as hidden failures, which are not taken into consideration in the model. To fill the void in relevant researches, power system cascading failure is extensively investigated, considering the coupled CPPS using the Complex Network (CN) theory and stochastic methodology.

Cyber-security is becoming extremely critical as its damage may lead to malfunction of power grid operation systems; for example, the severe cascading blackout initiated by cyber attacks on several substations in Ukraine in December 2015. Motivated by this, several attack strategies are simulated to test the vulnerability of a power network structure, so that the corresponding strategies could be implemented to protect the power system.

1.3 Objectives and Contributions

The first objective of the research is to provide a novel, proficient and simple method to model the cascading failure in the power system. To be specific, the balance between the accuracy and efficiency should be carefully considered. Therefore, a novel PSD-based cascading failure simulation platform is proposed. The proposed model allows precise modification of any parts of the model, and has a good trade-off between the complexity and the computational time. Furthermore, the CPPS is also investigated to find out the different characteristics under the cascading failure. The Markov Chain-based stochastic method is implemented in the model. Some useful metrics for assessing robustness and vulnerability of power networks are introduced. Various attack strategies are implemented to test the vulnerability of a power network structure. The cascading blackout distribution has also been carefully studied.

1.4 Thesis Outline

This thesis is supported by multiple academic journal papers and conference papers completed during my PhD study. The thesis is outlined as follows. Chapter 1 presents background information, motivation, objectives and contributions of the thesis, and also provides the thesis outline. In Chapter 2, a comprehensive and state-of-art literature review has been provided. The basic mechanisms of the cascading failure is introduced. Most existing valuable and useful models and analysis tools for cascading failures have been summarized and compared. Chapter 3 introduces a novel PSD-based Cascading Failure Simulator Platform (CFSP), which takes into consideration PSD, load models, relay models, and automatic load shedding mechanisms. The proposed PSD-based CFSP is tested using the IEEE 68-bus power system, and validated by studying the cascading failure propagation profile, which demonstrates great agreement with historical power system cascading failure patterns. The comparison between CFSP and a Complex Network (CN)-based cascading failure model has been conducted to vigorously study the impact of PSD. In Chapter 4, an improved topological model taking timescale into consideration is proposed. The model also incorporates the relay setting according to industrial standards, which reveals power system operation more accurately. Moreover, the model is validated using UIUC 150-bus and IEEE 39-bus test systems. In order to assess the vulnerability of a power network, the multi-step attack strategy has been proposed. Chapter 5 introduces a novel stochastic cascading failure model considering the interdependency between the cyber network and power network. A multi-agent system and a novel protection relay model are incorporated into the proposed model. To apply the proposed analytical method, a test power system, IEEE 68-bus power system, is used to study the impacts of a range of interdependencies and cyber network topological structures on the cascading failure. This thesis is comprised of a number of papers that have been published during my PhD study. The full publication list is detailed in the Authorship Declaration part.

Chapter 2

Overview and Literature Review

This chapter presents the overview and literature review of the cascading failure. Cascading failures have extremely complicated mechanisms, while its systematical investigation had not begun until approximate 2003 when several catastrophic cascading blackouts occurred all around the world [1, 4, 9, 10]. This chapter generally introduces the cascading failure, as well as some useful mathematical analytical models. The advantages and disadvantages are carefully compared between these various analytical simulators.

The content of this chapter is mainly based on and modified from the following academic papers:
***Hengdao Guo**, Ciyan Zheng, Herbert Iu, Tyrone Fernando, "A critical review of cascading failure analysis and modeling of power system", Renewable and Sustainable Energy Reviews, vol.80, pp.9-22, 2017.*

2.1 Overview of Cascading Failure

Cascading failure involves a large amount of complicated mechanisms, which makes the thorough understanding of the whole event a great challenge. This section demonstrates an overview on cascading failure from different perspectives, and highlights the challenges on its analysis and modeling.

2.1.1 Causes

There are various causes for a cascading failure [13]. Exogenous disturbances that initiate the event, and endogenous events that trigger the components can be generally classified as four groups:

• Nature disasters: Lightening strike, strong winds (tornado, hurricane), earthquakes;
• Human activity: Errors caused by human misoperations, inappropriate setting for protection devices, intentional physical or cyber-attacks on power grids;
• Unexpected component failures: Hidden failures (exposed during changing operation status), transmission line that contact vegetation;
• System failures: Distance relays trigger the transmission line due to overcurrent or undervoltage, voltage collapse, abnormal excitation in generators, abnormal speed in generators, generators tripped

by under-frequency, generators tripped by under-voltage, generators tripped by out-of-step, insufficient reactive power, small signal instability.

The first three groups are common failures that initiate the events, whereas the fourth group are failures that commonly exacerbate the cascading. The first and second group of failures are unlikely to be prevented with modern technology. The third and fourth groups are dependent failures that would be involved in most cascading failure blackouts.

The various and stochastic causes are challenging cascading failure analysis models on simulating all of these mechanisms. Quite a lot of approaches and models have been provided trying to take them into consideration [5, 7, 19, 22–107]. Some are using stochastic approaches to simulate uncertainties in a cascading failure. Some are modeling dynamics of the system to involve voltage and machine problems. High-level statistical models have also been provided in order to estimate the average cascading propagation and blackout distribution sizes. Interdependent infrastructures are modeled to analyze the interactions between power networks and cyber networks during the cascading failure.

2.1.2 Procedure

During the cascading process, the complex procedure and the complicated mechanisms are also great challenges. According to [22], cascading failure can be divided into two phases, namely slow cascade and fast cascade.

For the slow cascade phase, the failure cascades slowly and gives rise to little effect on power system stability. This phase ranges from several minutes up to several hours. Most failures during this phase are common problems that operators are hardly aware of their impact, thus missing the chance to prevent the cascading. Besides, some hidden failures may be exposed during this phase, which trigger some important devices or transmission lines. In [108], the author showed that the relays with hidden failures work normally under stable conditions. When a transmission line is tripped, all the transmission lines connected to its ends are exposed to incorrect tripping. Overload transmission lines that eventually touch trees or ground due to high temperature also have high probability to occur during this phase.

During the fast cascade phase, power systems become unstable. The overload transmission lines trigger successively in a short time, with possible voltage collapse, frequency oscillation occur in the system. Meanwhile, the dynamic instability may trip the generators, which lead to further load and generation imbalance and exaggerate the dynamic instability. It is almost impossible for operators to stop the cascading during this phase manually, because the short time period between each sequential event ranges from only milliseconds to tens of seconds [22]. Normally, the transmission lines can keep working under overload conditions for hours, and can be reclosed manually if no fault is found. Nevertheless, when there is a large power swing in the system, the distance relay setting on zone 2 or zone 3 may directly trip the transmission line with short time delay, and they will be acting so quickly that the operator has no time to reclose the line or stop the cascading. Thus, the cascading starts tripping like a domino and eventually cause a large blackout.

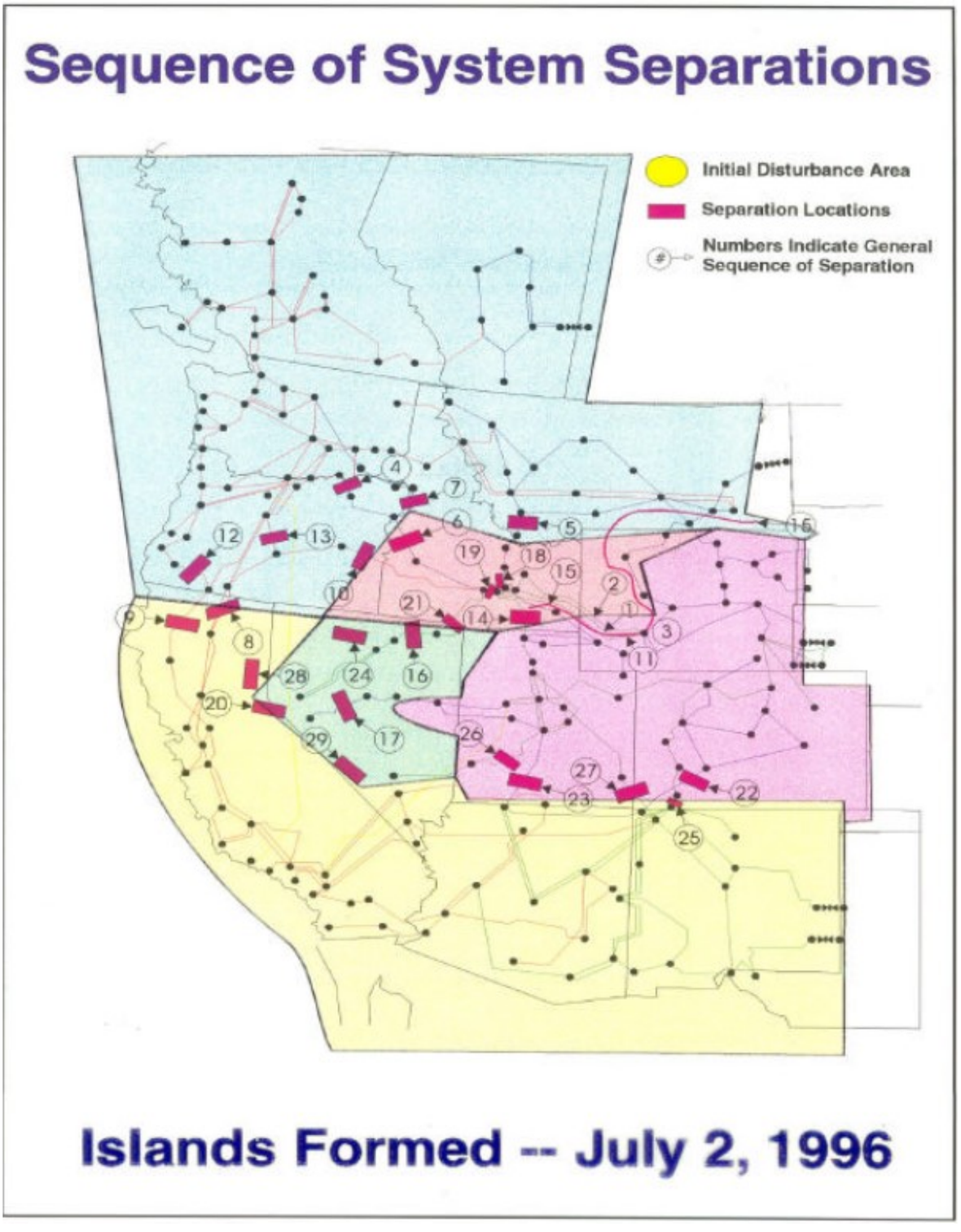


Figure 2.1: The figure illustrates the non-locally propagation of cascading failure, with triggering events labeled

Another characteristic is that cascade events propagate both locally and non-locally in a power system [109]. An example is the blackout occurred in U.S in 1996, as shown in Fig. 2.1 [107]. The sequence of the triggering events have been labeled, illustrating the non-local propagation. This property makes the prediction and prevention of cascading failures more difficult.

The procedure during a cascading is the most critical and difficult part to be simulated. Consequently, most emphasis are putting on this, with a large number of mathematical models and analysis tools provided [5, 7, 19, 42–107]. Some stochastic models are provided to simulate the non-locally propagation of cascading. Deterministic models aims at simulating detailed mechanisms during the cascading.

2.1.3 Impact and Restoration

After the cascading failure stops, the measure of the impact and the restoration of the power grid are two critical points. The impact of the cascading failure are hard to measure accurately due to many indirect costs. Based on power grid only, the impact of the cascading failure are commonly measured in terms of the loss of load [13]. Studies on different metrics have been proposed in order to assess the impact of a cascading failure blackout from a complex network perspective [23–38].

The restoration of the system after the blackout is an essential step. The faster the restoration, the less impact to society from the blackout. Thus, an efficient restoration approach is necessary. A graph-theoretic distribution system restoration-based spanning tree search strategy has been provided for this goal [110]. Another recent trend is the self-healing characteristic based on complex network analysis [111–113].

2.1.4 Risk Assessment

Risk assessment is another way to understanding cascading failure more deeply. Small disturbances in a power network occurred frequently due to severe weather or failure caused by aging devices, while the effect is relatively small compared with cascading failure blackouts. Thus, it is necessary to distinguish disturbances that could lead to cascading failures from small common failures. Despite the N-1 criterion that is regulated for most power system, large blackouts do occasionally happen due to failure of a single transmission element [111]. The utility companies are using various approaches and tools in order to assess, predict, monitor and mitigate the cascading failure, while there are no developed standard analysis tools. Meanwhile, there are many researchers focusing on the field, demonstrating helpful approaches [13].

2.1.5 Dynamic Analysis

Despite the significant contribution from the tripping of the overloaded transmission lines, dynamic instability of the system is another critical factor that cannot be ignored in the cascading failure [112]. In fact, power system dynamic instability will cause incredible effect on the whole system when generators get tripped. Analysis of dynamic phenomena, such as voltage collapse and frequency oscillation, is a conventional field of study in power system analysis [113]. However, most common approaches fail to analyze system dynamic instability under multiple contingencies. In the cascading failure, the balance between generation and load under no overloaded condition is a possible way that the cascading stops, while the dynamic instability may destroy such balance, thus exacerbating the cascading. How to accurately simulate such phenomena during a cascading failure still remains a challenge due to so many complex mechanisms. Existing methods are focusing mainly on a small part of the dynamic problems that is involved in a cascading [64–72]. Recent models have involved timescale of dynamic during the

cascading [73, 76].

2.1.6 Robustness and Vulnerability Analysis

Diverse studies have been proposed to focus on robustness and vulnerability of a power system, so as to increase the reliability of the grid against the cascading failure [23–38]. Many approaches analyze the robustness and vulnerability of power system by removing components from the system and evaluate the impact under different removing strategies. Some studies are using Monte Carlo method to simulate all possible contingencies so that the most likely failed components can be obtained. A variety of robustness metrics have been proposed in order to quantitatively measure the robustness [114–119].

2.1.7 Mitigation and Prevention of Cascading Failure

Mitigation and prevention of cascading play a critical role once the cascading starts. Great efforts have been made in this field [61, 70–72, 96–98, 105]. Mitigation approach aims at reducing the impact of cascading failure by load shedding, line switching or some other remedial actions. Prevention of the cascading utilizes some fast response remedial actions at each stage of the cascading to prevent or at least mitigate the spread of the cascading. The computation and action speed of such methodologies are crucial, as such automatic reaction should be faster than the spread of the cascading in order to take effect. Mitigation and prevention strategies are normally based on a given model, whereas the validation should also be provided under other similar models.

2.1.8 Grid Structural Improvement

The analysis on the power system structure in order to prevent the cascading failure has also been an increasingly important topic. Due to the rapid evolution of the power grid, the integration of the renewable resources and the replacement of the aging devices and transmission lines, it is necessary to find a more reliable construction for a power grid. Some studies are thus arising to deal with such significant issue from different perspectives [120–128]. The author in [120] provided an "effective augmentation" algorithm based on complex network analysis that allowed rapid and efficient augmentation of nodes in one time-step. Power system expansion with renewable energy was studied in [121]. A review was presented in [122], summarizing the expansion analysis from various perspectives. In [123], reliability has been taken into consideration with respect to power system evolution. There were some papers investigating the grid structural improvement against a cascading failure [124–128].

2.1.9 Independent Infrastructures

In recent years, the conception of a smart grid came up due to the increasing demand for reliable energy and numerous technological advancements in power grids. By definition from Department of Energy (DOE) in the United States, smart grid generally refers to a class of technology people are using to bring utility electricity delivery systems into the 21st century, using computer-based remote control and automation [129]. Smart grid enables generation, transmission and distribution of the electricity more conveniently and efficiently. Technologies such as phasor measurement unit (PMU), wide area measurement units and Supervisory Control and Data Acquisition (SCADA) will be deployed to help

achieve these goals, while they also present an increased dependency on cyber networks. Despite the benefits the smart grid brings, the security and reliability of the power system become more vulnerable because of the coupling with cyber networks. The cascading failure blackout, which affected most part of Italy on September 28, 2003, was an example how coupling network contributed to cascading failure [10]. Buldyrev analyzed the case by providing a coupling model to simulate the event [130]. Another case is the blackout caused by cyber-attack on December 23, 2015 in Ukraine [131]. Due to the complexity of the interdependent infrastructures and the difficulty on validation, there has been little progress on this topic. Most existing interdependent models related to cascading failure were based on complex network analysis [92–99].

2.1.10 Complex Network Perspective

Complex network analysis has a relative short history, while its rapid development greatly attracts attention from many scholars. It was not until late 1990s when some significant progress appeared. In 1998, Watts and Strogatz first introduced small-world networks, which were highly clustered and having small characteristic path lengths [132]. They demonstrated that the electrical power grid of the western United States were such kind of networks. Later in 1999, Barabasi and Albert proposed an improved network called scaled-free network, whose vertex connectivities followed a scale-free power-law distribution [133]. Most power grids were also proved to follow the power-law distribution [134–136]. Many researchers began to explore how to adopt complex network analysis to power system [23–41].

From a complex network point of view, a power grid can be simply demonstrated by a graph, where nodes or vertexes represent generators, transformers or loads, and links or edges represent transmission lines. Based on this structure, many different intrinsic characteristics have been applied to nodes and edges, in order to better reveal electrical properties in the power grid [137–140]. Studies focused on different countries got the similar conclusion that the power network belongs to small-world networks [127, 141–147]. While most models considered only high voltage system, reference [136] took medium voltage and low voltage grid into consideration. A survey of studying complex network theory for modern smart grid applications has been proposed in [148].

2.1.11 Other Features

Most people hold the opinion that cascading failures occur more often during summer or winter when there is a heavy load on transmission lines, while it should be noted that many large blackouts actually happened in spring and fall, as showed before in Table 1.1. The reason is that during these periods, many devices are under routine maintenance due to the relative low consumption compared with that of summer and winter. The unserved facilities seem reasonable during this period of time, while the combination of them with some other disturbances may increase the chance of an unexpected cascading failure. It can be found that the facilities under maintenance existed in most major blackouts [1, 4].

Islanding is a common phenomenon found in most major blackouts. Mostly, the islanding area will suffer the unbalanced generation with load, thus leading to dynamic instability. Occasionally, there will be the balance between the generation and load, and the islanding grid will eventually survive. Studies concentrated on this field mainly focused on development of the islanding optimization and control strategies [149, 150].

Climate change, another critical factor related to power system reliability, contributed significantly to major blackouts over the last 30 years [151]. The increase in the temperature will greatly affect the possibility of transmission line failure due to overcurrent or possible sagging into vegetation. Moreover, to deal with climate change, the integration of renewable energies has been widely implemented during these years. However, it also brings system reliability issues such as fault ride through (FRT) of wind generators and intermittency of power generation from renewable sources. Consequently, some mitigation strategies have been proposed to deal with such issues. As have been summarized in [152], there are many potential strategies, such as accurate weather location and severity estimation, reserve plans, demand side management, system configuration, undergrounding distribution and transmission lines, redundant transmission routes, microgrids, advanced control systems, etc. Despite the difficulty in modeling such phenomenon, it is worth putting emphasis on related issues, simulating the possible impact and proposing potential mitigation strategies.

The need for comprehensive cascading failure analysis tools is urgent. Most utilities do have reliability analysis tools, such as Real Time Contingency Analysis (RTCA), Static Security Assessment (SA), Transient Security Assessment (TSA), Voltage Security Assessment (VSA), Small Signal Analysis (SSA), etc., to simulate various conditions and possible outages [153]. Nevertheless, most of these analysis tools are only able to analyze a specific given contingency. The cascading events with so many mechanisms are totally beyond their capabilities. Moreover, common analysis tools focus simply on electrical problems rather than protection and human operations that can also lead to cascading failures, such as 2003 North America Blackout [1]. Most existing approaches and simulations tool can provide only part of the characteristics in a cascading failure. An exhaustive framework of the whole procedure is far from accomplished.

2.2 Literature Review of Cascading Failure Analysis Models and Methodologies

A great diversity of methodologies has been proposed aiming at modeling the cascading failures [5, 7, 19, 23–107]. However, to the best of author's knowledge, no existing approach is able to capture all mechanisms during the cascading failure. Each model has its own concentration and advantages, while the information of the overall phenomenon are still needed in the simulation. In this section, a brief review and summary of the state-of-art cascading failure analysis models and methodologies will be presented. Based on their different characteristics, the models can be roughly classified as six groups:

- Topological models
- Stochastic simulation models
- High-level statistical models
- Dynamic simulation models
- Interdependent models
- Other models

Note that each model may have characteristics involved in several groups, while it is only classified

by its main contribution and feature. This is just a rough division and there are other different classifications, such as component dependent models or probabilistic models.

2.2.1 Topological Models

Complex network analysis progressed rapidly these years and contributed significantly on cascading failure analysis. Since cascading failure almost exists in all complex networks, researchers are thus trying to adopt it to analyze power systems. Early studies proposed various models on cascading failure of a power system, which were relatively simple and failed to explicitly reveal the physical properties involved in the power grid [154–156]. At first, betweenness centrality models were introduced to describe cascading failure, where betweenness centrality was used to assess loads and capacities of nodes and lines [157–159]. The recalculation will be done each time the capacity has been exceeded. Later, a node capacity model has been proposed [160]. In this model, the node capacity is proportional to its initial load. Such capacity definition has been widely used in many models, especially when node or line capacity is not given.

Most of the early works concentrated only on purely topological properties, such as node degrees, betweenness centrality, distribution analysis and path length analysis [141–147, 154–161]. These works focused on more general complex networks rather than electric power grids. However, these purely topological models may be misleading in their results due to lack of consideration in electrical features [21]. Later, to tackle this problem, many modified topological models, with electrical properties involved, have been proposed. There were some existing reviews on complex network applications for cascading failure in power grids, while the models discussed were only classified by their analysis aspects (such as vulnerability, robustness, risk assessment, distribution, node and line criticality, attack-based resilience, evolution, etc.), rather than explaining how various models differ from each other and how these models evolved [114–119]. A rough classification of three different groups of topological models have been summarized below as a reference, namely modified topological models, maximum flow models and complex network-based interdependent models. Note that the last group will be discussed later in interdependent models instead of this part.

Modified Topological Models

Modified topological models typically consider part of electrical properties such as Kirchhoff's law, line impedance or reactance, line capacity and flow-based analysis. These models can partially reflect the characteristics of cascading failure in the power system. An early study modeled the power network in terms of generator nodes, transmission lines and distribution nodes (load), with vulnerability analyzed [23]. Another model given in [24], considered the dynamical redistribution flow and the effect from failed nodes towards the network. Later, [25] demonstrated a modified topological model considering resistance distance instead of common topology distance to reveal electrical properties. There were models that took into consideration the load flow, such as [26–30]. Electrical features and complex networks have been combined in [137], which provided equivalent impedance and the adoption of power transfer distribution factors (PTDF) to new metrics, such as entropy degree, electrical betweenness and net-ability [137–140]. These three metrics have been widely used in analyzing the vulnerability of power networks and identifying the critical components [31–34]. Robustness has been

analyzed using the model that is derived from admittance model provided by Grainer and Stevenson [89], with two new robustness metrics percentage of unserved nodes (PUN) and percentage of noncritical links (PNL) introduced [35]. The paper also investigated the structural features against cascading failure in terms of generator distributions. Fig. 2.2 illustrates the structural topology of power networks of the IEEE 118-bus test system with decentralized generator location in (a) and a centralized one in (b), where red squares represents the generators [35]. The result showed that for a given number (percentage) of available generators in a power system, the decentralized generator locations may greatly increase the robustness of the power grid. Another modified topological model was recently presented in [36], which considered node overload failures and hidden failures. The authors in [37] used an AC power flow-based model to analyze the impact of topology on the cascade propagation. By depicting the power network as a weighted graph based on the reactance matrix, [38] introduced an improved model to analyze vulnerability. The concept of load has been redefined by using power angle information with power flow constraints considered.

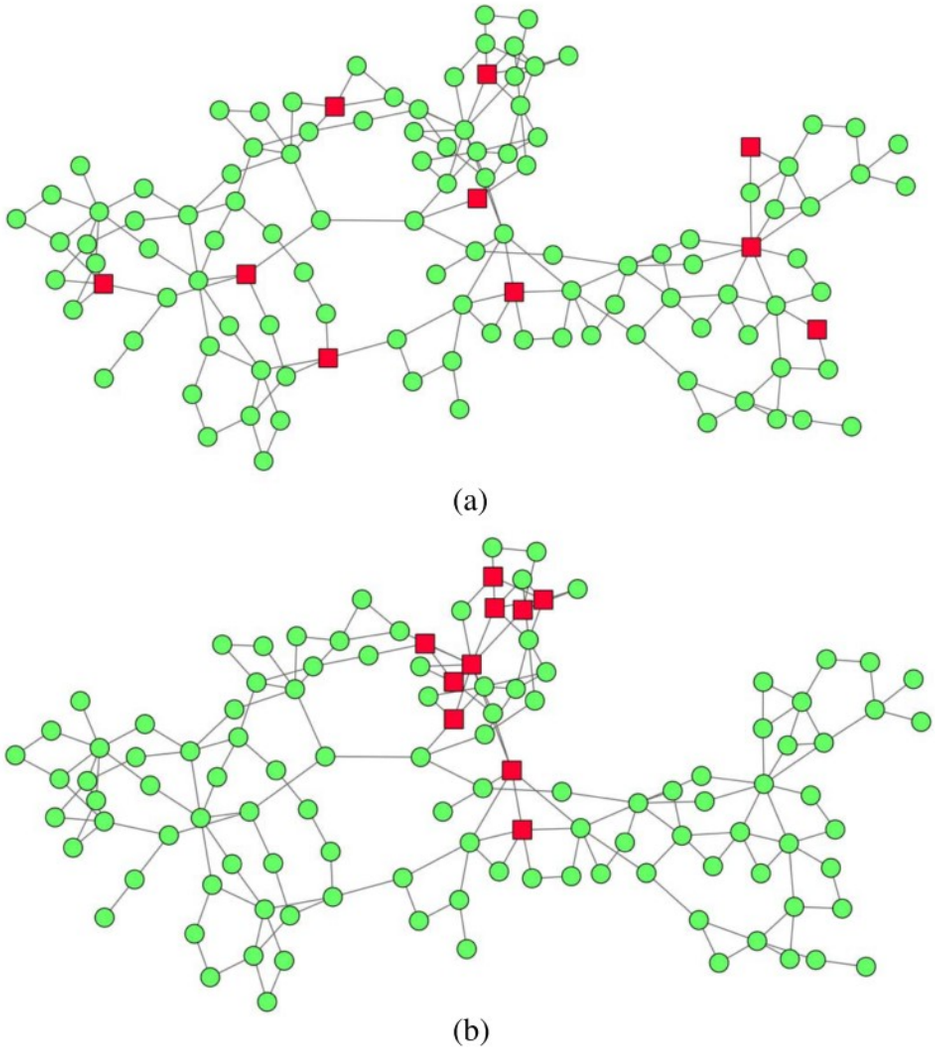


Figure 2.2: The figure illustrates the topology of power networks. (a) IEEE 118-bus with decentralized generator locations; (b) IEEE 118-bus with centralized generator locations. Red squares represent generators.

Maximum Flow Models

The maximum flow theory was originally raised by Harris in 1955 for maximum traffic in railway, and then studied by Ford in 1956 [162]. Its adoption to power grid first appeared in [39], which took into consideration the maximum flow from source (generator) nodes to the sink (load) nodes with respect to the line impedance. The model was used to identify the vulnerability of the power network. Later, the authors in [40] proposed an improved maximum flow-based model with node weight considered. The model also adjusted the max flow problems with multi-source/sink-node into single node pair. The model was validated by simulation in western Danish power system and the results were compared with the former model. Recently, an introduction to a maximum flow-based model to analyze the vulnerability of the power grid has been presented [41]. The initial load distribution was controlled by a tunable parameter, where the vulnerability was proved to be greatly affected by this parameter. A multi-attribute node importance evaluation method based on Gini-coefficient had been simulated using this model [42].

2.2.2 Stochastic Simulation Models

Many simulation tools emphasis on deterministic approaches, which can illustrate the sequence of the triggering events in details. However, cascading failure will rarely proceed as you expect. Due to the large uncertainties that initiate and exacerbate the cascading, the stochastic simulation, also known as probabilistic simulation, which takes all possible factors into consideration is necessary. Moreover, some factors that contribute to the cascading are hard to simulate, such as misoperations by human or transmission lines that contact overgrown trees due to relatively high current flow. It is thus essential to use stochastic approaches to simulate more possible events.

PRACTICE Models

Reference [43] provided a useful stochastic cascading simulation techniques that take probable factors into consideration. In this model, two cascading modes are allowed, namely "single-path" mode and "multi-path" mode. Single-path mode allows uncertainties only relevant to the initial events, after which the system will behave "as expected" during the cascading. Multi-path mode simulates uncertainties, as well as the response from protection systems, through the whole procedure during the cascading. The hidden failure probabilistic model, the overcurrent relay probabilistic model and the event tree-based probabilistic cascading approached was adopted in the techniques. The model was tested on Italian EHV transmission grid, which showed the data on peak and off peak load in early 2000's. A comparison was made with a detailed overload-based dynamic time domain simulator. The result reveals a good consistency between the two models, at least during the slow cascade phase [19]. The mismatch result in the fast cascade phase highlights the challenges on simulating so many mechanisms involved in the cascading failure.

Markov Chain Models

A Markov chain is a type of stochastic process used to describe a system that follows a chain of linked events [44]. It is adopted to modeling the stochastic factors in cascading failure of a power grid, such

as hidden failure or misoperations. The result of this model can be used as an evaluation of overall probabilities of all states that depicts the cascading failure. The model can normally have a very large size.

In [45], a stochastic Markov chain model has been introduced. The model was based on power flow redistribution. It took into account the uncertainties in the load setting, generation and line flows. The model also captured the cascading events with regard to real time signals. The critical components has been identified using the metrics provided. Recently, [46] presented a network-based Markov chain model to study the propagation dynamics of the entire power networks. Robustness of the power network has been analyzed through the model. An extended Gillespie method was adopted in the model [47]. It also showed that small-world network structure would propagate cascading failures more widely and rapidly compared with a regular network structure. A Markovian tree-based multi-timescale cascading outage simulation model has been provided in [48], aiming at risk assessment of cascading failure. The paper also proposed a novel forward-backward Markovian tree search scheme based on a risk estimation index. In [49], the author introduced a continuous-time Markov chain approach modeling the system dynamics. The model considered loading level, error in transmission-capacity estimation, and constraints in performing load shedding. It also allowed real-time prediction of blackout evolution probability.

Recently, an influenced graph model using Markovian chain was described in [50]. Large amounts of data from cascading failure simulations were synthesized into a Markovian network model. The distribution of the cascading failure results achieved from this model matched those from cascading failure simulators. A methodology based on this model has been demonstrated to identify the probability of risk when a component was upgraded in the power system.

2.2.3 High-level Statistical Models

Despite the accuracy and detailed mechanisms simulated by common cascading failure analysis tools, the computation speed remains a challenge when the simulator aims at predicting the real time propagation tendency and the distribution of the blackout size. High-level statistical models are advanced in their simulation speed because the detailed mechanisms of the cascading failure are ignored. Such models aim at providing an overall view of the cascading failure and are simple and tractable.

CASCADE Models

CASCADE model is an analytically tractable model based on load of the component [51]. It assumes a random initial load on all identical components and a given disturbance load on each component to initiate cascading. Some components may fail when load exceeds a certain threshold, where the load of other components will be redistributed, thus forming a cascading. The cascading only stops when no overloaded component exists or the whole system fails. The redistributed formulas are much simpler when compared with models that simulate detailed cascading failure mechanisms, so that the total number of failure components as well as the distribution of the blackout sizes are easier and faster to obtain. The model shows how system loading affect the risk of a cascading failure [7]. When load is in a low level, the approximate exponential tail of the failure components is found and the probability of a large cascading failure is low as well. However, when a critical loading is exceed, the distribution of

the failure components follows the power law and the risk of a large blackout increases sharply. The deficiency of the model is that it disregards all physical properties and the power network internal interactions. Thus, it can only provide a general view of the cascading failure.

Later, the model has been modified and used to analyze some other factors related to cascading failure. In [52], the cascading motor stall has been analyzed using CASCADE model. It demonstrated that the results from the CASCADE model, with a faster computation speed, matched well with that of other dynamic models. Reference [163] showed a high risk on voltage collapse when a failure triggered a cascade of motor stalling. In [53], the focus is on power system reliability through a modified CASCADE model, with the time of the failure propagation considered. Another important model, which can be regarded as an improved CASCADE model, is branching process model that will be discussed below.

Branching Process Models

Branching process methodology is widely utilized in probability theory, especially in simulating the reproduction according to some specific probability distribution [164]. It has been applied to many research areas such as the spread and disappearance of surnames in genealogy and Y chromosome transmission in genetics. The adoption of branching process models to cascading failure analysis is firstly introduced in [161], [54]. After that, some improvements have been made upon the models in [58, 63]. [55] highlighted the importance of the application of branching process models. More recently, some new applications on such models were presented in [37, 56, 62].

In the branching process model, each failure component interdependently affects the next stages through a given distribution. By considering the cascade propagating probability through a branching process, the results showed an approximation with that simulated by CASCADE model as well as the historical data [57]. The branching process model performed a simpler procedure and faster computation speed in estimating the propagation and blackout size distribution. The model has also been utilized in order to find the impact of topology on the average propagation of cascading failure in power system [58]. A novel and systematic approach to discretize load shed data has been introduced so that the Galton-Watson branching process with a Poisson offspring distribution can be applied on it [59]. Recently, a multi-type branching process has been applied to analyze the statistics and interdependencies of cascading outages [60]. It also demonstrated that the method can effectively predict the distribution of the load shed, the isolated buses and their conditional largest possible total outages (CLO) without detailed data provided.

The deficiency of this model is the lack of detailed mechanisms of the cascading failure in a power system. It only provides a general estimation of the propagation and blackout size distribution of the cascading failure. In order to deal with this problem, [61] tried to apply branching process to an interaction model to simulate and mitigate the cascading failure.

2.2.4 Dynamic Simulation Models

Dynamic simulation models are similar to conventional methodologies that focus on power system dynamic characteristic analysis, while the difference is that conventional methodologies are hard to simulate interactions under multi-contingency cases during the cascading failure. New dynamic

simulation models perform well on capturing some specific system dynamics during the cascading. In addition, most mechanisms can be included in dynamic simulation models under a variety of outages so that a relatively accurate prediction can be made. However, due to the large amount of details to be taken into consideration, the computation speed will be challenging. Right now, such models will mainly contribute to deeply understanding the cascading failure mechanisms rather than real time prediction and analysis for industrial utilization.

OPA Models

The ORNL-PSerc-Alaska (OPA) model is a DC power flow-based dynamic model proposed by researchers at Oak Ridge National Laboratory (ORNL), Power System Engineering Research Center of Wisconsin University (PSerc) and Alaska University (Alaska) [64]. At first, a solved initial case was given. A random line was supposed to fail and then the generation and load would be redistributed using standard linear programming approaches. The cost function has been taken into consideration to avoid unnecessary load shedding. The model presented a simplified dynamic process of cascading failure and allowed the analysis of self-organization of an evolving power grid. OPA was validated on a 1553-bus WECC network and the simulation results were compared with historical WECC data with a reasonable agreement [65]. However, the deficiency is that it does not provide an accurate simulation on real transmission line outage and update, as well as the probability distribution of blackout size.

Later, an improved OPA model has been presented to overcome these limitations in original model [66]. The improved OPA model took into consideration the effects of dispatching, automation, communication, relay protection, operation mode and planning. Two metrics, Value at Risk and Conditional Value at Risk, were defined in order to evaluate cascading failures risks quantitatively. The model was validated on the Northeast Power Grid of China (570 buses).

An AC OPA model was introduced in [67]. It simulated two types of dynamics, one was fast dynamics involving series blackouts in power systems, and the other one was slow dynamics that reflected the tendency of the power system time evolution. The model also analyzed the voltage stability problem and the enhancement function. The simulation results on IEEE 118-bus test system showed that self-organized criticality (SOC) could be interpreted from the ratio of total load demand to system effective transmission capacity, as well as the fractional overloads. The model was modified in [68], aiming at the slow process during the cascading failure. It focused on tree contact and failure of lines due to line heating, and the utility vegetation management (UVM).

Manchester Models

Manchester model is an AC power flow model developed by the University of Manchester in UK [69]. It allowed analysis of transmission line tripping, generator instability, under-frequency load shedding, post-contingency re-dispatch of active and reactive power, emergency load shedding, etc. It also took into consideration probabilities of generation, transmission and hidden failures. The model used Monte Carlo method for risk assessment of cascading failure. Several researchers used Manchester model to evaluate the blackout cost and mitigation methods [70–72].

COSMIC Models

Recently, a new nonlinear dynamic model of cascading failure in power systems (the Cascading Outage Simulator with Multiprocess Integration Capabilities, also known as COSMIC) has been introduced [73]. As a quasi-steady-state (QSS) model, COSMIC was able to simulate power system with a set of hybrid discrete and continuous differential algebraic equations, as well as protection systems and machine dynamics. The model involved a wide variety of mechanisms including rotating machines, exciters, governors, power flows using nonlinear power flow equations, load voltage responses, discrete changes (e.g. components failure and load shedding) and etc. COSMIC used a recursive process to compute the differential algebraic equations, which represented various mechanisms. Four kinds of loads, namely constant power (P), constant current (I), constant impedance (Z), exponential (E), or any combination thereof (ZIPE) have been modeled [74]. The model was validated by comparing the simulation results on IEEE 9-bus test system with PowerWorld, which was a widely used comprehensive commercial software [75]. Another simulation has been made and the results were compared with a simple dc-power-flow QSS model. The cascading in two models agreed in early stages, while the simulation were substantially different during later stages.

Multi-Timescale Quasi-Dynamic Models

The multi-timescale quasi-dynamic model is another recently published dynamic simulation model [76]. This model employed quasi-dynamic approach to overcome ambiguity of time in conventional cascading failure analysis models. It provided dynamic simulations such as load variation and generator excitation protection. An improved re-dispatch model based on sensitivity analysis has also been proposed due to the reason that, in reality, transmission loading relief (TLR) and re-dispatch normally took 10 to 30 minutes. Impacts from dispatchers' involvements were analyzed. The model was tested on IEEE 30-bus test system for generator protection and cascading failure stage characteristics. It has also been validated using data from US-Canada Northeast power grid. The test system had 410 buses, 882 branches, 200 generators and 162,121 MW load. The result showed consistency with the cascading failure happened in 2003.

ASSESS Models

ASSESS is a commercial cascading failure analysis tool developed by Reseau de Transport d'Electricite (RTE) in France, in collaboration with National Grid Company in UK [77]. It allowed a wide range of uncertainty input. Four main facilities have been provided in the model. The first one was a security-constrained AC optimal power flow [78]. The second one was a quasi-steady state simulator that could model dynamics of systems [79]. The third one was a full time-domain simulator that allowed modeling many controls in the system, such as field current limiters on generators, governors, or zone 3 relay protection on overloaded transmission lines [80]. The last one was the access to many statistical tools. ASSESS provided a wide range of models including sequences of events, protection settings, line ratings, fault clearance time, etc. The deficiency is that it requires numerous data and time to simulate, as well as the professional users to operate.

TRELSS Models

Transmission Reliability Evaluation of Large Scale Systems (TRELSS) is also a commercial cascading failure analysis tool, which is developed by EPRI in cooperation with Southern Company Services [81]. The model can simulate the cascade process as a sequence of quasi-steady state system conditions caused by a sequence of tripping events. The Protection and Control Group (PCG) has been modeled as actions of breakers. The voltage problems have also been taken into account using a quasi-steady state AC power flow model. The model was used to identify the most extreme initiating events simulated on Western Interconnection power flow model (approximate 16,000 buses) [82].

Dynamic PRA Models

A two-level dynamic probabilistic risk assessment (PRA) model was introduced in [22]. The paper divided the cascading failure procedure into two phases. The two different models were supposed to use to analyze the cascading failure. The impact of changes in power system, such as variation of cross-border power flows, wind generation penetration, maintenance and shut-down of power plants, were simulated in the model. The model was tested on New England Test System (NETS) and New York Power System (NYPS) 69-bus test system, using Monte Carlo methods. The thermal effects had been proved to contribute significantly to cascading failure.

Recently, an improved PRA model has been proposed [83]. It analyzed slow and fast cascade phases separately in two decomposition models. The results led to the estimation of the frequency of dangerous scenarios and their magnitudes in terms of energy not served. A clustering methodology was proposed to group scenarios at the end of slow cascade phases so that the dynamic analysis was manageable from a computational viewpoint. The dynamic models of generators were considered [84].

2.2.5 Interdependent Models

Interdependent networks have been studied for years in many areas [85]. The emergence of smart grid accelerates the coupling between the conventional power systems with computer cyber network, forming an interdependent network. However, smart grid brings not only the convenience, but also more risks from cyber network. The possible failures of control systems that connect to the cyber network, such as SCADA, may contribute to the cascading failures to some extent. Cyber attack that aims at smart grid can be various [86]. One such incident occurred in 2003, when the "Slammer" Internet worm took down monitoring computers and successfully blocked commands used to operate other power utilities [1]. Another case, as mentioned before, was the cyber network attack happened in Ukraine in 2015 [131]. The SCADA distribution management system has been remotely controlled by attacker, which led to disconnection of several substations for hours, and the failure cascade to more areas across Ukraine. Actually, once failure begins, due to the coupling effect, the failure in cyber network may directly affect the status of physical devices, thus aggravating the cascading. So far, many studies on assessing risks from cyber network appeared [87–91], while only limited researches have been proposed in studying the interdependent models due to the complexity mechanisms and difficulties in validation in reality.

Complex Network-Based Interdependent Models

Interdependent infrastructures are common in many real world networks, such as transportation networks and economic networks, and have been widely studied using complex network theory [92, 93]. The application on power system coupling with cyber network appeared later in [94]. The reason that complex network-based interdependent networks are divided into this group rather than complex network models is to highlight the importance and future trends of such models.

Buldyrev et al. introduced an interdependent models in [129], analyzing the robustness of interacting networks subject to cascading failures. The model used real-world data from a power network and an Internet network that were implicated in the cascading failure in Italy in 2003. An analytical solution has been illustrated that the removal of the critical nodes would lead to cascading failure and a complete fragmentation of two interdependent networks. The paper also concluded that a broader distribution will increase the vulnerability to random failure, which was opposite to how a single network behaved. The model has been improved by Shao in [95]. The model, based on random multiple support–dependence relations, analyzed the robustness of the power grid. The results were proved to be consistent with those simulated on other single networks. Another model has been proposed by Schneider in [96], whose data were also based on Italy communication system and power network in 2003 blackout. The model showed how to mitigate the cascading failure in a coupled network by selecting a minimum number of autonomous nodes. The disadvantages of these models are that they fail to consider enough electrical property, while focus only on purely topological parameters.

The interdependent model that took electrical property into consideration has been presented in [97]. The model also consisted of power grid and the supporting Control and Communication Network (CCN), where the substation, the generator and the router have been involved. The authors analyzed the minimum nodes that should be removed from both networks to totally fail the network, and proved it to be equivalent to NP-hard problem. A near optimal solution has been provided and tested using 2003 Italy blackout data. Later, a new coupling model that consisted three subnetworks, namely power grid, communication network and interdependency network, has been presented by the same author [98]. A sensitivity analysis was performed to evaluate the mitigating method in the model. A load control policy has also been simulated on the model, focusing on load factor and interdependence degree.

More recently, a more comprehensive model has been proposed in [99]. The model used a mesh network that considered power system characteristics. Moreover, bidirectional links including data uploading and command downloading channels were modeled. These links were supposed to connect all cyber network nodes as well as a corresponding physical node in power grids. The fragility of the coupling model has been studied under various cyber-attacks, such as denial-of-service (DoS) attacks, replay attacks and false data injection attacks. Load shedding and relay protection have been involved in the model.

Inter-Dependent Markov Chain Models

Inter-Dependent Markov Chain (IDMC) model provided a probabilistic cascading failure analysis framework to capture the effects of interdependencies among physical networks and power system [100]. The IDMC model enabled a system-level prediction of the interdependent systems with trackable

details of the system. It assumed that the communication system was more vulnerable when a failure occurred, thus increasing failure probability in the power grid. On the other side, when power system components failed, it may trigger cyber networks as well with a given probability. The IDMC model demonstrated that interdependencies between two systems can greatly affect each other on failure distribution sizes. The simulation illustrated that systems with exponentially distributed failure sizes tend to be less robust as evidenced by the power-law distributed failure sizes for both systems.

Flocking-Based Hierarchical Cyber-Physical Model

A hierarchical cyber-physical multi-agent model of smart grid based on flocking theory has been presented in [101, 102]. The model considered dynamic nodes (generators in this model), PMU and local cyber-controller. The frequency, phase angle and other related parameters were involved in the generators which were regarded as physical parts. PMU and local cyber-controller served as cyber elements. The model concentrated on control strategies for robustness and resilience of a coupling system. The potential performance improvements has been tested using New England 39-bus power system with various faults and communication delays.

2.2.6 Other Models

Potential Cascading Models

The Potential Cascading Models (PCM) is an integrated function of Physical and Operational Margins (POM) [103]. POM is an AC power flow analysis application which utilizes the full Newton-Raphson method to solve a nonlinear power flow. PCM, integrated in POM, aims at predicting potential cascading failures. It allows the identification of initiating events and cascading chains, the ranking and visualizing of cascading outages. It utilizes the "cluster" approach to get the initial N-1 or N-2 contingencies. The cascading failure analysis was performed using US 2007 Eastern Interconnection model summer peak case consisting of approximate 50,000 buses and 65,000 branches. Most results were consistent with prior manually analysis, while some unidentified potentially cascading-initiating possibilities were found. Later, the model was used to test remedial actions with the same data [104]. Optimal remedial actions were applied at each stage during the cascading until the cascading has been fully stopped. These remedial actions included MW dispatch, MVAr dispatch, transformer tap change, phase-shifter adjustment, capacitor and reactor switching, emergency load curtailment, line switching in and out and optimal capacitor placement (new sources of reactive power) [105]. It showed that all identified potential cascading failure can be prevented using the proposed remedial actions. The paper also studied the minimum amount of load shedding needed to mitigate the cascading.

Hidden Failure Models

The Hidden Failure models were first introduced in [106]. As mentioned before, hidden failure plays a critical role in cascading failure. To be specific, if one line suffers failure, all the lines that share the same bus of that failed line are exposed to the unexpected tripping. The model, based on DC load flow, focuses on hidden failure modeling and generator re-dispatch. Parameters, including system loading level, hidden failure probability, spinning reserve capacity and control strategy, were analyzed in the

model. The model used fast simulation technique and heuristic random search to find critical relays. The model was tested on WSCC 179-bus test system with mitigation approaches and risk assessments being studied.

Historical Data-Based Models

Historical data-based models may not be the best models to analyze the cascading failure. However, these models focused on accurately reproducing the history events by demonstrating how to modify the models and data. Blackout events can only be better and deeper understood with such an exactly blackout reproduction.

On August 10, 1996, a major blackout occurred in western North America with loss of 30,390 MW of load. Approximate 7.49 million people were affected during the outage. In [107], a simulation of the event has been made in order to better understand the blackout by using Electric Power Research Institute (EPRI) ETMSP models. At first, the authors used standard WSCC dynamic data base to get a result, which mismatched the real disturbance data. Afterwards, they showed a series of modifications in data and models to make the simulation results approximate the history data. The paper proved that there were many factors, which contributed to the severity of the cascading failure, need to be taken into consideration in a cascading failure analysis model.

2.2.7 Comparison Between Models

A brief comparison between different models mentioned above has been summarized in Table 2.1. The advantages and disadvantages are presented. It can be concluded from the table that no existing models can capture all the mechanisms during the cascading failure, while each model has its specific concentration and application. To sum up, topological models can be used to analyze vulnerability, component criticality, robustness, and structural view of a power network, while these models still need further improvements on including more electrical properties so as to achieve more accurate results. Stochastic simulation models take into account most possible uncertainties, while they fail to simulate dynamics of power system. High-level statistical models are able to predict average propagation of cascading and blackout distribution size quickly by ignoring cascading details. Dynamic simulation models capture dynamics of the system with a relatively high computation cost. Interdependent models consider both physical and cyber network as well as their interdependencies, while the validation remains a problem. Other models concentrate on a specific area rather than the whole cascading failure mechanisms, so that they can serve as complementary tools.

Table 2.1: Comparison of Cascading Failure Analysis Models with Advantages and Disadvantages Summarized

Models	Advantages	Disadvantages
1 Topological Models	Analyze the power system as a whole. Quickly discover the unexpected emergence of collective behavior.	
1.1 Modified Topological Models	Consider some essential electrical features. Electric-topology-combined metrics provided. Fast evaluation on robustness, vulnerability, risk probabilities, etc.	Lack of electrical features.
1.2 Maximum Flow Models	Maximum flow method applied.	

	Node and line weight considered.	
2 Stochastic Simulation Models	Take into consideration most uncertainties during the cascading.	Fail to consider dynamic instability and cascading details
2.1 PRACTICE Models	"Single-path" mode and "multi-path" mode introduced.	
	Event-tree-based approach adopted.	
2.2 Markov Chain Models	Non-locally propagation illustrated.	
3 High-Level Statistical Models	Enable quantitative risk assessment.	Ignores all details of cascading.
	Simple and tractable.	
3.1 CASCADE Models	Failure probability are related to load level.	
3.2 Branching Process Models	Can be regarded as an improved CASCADE model.	
	Consider each failure component from early stages through a given distribution.	
4 Dynamic Simulation Models	Simulate most dynamic mechanisms involved in cascading.	Slow simulation. Detailed power system data required.
	Provide deeper understanding in cascading failure.	
4.1 OPA Models	Takes into consideration the effects of dispatching, automation, communication, relay protection, operation mode and planning.	
	Tree contact, failure of lines due to line heating, and UVM modeled	
4.2 Manchester Models	AC power flow adopted.	
	Monte Carlo methods is applied to risk assessment.	
4.3 COSMIC Models	Consider non-linear dynamic mechanisms.	
	Various load and relay models involved.	
4.4 Multi-Timescale Quasi-Dynamic Models	Employ quasi-dynamic approach.	
	Approximate time evolution considered.	
	Improved re-dispatch simulation.	
4.5 ASSESS Models	Security-constrained AC optimal power flow provided.	
	Using quasi-steady state simulator to model dynamics of system.	
	Modeling controls in system through full time-domain simulator.	
	Access to many statistical tools.	
4.6 TRELSS Models	Take into account actions of breakers.	
	Voltage problems modeled using quasi-steady state AC power flow.	
4.7 Dynamic PRA Models	Two levels of cascading failure simulated using two different models.	
	The impact of various changes in system simulated.	
5 Interdependent Models	Analyze interactions between coupling networks.	Difficult to validate. Detailed mechanisms ignored.
5.1 Complex Network-Based Interdependent Models	Vulnerability of the whole coupling system analyzed.	
	Interdependencies depicted.	
	Computer and cyber risks considered.	
5.2 Inter-Dependent Markov Chain Models	Enable a system-level prediction with trackable details of the system.	
5.3 Flocking-Based Hierarchical Cyber-Physical Models	Dynamic nodes, PMU and local cyber-controller modeled.	
	Frequency, phase angle, and other related parameters involved.	
	Control strategies provided.	
6 Other Models	Concentrate on specific parts of mechanisms.	Focus only part of cascading failure mechanisms.
6.1 Potential Cascading Models	"Cluster" approach employed.	
	Aim at predicting potential cascading failures.	

6.2 Hidden Failure Models	Hidden failure and generator re-dispatch considered.
6.3 Historical Data-Based Models	Accurately reproduce history events. Complementary to existing models.

2.3 Interim Conclusion

This section focuses on cascading failure in power systems, presents various features related and reviews the current progress on cascading failure analysis tools and models. Cascading failures can be initiated by various causes, among which many are unexpected and uncontrollable, so that a total prevention is beyond the modern technology. However, by understanding these causes, corresponding strategies, such as suitable N-1 or N-2 criterion and more appropriate relay setting regulations, can be applied. Moreover, a deeper understanding of these causes can make the analysis models better simulate the cascading failure. The complicated mechanisms during the cascading are challenges in understanding and modeling. A variety of models have been provided, in which models are able to capture only a subset of the many mechanisms. The impact of the blackout should be carefully evaluated from both electrical perspectives, and economical and social perspectives. The restoration approaches play significant roles after the blackout occurs. A fast system restoration will greatly reduce the effect and cost of the blackout. Risk assessment can be done through detailed cascading failure simulation or probabilistic approaches. The dynamics of the system during the cascading draw wide attention due to complex mechanisms. There have been many conventional power system dynamic analysis tools, while most of them fail to consider so many interactions between components in the whole system during the cascading. Robustness and vulnerability have also been studied in order to find the most critical component in the power grid, as well as construct more reliable system. Once cascading begins, the mitigation and prevention strategies will be applied. These strategies require fast computation speed and accurate cascading prediction in order to act before the cascading events triggering. The emergence of smart grid brings not only benefits to power system, but also more risks from cyber network. The interdependency between coupling networks will be an important topic in cascading failure analysis. The impact of climate change should be taken into consideration as well.

This section also reviews most existing cascading failure analysis models by dividing them into six groups. Comparison and discussion have been presented. Each model has its own advantages and disadvantages, and the different combinations of these models will contribute to a more comprehensive and useful model.

Despite the existence of many complex models presented, the analysis on cascading failure is still an open question. To gain a more complete idea of the cascading failure in the future, it is worth investigating how different models can be combined into a more comprehensive one. To choose appropriate models under different situations and cascading phases also need to be carefully considered. Topological models can include more electrical features. Dynamic models can simulate more mechanisms, such as uncertainties of components triggered, and more details on generator configurations. Possible mitigation strategies can be adopted to models to better reveal real power system control. In addition, validation of many stochastic models and interdependent models are essential. The impact from renewable energies during the cascading failure should also be carefully studied.

Chapter 3

A Power System Dynamics-Based Cascading Failure Simulation Platform for Power Networks

Large blackouts around the world are mainly caused by cascading failures; it is thus of great importance to study the mechanism of cascading failures in power systems, so that the mitigation and prevention can be achieved. While a variety of analysis tools have been proposed in order to investigate this phenomenon, Power System Dynamics (PSD) are rarely considered in the existing studies. In this chapter, a novel PSD-based Cascading Failure Simulation Platform (CFSP) is proposed, which takes into consideration PSD, load models, relay models, and automatic load shedding mechanisms. The proposed PSD-based CFSP is tested using the IEEE 68-bus power system, and validated by studying the cascading failure propagation profile, which demonstrates great agreement with historical power system cascading failure patterns. The comparison between CFSP and a Complex Network (CN)-based cascading failure model has been conducted to vigorously study the impact of PSD. The failure propagation profiles of the two models match well for the slow cascade stage when PSD do not dominate the cascading process, but substantial difference appears when large-scale failure starts, i.e., fast cascade stage. Moreover, the impact of varied transmission line capacity on blackout sizes is also investigated for both CN-based model and the proposed CFSP, with comparison presented and analyzed.

The content of this chapter is mainly based on and modified from the following academic papers:
***Hengdao Guo**, Samson Shenglong Yu, Tat Kei Chau, Tyrone Fernando, Herbert Ho-Ching Iu, and Ciyan Zheng. "A Power System Dynamics-Based Cascading Failure Simulation Platform for Power Networks", IET Generation, Transmission and Distribution, submitted.*

3.1 Introduction

Power grids are critical infrastructure in modern cities as their reliability and vulnerability significantly affect people's daily lives. Cascading failures, one of the severest power system instability problems, may cause large-scale damages and massive economic losses to society. Cascading failures are defined

as "the uncontrolled loss of bulk electric system facilities triggered by an incident (or condition) at any location resulting in the interruption of electric service that cannot be restrained from spreading beyond a pre-determined area [5]." These large-scale cascading blackouts are infrequent but substantially risky due to their catastrophic consequences. For instance, 2003 North America blackout affected approximately 50 million people and caused losses of up to 10 billion US dollars [1]; further, 2012 India cascading blackout made 700 million people in 20 (out of 28) states live without electricity.

These large-scale cascading outages have drawn substantial attention from both academia and industry in recent years, with intensive studies conducted to investigate the mechanism of cascading failures [165]; however, an exhaustive simulation framework to assess cascading mechanisms is still far from being accomplished [19]. In [34] an Extended Betweenness Based Cascading Failure Simulator (EB-CFS) was introduced. The EB-CFS is based on topological models and uses modified Power Transfer Distribution Factor (PTDF) to evaluate the vulnerability of components in power systems. Similar topology-based model was presented in [35], which was derived from the admittance model proposed by the authors in [166]. The model focuses on grid robustness and structural features against cascading failures in terms of the generators' distribution topology. These topological models have been advanced in investigating the power grid structure at a holistic level, with a particular focus on the vulnerability and criticality of components, and the robustness of the overall system. However, it has been pointed out that solely focusing on the topological features without considering the characteristics of physical power systems can lead to erroneous and misleading results due to the omission of electrical properties [21].

In [46] a network-based Markov chain model was introduced to study the propagation dynamics of the entire power network. A stochastic cascading simulation technique was provided in [19] and [43], which takes probabilistic factors such as hidden failures into consideration. Such stochastic methodologies consider uncertainties, including human mis-operations and weather effects, which most deterministic models fail to incorporate. CASCADE model, proposed in [51], is a high-level statistical model with a high computation speed. It assumes that all identical components have a random initial load, and generate a perturbation to each load to initiate the cascading failure. The redistribution mechanisms of CASCADE are much simpler as they ignore many electrical properties and power network internal interactions. Recently, another high-level statistical model was introduced in [60], which used multi-type branching process to analyze the statistics and interdependencies of cascading outages. Both stochastic and high-level statistical models are advanced in risk assessment of cascading failures, while lacking important details regarding complex mechanisms during the cascading failure. Consequently, these models, although valuable, make limited contributions to the thorough study of cascading failures.

There are some distinctive perspectives in studying cascading failures. In [130], a coupled complex network was introduced in order to study the cascading failure propagation in both power networks and communication/Internet networks. Researchers in [167] investigated the coupling effects on cascading failure through a stochastic method. An "influenced graph" model was presented in [50], aiming to provide risk assessment of cascading failures. These pioneering ideas are innovative and useful in future studies; however, these models are inadequate to explicitly study the comprehensive mechanism of cascading failure in power system, i.e., the way cascading failures propagate, and require further

improvements.

In this chapter, a novel CFSP is established based on generator and power network models, for the purpose of explicitly demonstrating how PSD impact the propagation of cascading failures. The proposed CFSP has following advantages over most existing cascading failure models:

(i) In spite of numerous mathematical models proposed, researchers mainly focus on the case of DC/AC power flow model. This kind of model substantially lacks the consideration of PSD, which is the paramount mechanism that drives the power system operations, thus cascading failures in power grids [13]. Most cascading failure models are based on the premise that the system reaches the steady state after each failure, so that power flow re-calculation and load redistribution can be proceeded. Nevertheless, this assumption can be made only during slow cascade stages, where the failure propagates slowly, ranging from tens of minutes to several hours. During this stage, the voltage variation has limited impact to the system. In terms of fast cascade stages which usually have a time scale of milliseconds to several seconds, a large number of components are tripped within a very short period of time, during which PSD dominate the process. The proposed CFSP takes into consideration PSD and is thus able to provide useful and meaningful insight on its impact. In [76], the author describes a multi-timescale quasi-dynamic model for simulating cascading failures, while the model still fails to explicitly explain how PSD and transience affects the failure propagation or the final consequence of cascading failures.

(ii) Many cascading failure models assume generators as constant voltage sources, where their voltages remain unchanged when disturbances occur [34]. In reality, however, the output voltage of generators in power systems can change rapidly along with triggering events and thus dramatically affecting the cascading propagation.

(iii) Relay operation modeling is of critical significance in simulating the complicated cascading procedure. In most cascading failure models such as [76] and [41], however, component failures are over-simplified, and are instantaneously tripped the moment it is overloaded. In reality, relay protection system involves complex mechanisms; for example, many transmission lines are able to work appropriately for a period of time even when the current exceeds the preset threshold, according to different relay settings. Although there are some research articles that have considered the relay protection modeling, e.g., [34] and [73], they still utilized simplified and invalid assumptions by assigning a fixed time delay to protection relays after the electric component is overloaded. In this chapter, the International Electro-technical Commission (IEC) based Inverse Definite Minimum Time (IDMT) relay operation system for over-current protections are incorporated into the proposed CFSP in order to mimic a more realistic protective mechanism.

(iv) There are limited models considering PSD, such as [73] and [76]. The proposed CFSP adopts their advantages and overcomes their weak points. Meanwhile, some commercial tools (e.g., PowerWorld, PSS/E) have the ability to model the PSD, while they are not designed for the cascading failure. This makes these tools difficult to use. Moreover, due to the large computational demand during the modeling of cascading failure, the users also have to purchase the computing power in order to realize the simulations. The proposed CFSP is built specifically for modeling the cascading failure, with a well trade-off between the accuracy and efficiency of the simulation. Additionally, the proposed CFSP provides an easily modified framework that allows researchers to adopt their own models.

3.2 Mathematical Model of Cascading Failures and Power System Dynamics

In this section, a brief description of CFSP is presented, followed by a concise introduction of various mechanisms involved in the proposed platform.

3.2.1 Generator and Network Models

The generator model can be represented by a 4th-order Ordinary Differential Equation (ODE) as follows [168],

$$\dot{\delta} = \omega_b(\omega - \omega_s), \tag{3.1}$$

$$\dot{\omega} = \frac{\omega_s}{2H}\Big[T_M - (E'_d I_d + E'_q I_q) - D(\omega - \omega_s)\Big], \tag{3.2}$$

$$\dot{T}_M = \frac{1}{T_{CH}}\Big(- T_M + P_{SV}\Big), \tag{3.3}$$

$$\dot{P}_{SV} = \frac{1}{T_{SV}}\left(-P_{SV} + P_C - \frac{1}{R_D}\left(\frac{\omega}{\omega_s} - 1\right)\right). \tag{3.4}$$

where equations (3.1) and (3.2) together form a classical generator model, equations (3.3) and (3.4) are prime mover model and the governor model respectively. For additional detailed explanations and mathematical models, see [168]. Algebraic equations which consist of the stator algebraic equations and the network power balance equations can be expressed as follows,

$$0 = E'_d - V\sin(\delta - \theta) - R_a I_d + X'_q I_q, \tag{3.5}$$

$$0 = E'_q - V\cos(\delta - \theta) - R_a I_q + X'_d I_d, \tag{3.6}$$

$$0 = -P_i + \sum_{k=1}^{N} |V_i|\,|V_k|\,(G_{ik}\cos\theta_{ik} + B_{ik}\sin\theta_{ik}), \tag{3.7}$$

$$0 = -Q_i + \sum_{k=1}^{N} |V_i|\,|V_k|\,(G_{ik}\sin\theta_{ik} - B_{ik}\cos\theta_{ik}). \tag{3.8}$$

Functionally, equations (3.1) - (3.8) together form the set of Differential Algebraic Equation (DAE). The detailed notations of used parameters are introduced in Table 3.1.

If the power system consists of m generators and n load buses, then there will be $4m + 2m + 2n$ equations with $x = [x_1^T \cdots x_m^T]^T$ as the state vector where $x_i = [\delta_i \quad \omega_i \quad T_{M_i} \quad P_{SV_i}]^T$ for each generator. $u = [I_{d-q}^T \quad V^T \quad \theta^T]^T$ is the set of algebraic variables where $I_{d-q}^T = [I_{d1} \quad I_{q1} \cdots I_{dm} \quad I_{qm}]^T$, $V = [V_1 \cdots V_n]^T$, $\theta = [\theta_1 \cdots \theta_n]^T$. Therefore, the dynamic power system can be mathematically described as the DAE in the following compact form [169],

Table 3.1: Nomenclature of generator and network models

Nomenclature	
δ	rotor angle
ω	rotor speed
ω_b	base synchronous speed
ω_s	rated synchronous speed
H	shaft inertia constant
T_M	mechanical torque applied to the shaft
E'_d	transient voltage of d-axis
I_d	d-axis current
E'_q	transient voltage of q-aixs
I_q	q-axis current
D	additional damping constant
T_{CH}	steam chest output torque
P_{SV}	steam valve position
T_{SV}	steam valve torque
P_C	controlled input
R_D	speed regulation
R_a	armature resistance
X'_q	q-axis subtransient reactance

$$\dot{x} = f(x, u), \tag{3.9}$$

$$0 = g(x, u). \tag{3.10}$$

where equation (3.9) stands for "slower" dynamics of all generators, and equation (3.10) represents "faster" dynamics of transmission and distribution power networks with electricity transmission and electric loads. The generator dynamics are slower due to the fact that all generators in this study have mechanical procedures; whereas the fast dynamics primarily refer to the voltage and current signals in the network, which can vary significantly faster than the mechanical procedures [170]. It is shown that, during the fast cascade stage, i.e., when failures propagate quickly, the failure propagation is dominated by the transient states. In this study, more emphasis will be put on PSD in order to better understand the mechanisms during the cascading failure.

Constant impedance (Z), constant current (I), and constant power (P or Q) components, together form the ZIP load model, are incorporated into CFSP [171]. The frequency dependency effect in ZIP load model has been considered by incorporating the active and reactive power frequency dependency factor $(1 + K_{pf} \cdot \Delta f)$ and $(1 + K_{qf} \cdot \Delta f)$ [172]. Therefore, the load model can be described as follows,

$$P = P_0 \left[p_1 \left(\frac{V}{V_0} \right)^2 + p_2 \frac{V}{V_0} + p_3 \right] (1 + K_{pf} \cdot \Delta f), \tag{3.11}$$

$$Q = Q_0 \left[q_1 \left(\frac{V}{V_0} \right)^2 + q_2 \frac{V}{V_0} + q_3 \right] (1 + K_{qf} \cdot \Delta f). \tag{3.12}$$

where p_n and q_n are ZIP coefficients subject to $p_1 + p_2 + p_3 = 1$ and $q_1 + q_2 + q_3 = 1$, P_0 and Q_0 are the real and reactive power at V_0, $\Delta f = f - f_0$, K_{pf} and K_{qf} are the frequency sensitivity coefficients, f is the nodal voltage frequency and f_0 is the nominal frequency respectively. Load types can be modified by adjusting the configuration of p, q, K_{pf} and K_{qf} for each load, which makes the system more realistic. Detailed configuration introduction are explained in Section 3.3.

3.2.2 Protection Relays

Protection relay modeling and failure determination play a significant role for establishing the cascading failure platform. The mal-operation of the relay has been proved to be one of the most critical reasons that could cause a the cascading failure [1]. There are various types of relays for generators, transformers, and transmission lines. The protective relay model based on IEC standards in [173] is adopted in the proposed model. The relay follows IEC-based IDMT settings for over-current protections. Generally the relay tripping time is acquired by,

$$T_{\text{cc}} = \frac{K'}{(\frac{I_{\text{line}}}{I_{\text{cap}}})^{\alpha'} - 1} \cdot T_{\text{MS}}, \tag{3.13}$$

where T_{cc} is the tripping time for constant line current I_{line}, K' and α' are the characteristic constants, I_{cap} is the current capacity, or pickup current, of the over-current relay, and T_{MS} is the time multiplier setting [173].

In practice, transmission line power flow is monitored rather than current, and the ratio of transmission line power and transmission line capacity, i.e., $P_{\text{line}}/P_{\text{cap}}$ is used to calculate T_{cc}. In this case, equation (3.13) is still valid by manipulating the characteristic constants K' and α'. Then equation (3.13) becomes (3.14) with updated characteristic constants K and α, which is shown as follows,

$$T_{\text{cc}} \approx \frac{K}{(\frac{P_{\text{line}}}{P_{\text{cap}}})^{\alpha} - 1} \cdot T_{\text{MS}}, \tag{3.14}$$

The transmission line power flow in reality is time-varying, and the power flow $P_{\text{line}}(t)$ across the transmission line is approximated using the equation as follows [174],

$$P_{\text{line}}(t) = \frac{|V_S(t)|\,|V_R(t)|}{X_{\text{line}}} \sin(\delta_S(t) - \delta_R(t)), \tag{3.15}$$

where $V_S(t)$ is the voltage of the bus at the sending end, $V_R(t)$ is the voltage of the bus at the receiving end, X_{line} is the reactance of the transmission line, $\delta_S(t)$ and $\delta_R(t)$ are voltage angles of the sending-end voltage and the receiving-end voltages. With time-varying line power flow, equation (3.14) is no longer valid for computing the relay tripping time, which will be calculated in the next subsection.

3.2.3 Transmission Line Tripping Determination

Following the above discussion, in this study, the transmission line tripping in the CFSP is determined based on the extent of overloading of a particular transmission line, which is computed with monitored power flow, IDMT relay settings and a cumulative overload function. With regard to the threshold of transmission lines, the capacity tolerance parameter γ is used. To be exact, the value of transmission line capacity P_{cap} is defined as,

$$P_{\text{cap}} = P(0) \cdot (1 + \gamma), \tag{3.16}$$

where $P(0)$ is the initial power flow of transmission line, and the capacity tolerance parameter γ is able to display the relationship between transmission line capacity and load.

For time-varying power flow in transmission lines, the cumulative overload function between t_s and $t_s + \tau$ is introduced to calculate the relay operation time as follows, which is modified from (3.14),

$$C(t, \tau) = \int_{t_s}^{t_s+\tau} \frac{T_{\text{step}}}{K \cdot T_{\text{MS}}} \left[\left(\frac{P_{\text{line}}(t)}{P_{\text{cap}}} \right)^{\alpha} - 1 \right] dt, \tag{3.17}$$

where C is the cumulative overload index of the transmission line, t_s is the time when P_{line} starts exceeding its capacity, τ is the counter variable to record the time duration between the inception of over-current and relay action, T_{Step} is the time-step of numerical solver for DAE. Pseudo-code to calculate over-current counter variable τ for transmission line i (subscript i is omitted in the pseudo-code) and over-current relay activation time T_R is introduced as follows:

start pseudo-code for over-current relay action time T_R
$C = 0$; $\tau = 0$; record t_s; ····· %Initialize C_i and τ
while $C < 1$ **do** ····· %relay does not take action
    if $P_{\text{line}} > P_{\text{cap}}$ **then**
        Calculate C with equation (3.17)
    else
        $C = 0$ ····· %reset C when over-current disappears
    end if
    $\tau = \tau + T_{\text{step}}$ ····· %increment the counter variable
end while
$T_R = t_s + \tau$ ····· % relay triggers circuit breaker at T_R
end code

3.2.4 Generator Protection

In addition, off-nominal voltage and frequency relay protections are employed to protect generators. In practice, industry policies require that online generators stay interconnected and producing power for a while according to different overloading conditions before they get tripped. In the proposed CFSP, for simplicity, only the emergency threshold is set according to Western Electricity Coordinating Council (WECC) [175] and [176], below which the system may suffer great oscillations and possible collapse.

The voltage threshold is set 86% of its lowest normal operating voltage and the frequency threshold is set 95% of the system frequency, both with a time delay of 3.5 seconds. If the frequency or voltage recovers and settles at and above the threshold within the time delay, the timer will be reset. Only when the value keeps sitting below the threshold for over 3.5 seconds will the relay trip the generator (disconnected from the grid).

3.2.5 Automatic Emergency Load Shedding

Automatic emergency load shedding has been proved to be effective on mitigating the cascading failure [177]. Therefore, the Under Voltage Load Shedding (UVLS) and Under Frequency Load Shedding (UFLS) have been implemented in the proposed CFSP. To avoid the mis-operations of the relay during the transience when system suffers small disturbances, a pickup voltage/frequency and a time delay will be set. In order to comply with industry standards, according to WECC [175] and [176], the emergency pickup threshold for UVLS and UFLS are set the same with generator protection settings, which is depicted as T_{LS}. In this study, assumptions are made that each bus can only shed load once during the entire cascading process, and 40% of the initial load P at the bus that activate the relay is shed. Industries use much more complex load shedding schemes for different power systems and utility companies. The percentage of load to be shed normally ranges from 25% to 65% [178].

To sum up, the proposed CFSP involves generators' and transmission system's phasor model, and runs in time-domain simulation. The flowchart of the proposed CFSP is illustrated in Fig. 3.1, and the detailed algorithm is shown in **Proposed CFSP Algorithm**.

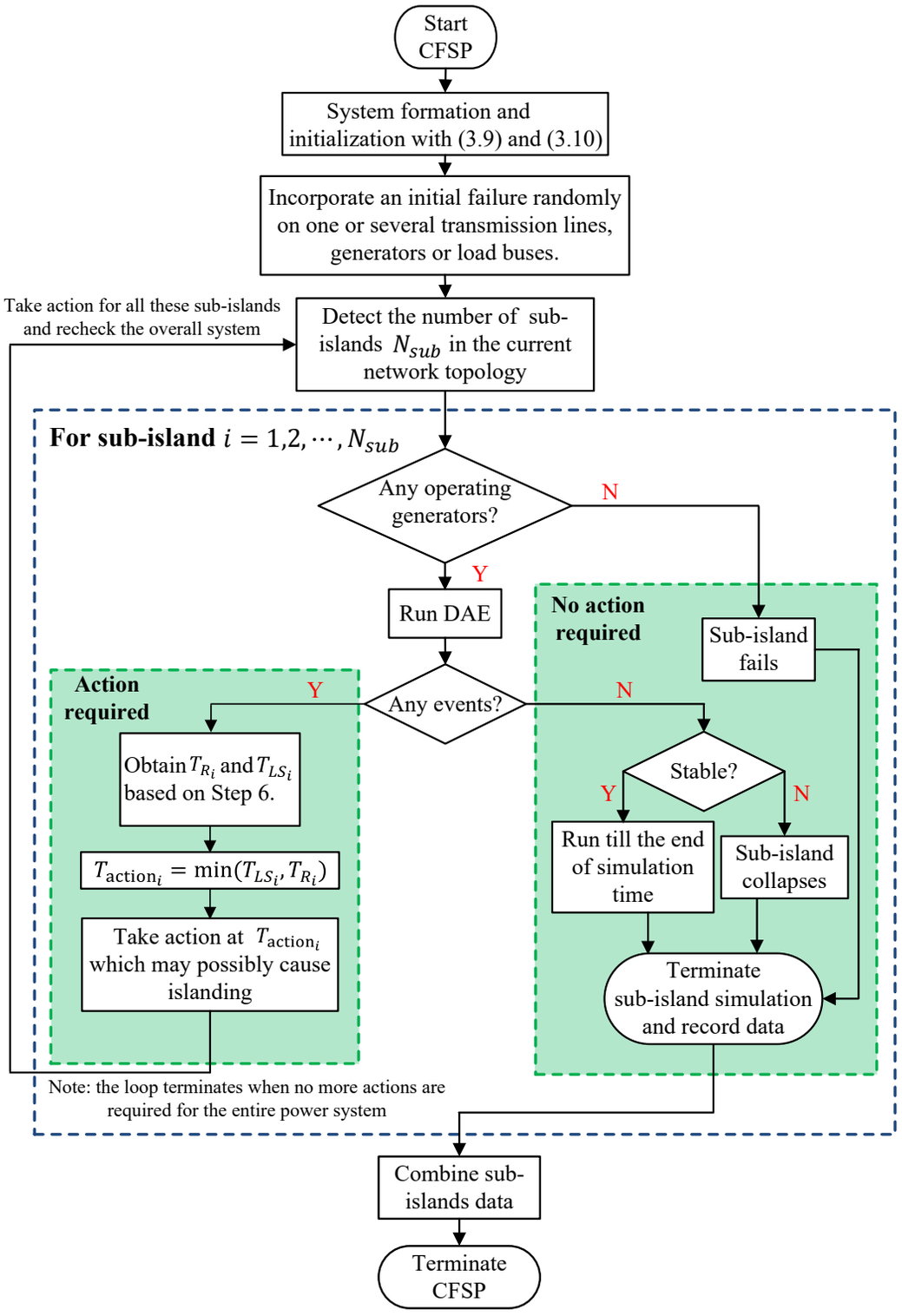


Figure 3.1: Flowchart of the proposed CFSP

Proposed CFSP Algorithm

Step 1: Initialization

- Initialize CFSP and DAE system with given information.

- Run dynamic simulation for the steady-state operation.

Step 2: Initial Failure

- Implement an initial failure, which can be the disconnection of one or several transmission lines, load or generator buses. For the case that a bus fails, all transmission lines connecting to it will be tripped.

Step 3: Detect the number of sub-islands

- The initial failure changes the topology of the network, and it is possible that the entire system is divided into several islanded sub-systems, or sub-islands. After the initial fault, the proposed CFSP identifies the number of all sub-islands, denoted by N_{sub}, and run simulations for them individually and simultaneously.

- Run from ***Step 4*** to ***Step 7*** for sub-island $i = 1, 2, \cdots, N_{\text{sub}}$.

Step 4: Check operating generators in each sub-island

- If there are any operating generators, go to ***Step 5***.

- On the other hand, if there are no operating generators in a sub-island, the sub-island is regarded "fail". Terminate simulation of this sub-island, record sub-island data and go to ***Step 8***.

Step 5: Run simulations and check events

- Run DAE for sub-island i with equation (9) and (10).

- Check if any event occurs, i.e., off-nominal frequencies or voltages, or over-current transmission lines. If yes, go to ***Step 6***, otherwise ***Step 7***.

Step 6: Proceed events

- There are two possible events in the proposed CFSP, namely shed load and trigger relay, which will alter the system condition and topology. Therefore, simulation needs to stop, check islanding, and run DAE with the updated system (i.e., Scenario 1 and Scenario 2):

- Scenario 1: When $V_{\text{Load}} < V_{LS}$ or $f_{\text{Load}} < f_{LS}$ for over 3.5 seconds continuously, the bus load will be shed instantly at time T_{LS_i}.

- Scenario 2: When $P_{\text{line}} > P_{\text{cap}}$ and $C \geq 1$ (the cumulative overload index, see Chapter 3.2.3), the line will be tripped at T_{R_i}, which is obtained by the pseudo-code in Chapter 3.2.3.

- For a sub-island with two events, obtain both T_{LS_i} and T_{R_i}. For a sub-island with only one event, i.e., either load shedding or relay operation, the time of the event is calculated, whereas the occurrence time of the other event is assumed infinity. For instance, if only load shedding occurs in a sub-island, the T_{LS_i} is obtained based on Scenario 1, and $T_{R_i} \to \infty$, vice versa. Then for all sub-islands with one or two events, the action time $T_{\text{action}_i} = \min(T_{LS_i}, T_{R_i})$.

- Take corresponding action at $t = T_{\text{action}_i}$, which may causing subsequent system islanding. Therefore, it is necessary to re-detect the number of sub-islands. Go back to ***Step 3*** and iterate the loop.
- The loop terminates when no more actions are required for the entire power system.
- Note that the CFSP is built in time domain for post-event analysis, which is not a chronological simulation.

Step 7: Check system state

- Check system stability when there are no events.
- If CFSP diverges, the sub-island is regarded unstable and collapsed. Terminate simulation of the sub-island, record sub-island data and go to ***Step 8***.
- If the system is stable, run simulation till the end of the simulation time, record sub-island data and go to ***Step 8***.

Step 8: Data Combination and Simulation Termination

- After simulations for all islanded power systems, combine the simulation results and terminate CFSP.

3.3 Model Validation

3.3.1 Power System Dynamics and Relay Operations

As mentioned before, most cascading failure models assume generators as constant voltage sources, where their voltages remain unchanged when disturbances occur. This false assumption dramatically affects the propagation of the cascading failure as well as the sequence of events being triggered. Particularly, the sequence of cascading failure propagation is proved to be dominant and can lead to entirely different simulation results according to [13]. Using the proposed CFSP, deviation of the generator output voltage can be simulated.

The test system used here is IEEE Standard New England 68-bus test system, which is a simplified equivalent power network of the inter-connected New England Test System (NETS) and New York Power System (NYPS) [179]. The simulations are conducted with MATLAB R2014a on an Intel(R) Core(TM) i7-6700 CPU @ 3.40GHz PC with 8GB RAM. The computational time for each simulation case on the IEEE 68-bus test system depends on the number of failures during the entire cascading failure simulations. The computational time ranges from 1.4 seconds to 10.6 seconds, with an average time of 5.8 seconds per simulation. Some important parameter for CFSP are listed in Table 3.2. The detailed mathematical models of the generation units can be found in [180]. Models of transmission lines and loads are introduced in Chapter 3.2.

Fig. 3.2 illustrates the frequency profile of 4 selected generator buses at the time when the first contingency and a triggering event occur. The first contingency, occurs at time $t = 1s$, is the

Table 3.2: Important Parameters for CFSP

p_1	0.3	p_2	0.3	p_3	0.4
q_1	0.3	q_2	0.3	q_3	0.4
γ	1.0	K	0.14	α	0.02
T_{MS}	1.0				

disconnection of the transmission line connecting bus 2 and bus 58. When $t = 7.88s$, the first cascading event occurs. The transmission line between bus 56 and 57 is tripped due to the over-current protection relay. There is no more event and the system finally settles to steady state over time. Fig. 3.3 demonstrates the change of output voltages of 4 generators under disturbances. It is noted that the system has observable oscillations during the two events, with corresponding output voltages and frequency deviating from their nominal values at each event. These varied output voltages will impact the power flow re-distribution, resulting in a totally different simulation results than the existing CN models, which proves the invalidity of the "constant voltage" assumption in power system cascading failure studies.

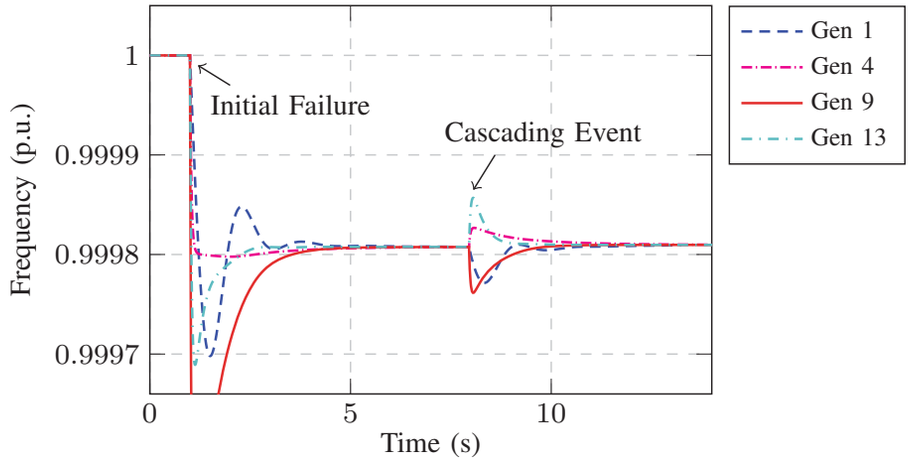


Figure 3.2: PSD validation with the proposed CFSP on generator frequency. Generator frequency deviates from their nominal values each time failures occur.

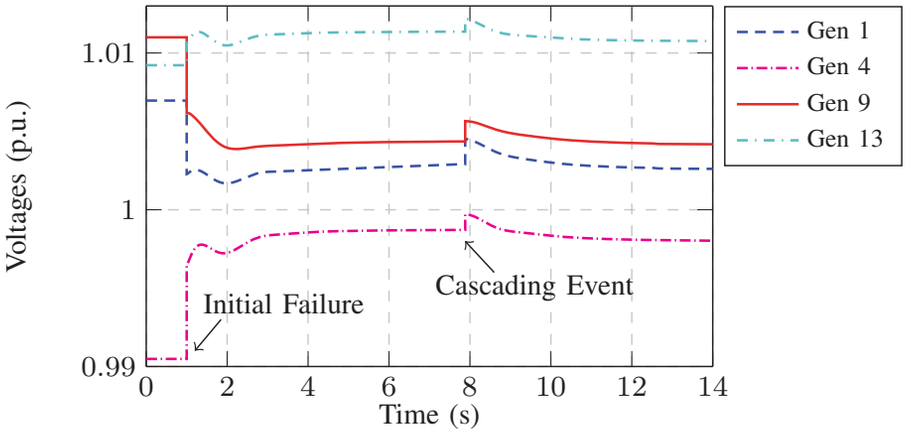


Figure 3.3: PSD validation with the proposed CFSP on generator output voltages. Generator output voltages deviate from their nominal values each time failures occur.

3.3.2 Automatic Load Shedding Mechanism

During the fast cascade stage, a considerable number of triggering events occur in a very short time period, which prevents human operators from manually taking actions to mitigate or stop the cascade. In this case, automatic load shedding mechanisms are rather critical in order to alleviate or prevent the possible cascading failures.

In the proposed CFSP, automatic emergency load shedding mechanism is incorporated in order to release the system stress. Fig. 3.4 shows a case that automatic emergency load shedding successfully prevents failures from propagating in a cascading manner, and brings the system back to a safe operating region. In this case, we also use the IEEE 68-bus test system and voltages of selected buses are illustrated in Fig. 3.4. The cascading failure is initiated by a transmission line disconnection between bus 65 and 66 at $t = 6s$. The second failure is triggered at $t = 13.16s$ with transmission line connecting bus 56 and 57 fails. After this failure, the voltage at bus 64 falls down below the pickup value and the timer for UVLS is activated at time $t = 24.52s$. After 3.5 seconds at time $t = 28.02s$, UVLS takes action and shed 40% of the load at bus 64. Afterwards, as can be seen in the figure, several buses that had falling voltages gradually settle to their nominal operating ranges. UVLS has been proven to successfully release the system stress and prevent cascading failures from propagating.

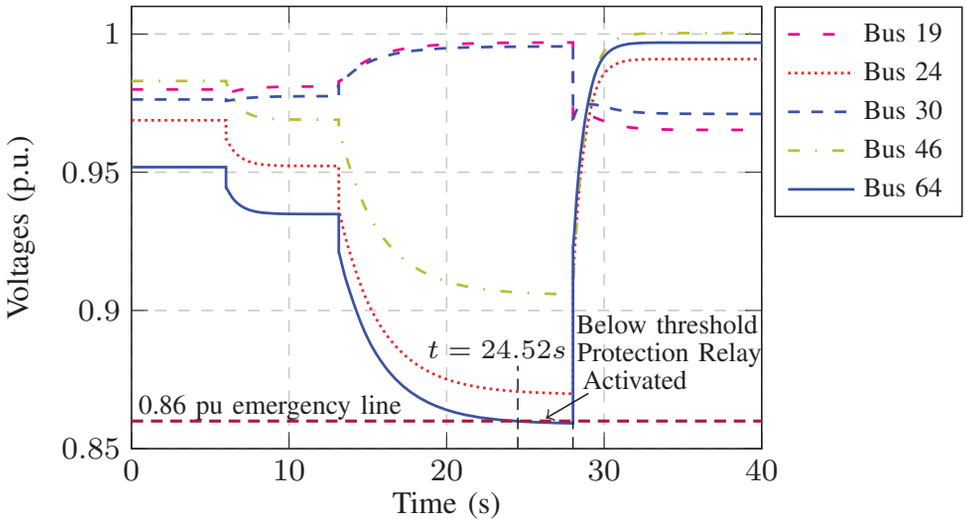


Figure 3.4: Load shedding validation with the proposed PSD-based CFSP. The voltage at bus 64 falls below $0.86p.u.$ at $t = 24.52s$, and the UVLS is activated $3.5s$ later at $t = 28.02s$.

3.3.3 Cascading Failure Propagation Profile Comparison with Historical Events

Comparison with historical data is conducted via two methods, 1) reproduce the historical cascading outages using exactly the same data; 2) produce similar characteristics and failure propagation profile. In this chapter, the second method is adopted to validate the proposed CFSP. One remarkable characteristic of cascading failures is that the cumulative failure components may go through slow cascade stage and fast cascade stage with different propagation patterns. During the slow cascade stage, the failure is triggered slowly and the cumulative number of failure components increases steadily; whereas, in the fast cascade stage, the failure triggers rapidly and the cumulative number of failure components grows sharply till the entire system fails or reaches another steady state. Although

not all cascading outages have slow cascade stages, there must be one or several fast cascade stages where failures propagate rapidly with timescales ranging from milliseconds to several minutes. This characteristic has been proven in many cascading outages [1], [181].

Fig. 3.5 demonstrates the failure propagation profile of the 2003 North America Blackout, which shows the typical characteristic of two cascade stages [1]. The slow cascade stage ranged from 16:05 to approximate 16:10, then the fast cascade stage occurred within the next 2 minutes with totally over 600 transmission lines, transformers, and generating units failed.

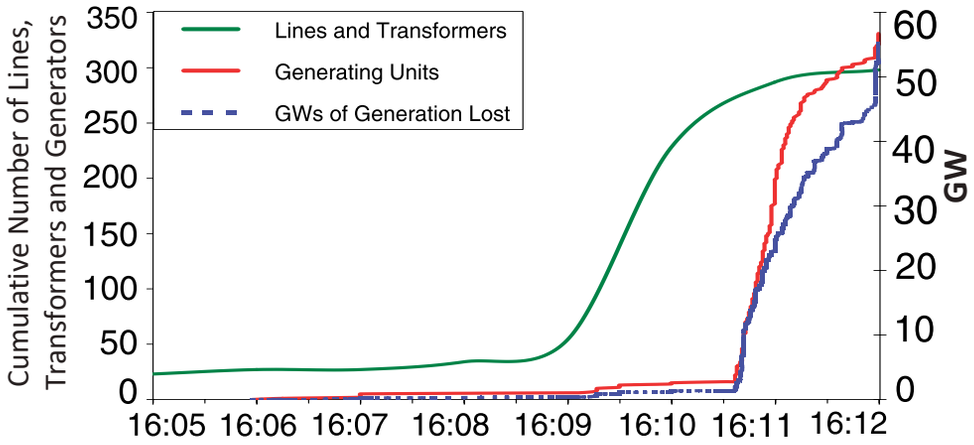


Figure 3.5: Historical data–2003 North America blackout propagation profile

In order to validate the proposed CFSP, the cascading failure propagation profile is achieved and compared with historical data. Particularly, the cumulative number of tripped generators, transformers and transmission lines are plotted. In this study, 68-bus test system is used, as shown in Fig. 3.6, and four cases are conducted with different initial failures. The simulation results are presented in Fig. 3.7. Simulations are conducted with different initial events with a same launching time at $t = 6s$. All four simulation results display the similar typical growing pattern with the historical blackout propagation profile that can be seen in Fig. 3.5.

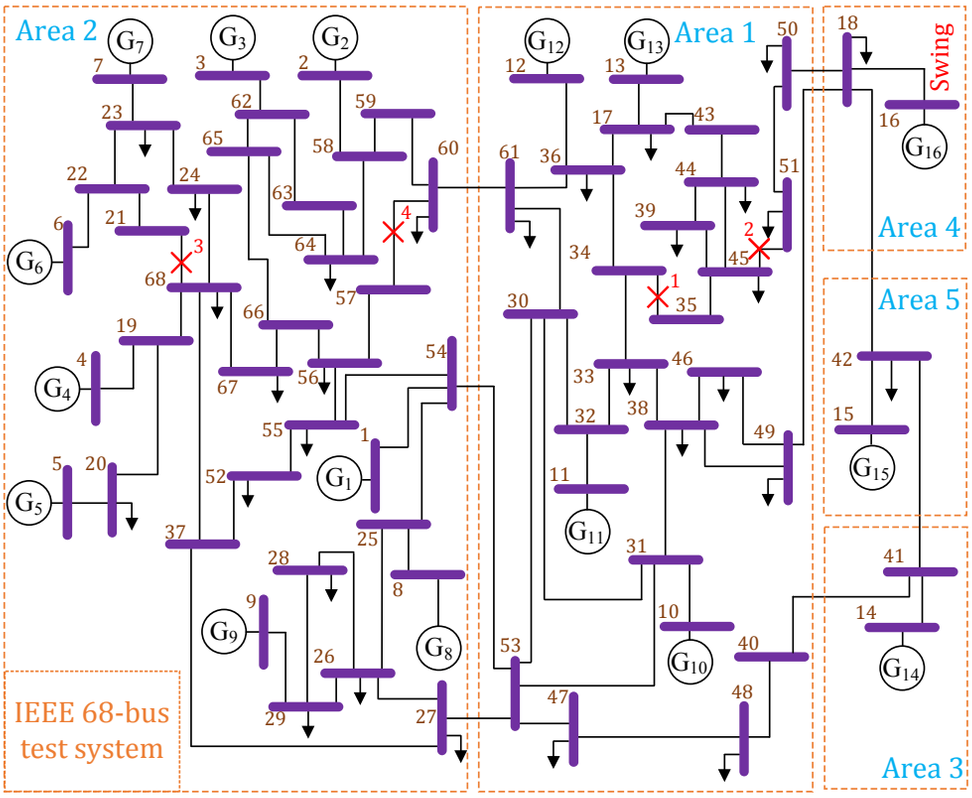


Figure 3.6: IEEE 68-Bus test system-initial failures of 4 cases are marked with 1,2,3 and 4

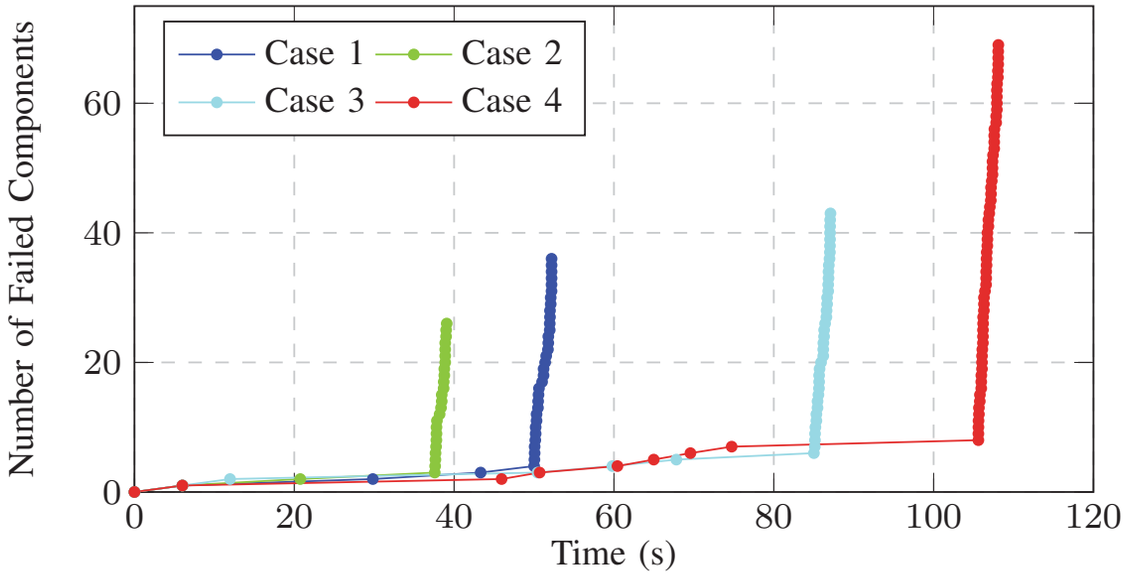


Figure 3.7: Propagation profile validation with the proposed PSD-based CFSP

3.4 Case Study

3.4.1 Sensitivity Analysis

Power grid is a complex system with a large number of configurations, whose settings may greatly affect the condition and operation of the system. In this part, sensitivity analysis of two important

parameters are studied.

Due to the economic consideration, most modern utility companies are trying to fully utilize the electric power transfer capability of transmission lines. This can be reflected through the relatively small γ we have defined previously. In order to investigate the impact of capacity on the cascading failure mechanism, sensitivity analysis of γ is conducted.

The test system is 68-bus test system with different input values of the capacity tolerance parameter γ, which ranges from 0.2 to 1.8, is able to reflect most realistic cases. Numeric results have been demonstrated in Table 3.3. In order to measure the extent, to which power system is damaged, the *Blackout Size* (BS) is introduced here as the damage index,

$$BS_i = \frac{N_{\text{fail}_i}}{N_{\text{total}}} \tag{3.18}$$

where BS_i represents the blackout size initiated by event i, $N_{\text{fail}(i)}$ is the accumulative number of failed loads, transformers and transmission lines, and N_{total} is the total number of all elements in the power system. In Table 3.3, BS is divided into 4 categories. It can be found that BS changes steadily with the increment of γ in categories of $BS < 10\%$ and $BS > 80\%$. However, BS falls rarely on the other two categories. Such phenomena are mainly caused by the inclusion of PSD and the load shedding mechanisms, proposed in this study. During the fast cascade stage, failures of a large number of elements are likely to cause the voltage or frequency collapse and finally fail the whole system. When some small disturbances occur, automatic load shedding mechanisms may effectively prevent the cascade from spreading so that BS is kept relatively small.

Table 3.3: *BS* Distribution with different γ using the proposed CFSP

γ	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8
BS<10%	5	20	31	35	45	48	56	67	73
10% - 40%	0	0	1	4	9	9	8	2	0
40% - 80%	0	0	0	0	6	8	4	1	0
>80%	78	63	51	44	23	18	15	13	10

A comparison simulation has been conducted using the conventional CN model [35] in order to better study the case and demonstrate the distinctions between the proposed CFSP and CN model. The simulation result is presented in Table 3.4, where BS demonstrates a similar trend. However, no BS is larger than 80%. Such phenomenon is due to the absence of PSD in the conventional CN models. In fact, in most cases, when failures propagate quickly, the system may suffer great oscillations and potential frequency or voltage collapses, which can lead to the failure of the entire system. Only those disconnected islands where power supply and demand are temporarily balanced can survive. In reality, disconnecting tie-lines between two areas manually is one of effective methods that have been utilized most commonly to prevent failures from propagating to adjacent areas.

Table 3.4: *BS* with different γ using CN-based Model where busbar voltages are assumed constant [35]

γ	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8
BS<10%	0	24	43	63	73	75	78	80	80
10% - 40%	3	50	36	20	10	8	5	3	3
40% - 80%	80	9	4	0	0	0	0	0	0
>80%	0	0	0	0	0	0	0	0	0

Note: γ is transmission line capacity tolerance parameter; *BS* is Blackout Size; Numbers in the table represents the times BS falls in the category among the total 83 simulation results for each γ

Automatic load shedding is an effective cascading failure mitigation strategy. Load shedding schemes in the industry are complicated and depending on different power systems and utility companies. Generally, it ranges from 25% to 65% [178]. In this chapter, simplified automatic load shedding, instead of complex load shedding schemes, are studied. Three different load shedding percentages, 25%, 40%, and 65%, are tested and compared with no load shedding situation in Table 3.5, in order to study the system response under cascading failures. We can observe that with automatic load shedding, the average blackout size is reduced from 69.74% to less than 45%. So we can conclude that automatic load shedding mechanism is able to dramatically mitigate the severity of cascading failure with a substantial reduction in the blackout size. However, different load shedding percentages have limited effect on BS caused by the cascading failure.

Table 3.5: Average *BS* with different load shedding rate

Load Shedding Rate	0%	25%	40%	65%
Average BS	69.7529	44.9771	44.3976	44.2217

3.4.2 Grid Robustness Analysis

Grid robustness assessment is useful to indicate the vulnerability of the grid so that corresponding protective measures can be executed.

In this study, transmission line criticality is identified through simulations on 68-bus test system using the proposed CFSP and conventional CN models. The criticality index Λ_i of the transmission line i is defined as the sum of BS_i with γ ranging from 0.2 to 1.8 divided by N_{total}. By definition, Λ is able to provide information about the average level of line criticality under different system stress conditions. The simulation result is shown in Fig. 3.8, where Λ of each transmission line is plotted. A larger Λ indicates a higher criticality of the specific transmission line. It can be seen that Λ varies significantly from line to line when using the proposed CFSP, while the difference in the line criticality is relatively small in the conventional CN model. This difference is caused by system dynamics and load shedding mechanisms, which may considerably affect Λ. From Fig. 3.8, we can see that the majority of critical lines determined by two models are unmatched.

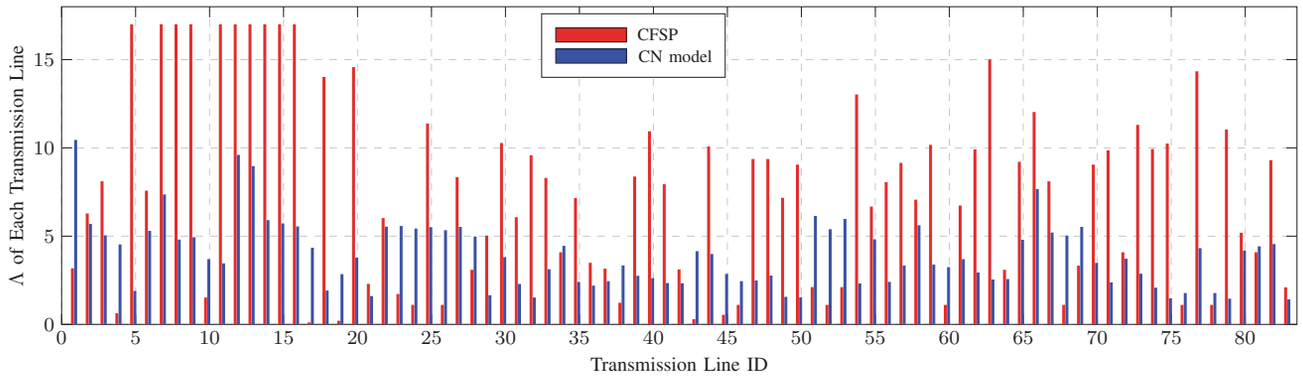


Figure 3.8: Line Criticality Assessment using the proposed PSD-based CFSP and the conventional CN-based model

3.4.3 Cascading Failure Propagation Profile Comparison

The sequence of cascading failures has been proven to have considerable effects on the consequence of outages, where different sequences of triggering events can lead to entirely distinct situations [13]. In this section, the cascading failure propagation profile is studied and compared with the conventional CN model. The comparison is able to provide useful information on the difference between two models, as well as how PSD affect the characteristics of cascading propagation. The matching index ζ is introduced in order to indicate the agreement of failure propagation. The definition is shown as follows,

$$\zeta = \frac{1}{N_{\text{events}}} \sum_{i=1}^{N_{\text{events}}} \frac{P_{i,\gamma} \cap P'_{i,\gamma}}{\max(P_{i,\gamma}, P'_{i,\gamma})} \tag{3.19}$$

where N_{events} is the total number of events that are simulated to initiate the cascading failure, $P_{i,\gamma}$ is the failure propagation profile simulated using CFSP with the initial event i and the specified γ, and $P'_{i,\gamma}$ is the failure propagation profile simulated using CN model. According to the definition, a larger ζ represents greater similarity between the two models. Table 3.6 shows that the value of ζ under simulation of the entire cascading process ranges from 0.1511 to 0.5113. This is a relatively small value indicating the considerable mismatch of two models. The table also presents that the agreement between two models of the entire cascading process is much lower than that of only the first 4 failures. This is mainly because during the early stage of the cascading failure, the system tends to be steady, i.e., at the slow cascade stage, and PSD have very limited effect during this cascade stage. Nevertheless, during the fast cascade stage, the system suffers great oscillations where PSD dominate the failure propagation. Furthermore, ζ increases with the increment of γ in both cases. This can also be explained by the PSD, where larger γ leads to less possibility of cascading failures and system oscillations.

Table 3.6: Failure Propagation Profile Comparison using ζ

γ	ζ of First 4 failures	ζ of Entire Process	γ	ζ of First 4 failures	ζ of Entire Process
0.2	0.6898	0.2759	1.1	0.6908	0.2978
0.3	0.6054	0.1511	1.2	0.6908	0.3035
0.4	0.5633	0.1227	1.3	0.6908	0.3395
0.5	0.6386	0.1275	1.4	0.7359	0.4005
0.6	0.6386	0.1313	1.5	0.7420	0.4536
0.7	0.6325	0.1807	1.6	0.7631	0.4840
0.8	0.6777	0.2123	1.7	0.7871	0.5152
0.9	0.6707	0.2341	1.8	0.8263	0.5113
1.0	0.6797	0.2566			

Note: Value of ζ represents the similarity of failure propagation

3.4.4 Impact of Load Modeling

Load models are critical in the study of PSD. Different load models will lead to different simulation results in the PSD studies. In this chapter, ZIP load models and frequency dependent load models are incorporated and investigated. The average BS under different load coefficient settings are compared in Table 3.7, which shows that various load models have certain impact on the average BS. It could be found that constant Z (impedance) and constant I (current) load models are more likely to cause larger blackouts. Meanwhile, the small differences demonstrated in the large BS ($BS > 80\%$) are critical in the cascading failure, especially in predicting the probability of large-scale blackouts ($BS > 80\%$).

Table 3.7: Comparison of BS Under Different Load Models

ZIP coefficients	frequency coefficients	γ	Average BS	BS>80%
$p_1 = p_2 = p_3 = \frac{1}{3}$ $q_1 = q_2 = q_3 = \frac{1}{3}$	$K_{pf} = 3$ $K_{qf} = -2$	0.2	65.0584%	39.76%
		1.0	44.9982%	26.51%
		1.8	39.5283%	24.10%
$p_1 = p_2 = p_3 = \frac{1}{3}$ $q_1 = q_2 = q_3 = \frac{1}{3}$	$K_{pf} = 1.5$ $K_{qf} = -1$	0.2	65.1746%	39.76%
		1.0	44.4454%	26.51%
		1.8	39.1688%	24.10%
$p_1 = p_2 = p_3 = \frac{1}{3}$ $q_1 = q_2 = q_3 = \frac{1}{3}$	$K_{pf} = 0$ $K_{qf} = 0$	0.2	64.4578%	38.55%
		1.0	44.3976%	26.51%
		1.8	38.9277%	24.10%
$p_1 = q_1 = 1$ $p_2 = q_2 = 0$ $p_3 = q_3 = 0$	$K_{pf} = 0$ $K_{qf} = 0$	0.2	66.3862%	44.58%
		1.0	47.3260%	30.12%
		1.8	40.8561%	24.10%
$p_1 = q_1 = 0$ $p_2 = q_2 = 1$ $p_3 = q_3 = 0$	$K_{pf} = 0$ $K_{qf} = 0$	0.2	65.4422%	42.17%
		1.0	46.3820%	28.92%
		1.8	39.5121%	24.10%
$p_1 = q_1 = 0$ $p_2 = q_2 = 0$ $p_3 = q_3 = 1$	$K_{pf} = 0$ $K_{qf} = 0$	0.2	64.5409%	39.76%
		1.0	44.4807%	26.51%
		1.8	39.0108%	24.10%

3.5 Interim Conclusion

In this chapter, a novel Power System Dynamics-based Cascading Failure Simulation Platform for complex power networks has been proposed to study the characteristics of cascading failure. The adoption of complex PSD make the model more realistic and accurate. In the proposed model, PSD, load models, relay models, and automatic load shedding mechanisms are taken into consideration. The IEEE 68-bus test system is chosen as the testbed, where validation of the proposed model is conducted by comparing the simulation results with the historical failure propagation profile. Extensive simulations of several cases have provided additional and meaningful insights on the impact of the PSD on cascading failure propagation profile. By comparing the simulation results from the proposed model with the conventional complex-network based model, the significance and necessity of considering PSD when analysing cascading failure propagation have been demonstrated. To be specific, the lack of consideration of PSD may lead to under-estimation of the probability and size of cascading blackouts. In addition, while the two models tend to have similar cascading failure propagation at early stages, the overall trend of cascading failure propagation analysed by these models differs significantly. This indicates the great effect of PSD during the cascading failure propagation especially during the fast cascade stage. Furthermore, with increasing computational capacity, such computationally demanding platforms are likely to operate on a real-time basis, allowing network operators to make more sensible and timely decisions to mitigate the adverse effects of cascading failures in power systems.

Chapter 4

Cascading Failure Model Considering Multi-Step Attack Strategy

Modeling and analysis of cascading failures draws wide attention recently due to frequent occurrences of large blackouts all over the world. Models based on complex network theory contribute significantly on analyzing the robustness of power systems and assessing the risk probability, while they fall short of producing the propagation of the cascading failure in exact time points. This chapter presents an improved topological model taking timescale into consideration, as well as the relay setting which reveals its operation more accurately according to industrial standards. The chapter then validates the model using UIUC 150-bus and IEEE 39-bus test system. In order to assess the vulnerability of a power network, the multi-step attack strategy has been provided. The results demonstrate that timescales and relay settings have a critical impact during the cascading failure and the proposed attack strategy may lead to larger blackouts than normal strategies.

The content of this chapter is mainly based on and modified from the following academic papers:
***Hengdao Guo**, Herbert Ho-Ching Iu, Tyrone Fernando, Ciyan Zheng, Xi Zhang, Chi K. Tse. "Cascading Failure Model Considering Multi-Step Attack Strategy" 2018 IEEE International Symposium on Circuits and Systems (ISCAS), pp. 1-5, Italy, May 2018.*

4.1 Introduction

Power networks, critical infrastructures in modern cities, are generally robust to commonly occurred small disturbances. However, it is still possible that combinations of small disturbances finally lead to large cascading failure blackouts. These major power outages, mostly caused by cascading failures, have a low occurrence probability while catastrophic effects. According to North American Electric Reliability Corporation (NERC), a cascading failure is "the uncontrolled successive loss of system elements triggered by an incident at any location" [13]. The emergence and rapid evolution of smart grids has brought not only the efficiency and advantage of power systems, but also new security challenges such as cyber-attacks that aim at communication and control systems . These attacks may greatly enlarge the vulnerabilities of power networks. Therefore, modeling and analysis of cascading failures becomes increasingly important these years.

A variety of mathematical models have been proposed aiming at studying the complicated mechanisms during the cascading failure. A comprehensive summary has been provided in [165]. Among them, models based on complex network contribute significantly on analyzing robustness and assessing the risk probability. Early complex network-based models emphasize only on pure topological characteristics while failing to explicitly reveal the necessary electrical properties [157]. The modified topological model, presented by Zhang et al., demonstrates a novel complex network-based model that is derived from the admittance model, which reflects the power flow across the power system [182]. Maximum flow-based models also attract researchers' attention [20]. However, these models mainly focus on analyzing the robustness and vulnerability of power networks statically, without considering the timescale during the cascading failure. Besides, the operation of relays are oversimplified to demonstrate real working conditions under contingencies.

This chapter first introduces the mathematical model of cascading failures that incorporates with timescales and relay settings. Validations of the proposed model have been conducted using the UIUC 150-bus test system by comparing the result with historical data. After that, the multi-step attack strategy has been introduced. In particular, what it differs from traditional simultaneously attack strategies is the consideration of timescales. The new strategy may possibly cause a larger blackout by launching attacks sequentially. Different metrics on selecting attacking targets are studied.

The rest of the chapter is organized as follows. In Chapter 4.2, a comprehensive mathematical model for cascading failures considering timescales and relay settings is presented. Model validations have also been conducted. The proposed multi-step attack strategy has been introduced in Chapter 4.3, as well as the case study and comparison of simulation results. Finally, a conclusion will be drawn in Chapter 4.4.

4.2 Comprehensive Mathematical Model For Cascading Failure

4.2.1 Model Introduction

The comprehensive mathematical model proposed in this chapter is based on the one provided by Zhang et al., which is built according to the admittance model presented by Grainger and Stevenson [166]. By Kirchoff's law equations, a power system with n buses can be expressed as

$$\begin{bmatrix} Y_{11} & -Y_{12} & \cdots & -Y_{1n} \\ -Y_{21} & Y_{22} & \cdots & -Y_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -Y_{n1} & -Y_{1n} & \cdots & Y_{nn} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_n \end{bmatrix} \tag{4.1}$$

where Y_{ij} represents the admittance of the transmission line between node i and j, and $Y_{ii} = -\sum_{j \neq i} Y_{ij}$. v_n and I_n are voltages and external injected currents at node n respectively. DC power flow is used for simplicity. Four different kinds of nodes are incorporated into the model, which are generation nodes, consumer nodes, distribution nodes, and transformer nodes. Generation nodes are assumed to be fixed voltage sources that can be expressed as

$$\begin{bmatrix} 0 & \cdots & 1 & \cdots & 0 \end{bmatrix} * V = v_k \tag{4.2}$$

For distribution and transformer nodes, they neither produce nor consume power. The voltage on both sides of the transformer nodes are equal when referring to per unit. Thus, the equation of node is

$$\begin{bmatrix} -Y_{i1} & \cdots & Y_{ii} & \cdots & -Y_{in} \end{bmatrix} * V = 0 \tag{4.3}$$

Consumer nodes dissipate power and sink current. The power is consumed so that the current should be negative. The equation is

$$\begin{bmatrix} -Y_{i1} & \cdots & Y_{ii} & \cdots & -Y_{in} \end{bmatrix} * V = -I_i \tag{4.4}$$

A combined equation can be achieved according to equations (4.2) - (4.4),

$$A * V = B \tag{4.5}$$

Therefore, given the power consumption, the generation information and the network topology, the voltage of each node will be obtained. The current of each node can be calculated using

$$I_{ij} = (v_i - v_j) * Y_{ij} \tag{4.6}$$

To better reveal the cascading failure procedure, timescales and relay settings are incorporated into this model. Relays play critical roles in power systems, whose operation mode has significant effects on the propagation of the cascading failure. There are various kinds of relays for generators, transformers, and transmission lines. For simplicity, this chapter adopts a unified protective relay model based on IEC standards [173]. The relay follows IEC based Inverse Definite Minimum Time (IDMT) settings for over-current protections. To compile with BS142 and IEC 60255 standards, the IDMT characteristics are derived from the equation

$$T(s) = \frac{K}{(\frac{I}{I_s})^{\alpha} - 1} * TMS \tag{4.7}$$

where T is the operating time in seconds, K and α are constants, I represents the actual value of current, I_s stands for the value of relay current settings, and TMS is the time multiplier setting. In this chapter, the normal inverse curve has been adopted, where α is set as 0.02, K is 0.14, and TMS is 0.1. IDMT relay curves will affect the tripping time according to the overloading level of current, the higher the overloading current, the shorter the tripping time.

Traditional topological models trip all failed elements simultaneously without delay that they fail to reflect how critical timescales affect the propagation and the final result of the cascading failure. With IDMT relay curves adopted, it is possible to mimic a more realistic protective mechanism, so as to track accurately the propagation of failures.

Cascading failures normally begin with one or two initial failures, which may cause the re-distribution of power flow across the whole power system after the removal of failed elements. Afterwards, this process may lead to new overload and trigger more elements. This triggering process iterates until the whole system fails or no more elements overloaded. A general flowchart of the cascading failure simulator is shown in Fig. 4.1 with detailed process described as follows,

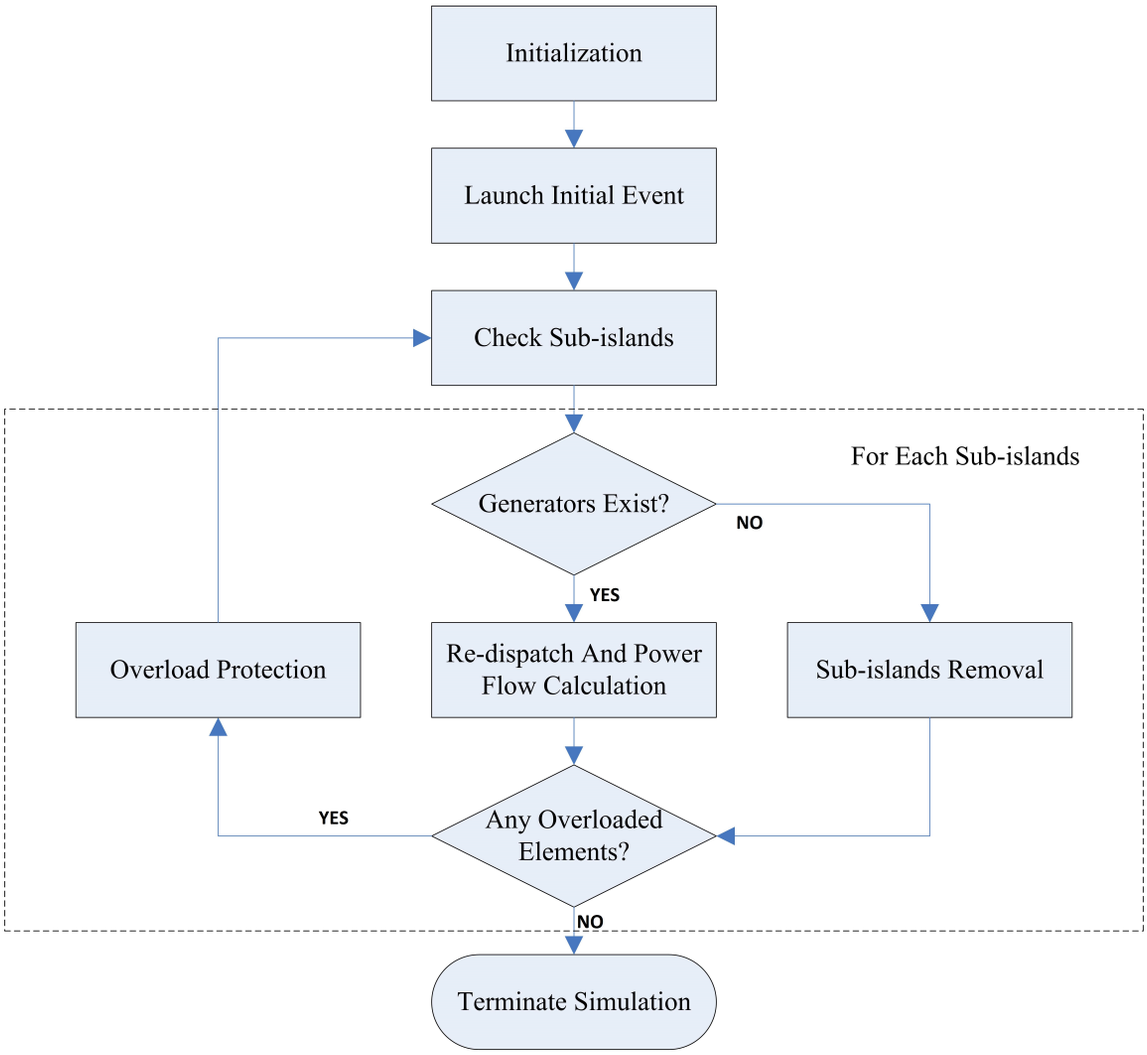


Figure 4.1: Cascading Failure Simulator Flowchart

Step 1: Initialization

Power system configurations will be initialized first, including the topology of power network, loads and generation at each node, the admittance of each transmission line, etc. For simplicity, voltages of generator nodes are set at 1 p.u. and currents of consumer nodes are 1 p.u. The value of relay current setting is defined as $1 + C_{link}$ times its initial value $I_{ij}(0)$ for transmission lines, and $1 + C_{node}$ for generator nodes. In this chapter, C_{link} is set as 0.2 and C_{node} as 0.5, which fits most cases in reality due to economic considerations. The initial system time t_0 is set as 0. The first contingency event is launched at time t_0 with corresponding elements removed without time delay. For line failures, only the failed transmission lines will be removed. For node failures, the node and all lines connecting to it will be removed.

Step 2: Check Sub-islands

After the initial contingency, the topology of the power network will be changed. The simulator will first search all sub-islands and check if generators exist on each island. If no generator exists, the whole sub-island will be removed. Remaining islands will go on step (3) and step (4).

Step 3: System Re-dispatch and Power Flow Calculation

All surviving islands will be re-dispatched automatically according to the equation (4.5). The renewed power flow will be calculated with new power network information. If any overload element observed, go to step (4) and start the overload protection. If no overloaded element exists, jump to step (5).

Step 4: Overload Protection

For all overloaded elements, the IDMT-based over-current relay is used to determine when and which element to fail. Record the system time at the moment as t_i for the ith component failure. The simulator then calculates the time delay Δt for each overloaded element and finds out the smallest value Δt_j, which means that component j would fail the earliest. Update the system time from t_i to $t_i + \Delta t_j$ and remove the jth element. Refresh the time delay Δt for all the other overloaded components accordingly. Go back to step (2).

Step 5: Simulation Termination

At this stage, the system reaches a new balance: no more overloading elements or the whole system fails. If no more attacks/contingencies occur, terminate the simulation and record the final system time t_{end}. Output final simulation results.

4.2.2 Model Validation

To validate the proposed model, simulation results are compared with historical data. The cumulative number of failed lines, transformers, and generators with timescales are shown in Fig. 4.2, which is extracted from the final report of the 2003 North America Blackout [1]. The figure demonstrates that the cumulative number of failed components grows slowly at first stage, and then accelerates sharply to its final state in a short time. Similar trends occurred in many other blackouts [4]. The simulation result on UIUC 150-bus test system with an initial failure of the node 128 is shown in Fig. 4.3, whose propagation profile displays the same pattern with historical blackout data. In the first 140 seconds after the initial contingency, the cumulative number of failed elements grows steadily, while in the following 3 seconds, it picks up the speed and reaches its final state soon.

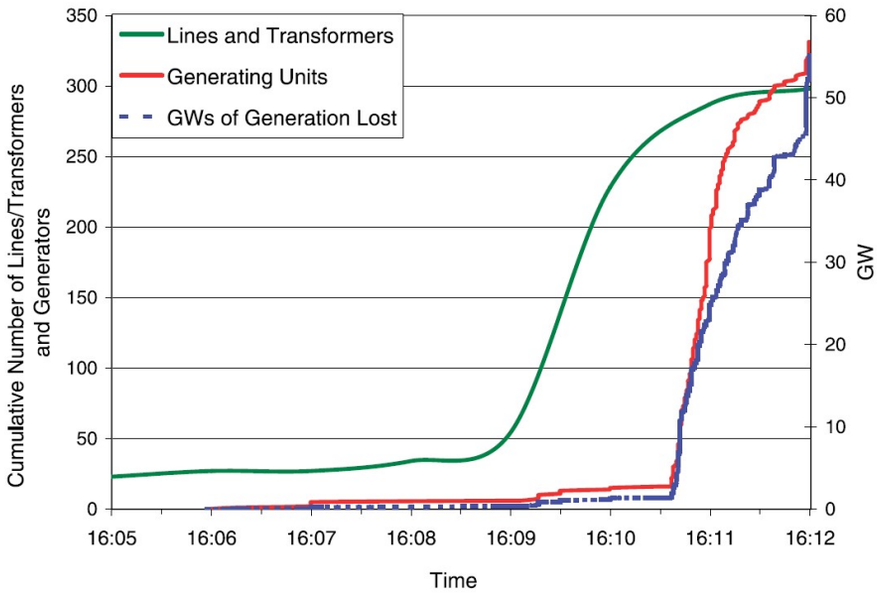


Figure 4.2: 2003 North America Blackout Data. Cumulative number of tripped lines, transformers and generators against time is given. Figure is extracted from [1].

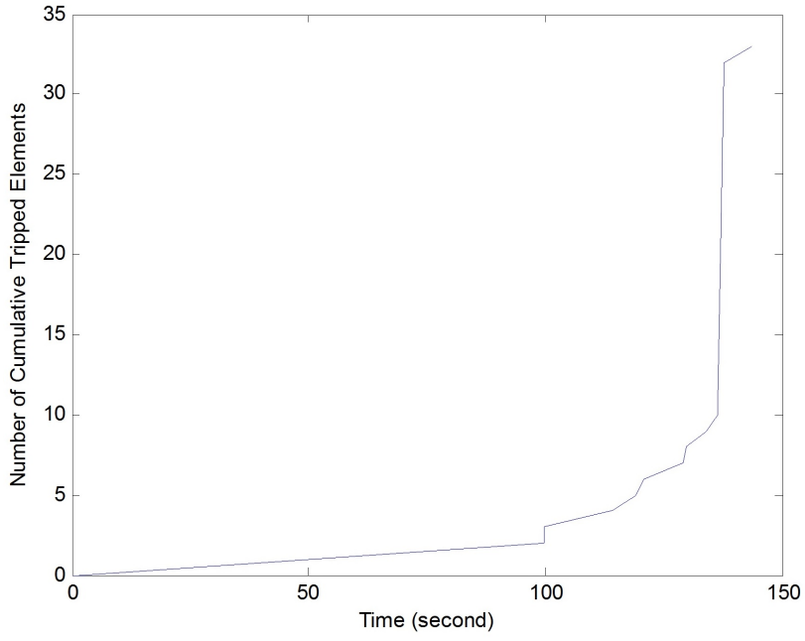


Figure 4.3: Simulation on UIUC 150-Bus Test System with initial failure of node 128

After that, validations on how timescales and relay settings affect the propagation of the cascading failure have been conducted using UIUC 150-bus test system [183]. The validation example has an initial event that two elements fail (failure of nodes 101 and 144). Three different cases are simulated based on this example, 1) the node 101 fails at time 0 and the node 144 fails 5 seconds later; 2) the node 144 fails at time 0 and the node 101 fails 5 seconds later; 3) both nodes fail simultaneously at time 0. Simulation results are shown in Fig. 4.4. The result highlights the significant influence caused by timescales. After taking node removal orders into consideration, the propagation and the final state

of the cascading failure differ between each other.

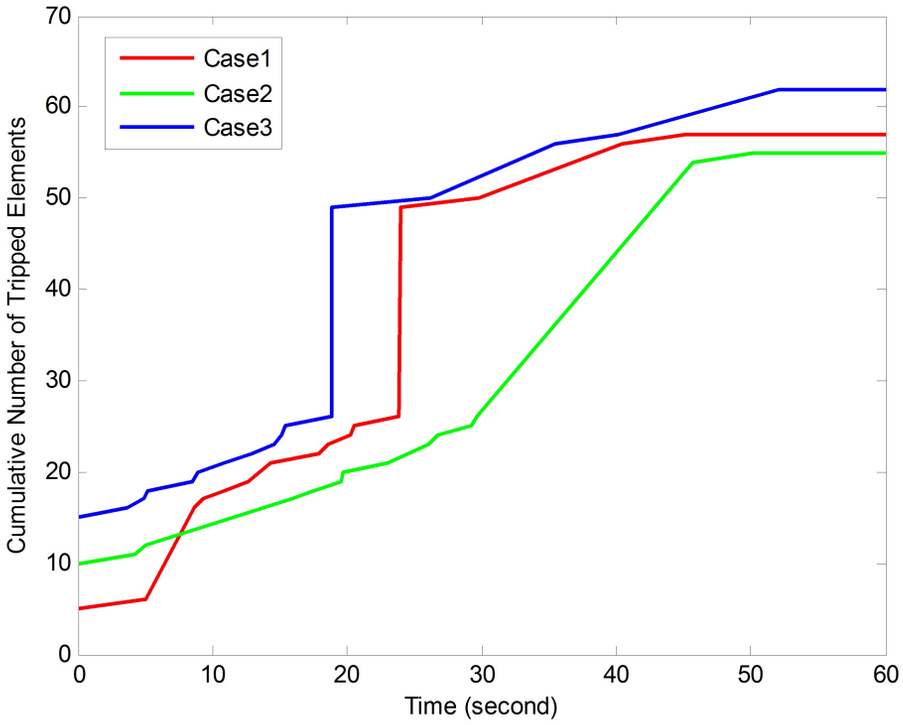


Figure 4.4: Simulation on UIUC 150-Bus Test System with 3 cases: 1) node 101 fails at time 0 and node 144 fails 5 seconds later; 2) node 144 fails at time 0 and node 101 fails 5 seconds later; 3) both nodes fails simultaneously at time 0.

4.3 Proposed Attack Strategy And Case Study

4.3.1 Robustness Index

Robustness and vulnerability are defined as the ability of a power system against the failure, whatever accidents or intentional attacks. A variety of metrics have been introduced in order to qualitatively or quantitatively measure the robustness of power systems. When a cascading failure occurs, it is impossible to directly calculate the total cost of the blackout. Therefore, from an engineering perspective, the cost can be regarded as the loss of nodes or loads. In this chapter, the percentage of unserved nodes (PUN), which provides a quantitative measure of robustness, is used as the index [182]. It is defined as

$$PUN(i) = \frac{n_{unserved}(i)}{n} \tag{4.8}$$

where i represents the initial event (one or several initial failures), $PUN(i)$ is the percentage of unserved nodes caused by the event i, $n_{unserved}(i)$ is the number of unserved nodes at final status after launching the event i, n is the total number of nodes in the power network. It can be seen that larger PUN may have severer damage to the system.

4.3.2 Proposed Attack Strategy

Various methodologies have been proposed to assess the vulnerability of a power network against intentional attacks using traditional complex network-based models [165]. However, without considering timescales, only limited attack strategies can be adopted. As shown in Fig. 4.4, adopting attacks one after another differs dramatically from simultaneous attacks. According to [184], by installing sensors or hacking into an existing power system monitoring infrastructure, the attacker is able to monitor the information of the whole system and launch either preset or real-time attack strategies.

In this chapter, a multi-step attack strategy has been proposed. Such attack strategy is launched under the situation that the real-time information is monitored by the attacker. Take a k-step attack strategy as an example, the first attack will be launched at time 0, and a second one right after the system reaches a steady state. Repeat the process until the kth attack adopted. Compared with the simultaneously attack strategy, such strategy may cause larger damage to the power network, which reveals some overlooked vulnerabilities of the system.

IEEE 39-Bus Test System is used as the test benchmark [185]. There are 39 nodes, 46 links and totally 85 elements in the system, with an ordered two-element combination of $P(85, 2) = 7140$ different cases. For a real power network with more elements and larger k, the number of cases would increase exponentially and are computationally expensive. Thus, some commonly used ranking metrics are introduced in order to find out the strongest attack:

1) Random selection

Select elements randomly from all components. For each k-element combination, an average result is calculated through 1000 simulations.

2) Degree-based selection

Select elements based on the degree value of nodes and links. Top 5 nodes and links are selected as candidates, where a total of $P(10, k)$ simulations for each k-element combination are conducted. The results are averaged.

3) Betweenness-based selection

Select elements based on the betweenness of nodes and links. Similar with previous one, an average result is given under $P(10, k)$ simulations for each k-element combination.

4) Load-based selection

Select elements based on the initial load value of nodes and links. $P(10, k)$ simulations for each k-element combination are conducted with an averaged result calculated.

The result is presented in Fig. 4.5, where k is set as 2,3,4. From the graph, it can be drawn that the average damage caused by the multi-step attack strategy is larger than that by simultaneously one. Besides, degree-based and betweenness-based metrics destroy more elements on average. This indicates that these two metrics have better performances on destroying the power network, and thus should be put more emphasis on. The simulation results prove the significance of the multi-step attack strategy.

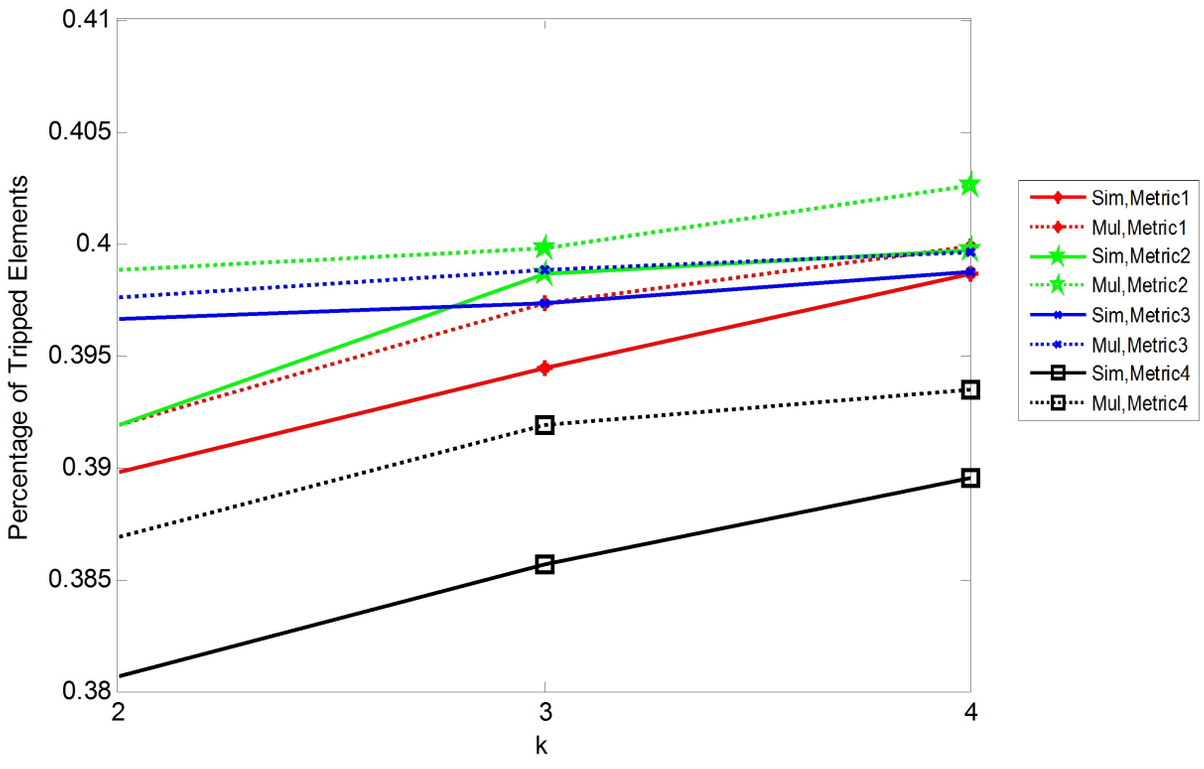


Figure 4.5: Comparison between simultaneously attack strategy and multi-step attack strategy with $k = 2, 3, 4$. IEEE 39-Bus Test System is used as test benchmark.

4.4 Interim Conclusion

In this chapter, a modified cascading failure model considering timescales and IDMT relay settings is proposed. Simulation results proved the significance of timescales and relay operations, which may greatly affect the propagation of the cascading failure. In addition, a multi-step attack strategy is introduced in order to better assess the vulnerability of power networks. Four different selection methods are simulated, where degree-based and betweenness based metrics play more significant roles during the cascading failure. Furthermore, the multi-step attack strategy is proved to cause more damage than the simultaneously attack strategy during the cascading failure. Future emphasis can be put on how generator dynamics affect the cascading failure.

Chapter 5

A Complex Network Theory Analytical Approach to Power System Cascading Failure - From a Cyber-Physical Perspective

Modern electric power grid is evolving rapidly into such a state that distributed controllers and two-way energy and information flow is replacing the traditional paradigm of electricity distribution and energy management. Therefore, power grid coupled with communication network is playing a pivotal role in establishing modern electric power systems. Previous cascading failure analysis in power systems focused more on the physical network, while falling short of investigations on the coupling effect of interdependency of the integrated electricity and communication networks, i.e., cyber-physical power systems. To address such pressing issue, this chapter introduces a novel stochastic cascading failure model considering the interdependency between the cyber network and power network. A multi-agent system and a novel protection relay model are incorporated into the proposed model. To apply the proposed analytical method, a test power system, IEEE 68-bus power system, is used to study the impacts of a range of interdependencies and cyber network topological structures on the cascading failure. Simulation results show the necessity and effects of consideration of cyber communication network when investigating power system cascading failures. The study also provides useful information on robustness and vulnerability of a particular power grid, given different communication topologies and interdependencies.

Cascading failure is one of the severest problems in electric power systems that may cause tremendous economic losses and large-scale damages to electricity customers. Previous studies relating to it focused mainly on the physical layer, which overlooked the possible impact caused by the cyber communication system. In practice, modern electric power grid is a typical cyber-physical system, whose robustness and wellbeing is greatly affected by the physical network, communication network and their interactions. To fill the void in relevant research, in this chapter, power system cascading failure is extensively investigated considering the coupled cyber-physical power system using complex network theory. In addition, the impact of various interdependencies and cyber network structures on

the robustness of the power network are carefully studied, analyzed and compared.

The content of this chapter is mainly based on and modified from the following academic papers: ***Hengdao Guo**, Samson S. Yu, Herbert Ho-Ching Iu, Tyrone Fernando, and Ciyan Zheng. "A Complex Network Theory Analytical Approach to Power System Cascading Failure - From a Cyber-Physical Perspective" Chaos: An Interdisciplinary Journal of Nonlinear Science, vol. 29, no. 5, 053111, May 2018.*

5.1 Introduction

The emerging interdependency between the electric power grid and Information and Communication Technology (ICT) contributes significantly to improving the performances of modern smart grids. This interdependent system can be regarded as a coupled Cyber-Physical Power System (CPPS), where cyber network is used to transfer useful data and control signals between the measurement points in the power system and the control center. Over the past decades, cyber network and power network have become more interdependent, driven by the emerging needs of information sensing and processing, intelligent controllers from ICT and other advanced technologies. The synergy of CPPSs enables electricity to be generated, transmitted, distributed and consumed in a more efficient, effective, economical and socially friendly fashion [3]. The electric power grid has been evolving to include more distributed control structures and bidirectional flow of electricity and information, as can be seen in Fig. 5.1. On the other hand, the increasing interdependency among CPPSs also brings challenges and potential risks to the power system, such as mis-operations, and cyber attacks that aim at vulnerable or critical components inside a particular CPPS. In some cases, these seemingly minor contingencies may lead to a large blackout across the entire CPPS; for example, the severe cascading blackout initiated by cyber attacks on several substations in Ukraine in December 2015 [131]. Normally, cascading failure, one of the severest power system instability problems, may cause large-scale damages and massive economic losses to society. Cascading failures are initiated by one or several contingencies, propagate through the physical network causing power outages, and finally result in the interruption of electric service in a large area. Such cascading failures are infrequent but substantially risky due to their catastrophic effects, which thus have drawn great attention from researchers and power system operators all over the world.

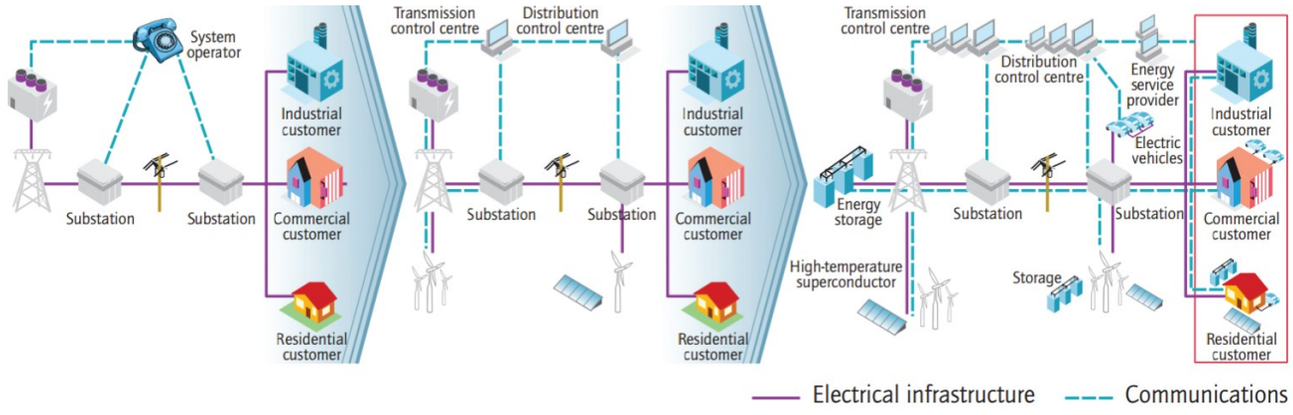


Figure 5.1: Evolution of the electric power grid [2]. Interactions of physical and communication systems in modern power networks.

In order to tackle with this issue, many researchers use Complex Network (CN) theories to investigate cascading failures in the power system. The incorporation of CN theories contributes to the study of connectivity, topological structures, and strengths of connections as key factors to understand, model, and manage the networks without losing sight of the huge dimensionality and complexity [186]. In [160], the authors proposed a node capacity model to analyze the cascading failure in a complex network. Despite its popularity, it did not thoroughly consider the essential physical and electrical properties of the power grid and may possibly lead to erroneous and misleading results [21]. After that, some modified topological models have been introduced to investigate cascading failures in the power system, such as POWERmap-based model [23], maximum flow models [41]. However, most existing cascading failure models suffer from a range of problems, i.e., most of them focus mainly on the physical power network and fail to consider the impact from the coupled cyber network. Recently, researchers began to realize the importance of understanding the behavior of CPPSs. Buldyrev et al. studied the cascading failures by modeling the Italy power grid as a set of interdependent power and communication networks [130]. The vulnerability of CPPS was investigated by considering network topology and power flow properties in [187]. Zhang et al. analyzed the coupling effect between CPPSs, where both physical and cyber systems suffer cascading failures, due to power overloading and malware contagion [167]. These interdependent cascading failure models have provided useful and helpful ideas on analysis of CPPSs; nevertheless, more improvements on models and mechanisms are needed for the deeper understanding of cascading failures. For instance, these cascading failure models over-simplified the protection relay and cyber communication systems.

Cascading failures are becoming increasingly complicated with rapid evolution of smart grid, in which the failure could be initiated and aggravated by various factors, i.e., cyber security and Operational Technology (OT) issues. OT system differs from traditional ICT with regard to the data availability aspect, such as the requirement of high-speed data transfer for reliable operations. In practice, communication networks are robust as they are supported by standby telecommunication equipment and lines. Nevertheless, time latency of the control signal affects the occurrence of cascading failure significantly, because the control and mitigation strategies are required to be operated prior to the next cascading failure being triggered in order to take effect. Furthermore, communication error and communication interruptions may also lead to control delays or errors. One typical example is the

failure in El Paso Electric system, where falsely tripped phase angle comparison relays caused the loss of loads. This improper operation resulted from the miscalculations with an incorrect communication latency value [188]. Another example occurred in the Hydro-Quebec system in 1988, where the Special Protection System (SPS) experienced a single point failure in the supporting communication system [188]. Existing studies only offer limited models and tools for the systematic analysis of cascading failures in the CPPS. Motivated by this, it is worthwhile taking advantage of complex network theory to carefully investigate the cyber network, as well as its interdependent effects with the physical network in CPPSs when studying the cascading failure.

For the purpose of modeling a more realistic CPPS to study the cascading failure without losing generality, efficiency and accuracy, it is necessary to understand the structure of both power and cyber networks. Generality means that the model should be able to reflect the necessary characteristics of CPPS, and efficiency and accuracy show the trade-off between the computational speed and the complexity of the model, i.e., a more complex model provides more details of CPPS with a higher computational burden. Existing technologies in the smart grid, such as Energy Management System (EMS), Distribution Management System (DMS), Phasor Measurement Unit (PMU), Remote Terminal Units (RTUs), and Supervisory Control and Data Acquisition (SCADA), are critical frameworks of the cyber network. These technologies help the Energy Control Center (ECC) to gather and process power network data and subsequently control the electric grid. Multi-Agent System (MAS) is incorporated to achieve a more efficient power system operation and control. The control centers could be divided hierarchically into several levels, such as local control center, area load dispatch center, state load dispatch center, and regional control center [189]. MAS, together with other technologies mentioned above, can be regarded as the backbone of the cyber network. A variety of models have been proposed to represent the cyber network [130, 167, 187], while these models have not adequately provided enough information to investigate the cascading failure in the CPPS. Traditional power system dynamic-based cascading failure model [73] is advanced in demonstrating generator dynamics during the cascading failure, but the efficiency of the model needs to be improved to reduce its high computational cost.

In this study, a novel stochastic CPPS model is proposed to investigate the cascading failure. The proposed model has following novelties and advantages over existing cascading failure CPPS models: (i) MAS is taken into consideration to mimic a more realistic communication and control cyber system. (ii) The stochastic model of cascading failure propagation in CPPS is incorporated in order to make a trade-off between the complexity and computation speed. (iii) A novel transmission line protection system is introduced based on the electric transmission line relays, where circuit breakers are placed at both ends of the line. Previous transmission line protection systems were over-simplified, where only one protection relay is modeled for each line. However, an electric transmission line has two circuit breakers placing at both ends of the line, which should be carefully considered when cyber communication network is involved. (iv) the cascading failure in CPPS is investigated with the proposed model from various perspectives using the complex network theory, where a systematic approach using the model is provided. (v) The simulation results provide useful information for the robustness, vulnerability and economic aspects for CPPS given various network conditions and topological features.

The remainder of the chapter is organized as follows. Section 5.2 introduces the proposed stochastic CPPS cascading failure model, including physical power network model, cyber network model, interde-

pendencies, cascading failure mechanisms, and the state transition for the Markov process. Section 5.3 presents the case study, which involves the analysis of CPPS network structures from a topological point of view, and the simulation results and their comparisons. Section 5.4 draws the conclusion of the study.

5.2 The Proposed Model

In this section, an elaborate description of the proposed stochastic CPPS cascading failure model is presented, followed by a concise introduction of various mechanisms inside the model.

5.2.1 Physical Power Network Model

The function of physical power network includes power generation, transmission, and distribution. The physical power network model used in this chapter is a modified model based on the one introduced by Zhang et al. [46]. In this model, electric substations are abstracted as nodes – consumer nodes, distribution nodes, generation nodes, and transformer nodes. Edges are the abstraction of the transmission lines. Deterministic power flow equations are used to calculate the power flow distribution across the network. More detailed explanations and mathematical models can be found in the [35].

5.2.2 Cyber Network Model

Cyber network of the power system is normally an Ethernet-based communication network, composed of various functional cyber devices in order to collect data and send control signals. The cyber network model proposed in this chapter is structured based on MAS and its corresponding power network. MAS is a widely used computerized system incorporated in many industrial fields, especially in the power industry [190]. Its potential benefits include fault tolerance, agent encapsulation, and the ability to serve as a platform for distributed systems. MAS in the proposed model is divided hierarchically into three levels, which are local control center, area load dispatch center and regional control center. Local control centers can be regarded as cyber nodes. The locations of area load dispatch center and regional control center depend on the test system, and are affected by both geographical factors and physical network topological features. To model the cyber network, all functional cyber devices, i.e., PMUs and RTUs, in a single bus (power node) are abstracted as a cyber node. The edge represents the data transmission medium. Although in practice, various communication media are applied to power industry, many of them, especially the wireless ones, do not have the quality of service function to guarantee the time latency requirements by fast control and protection functions [191]. In view of the practical situation, the fiber optics communication system is modeled as edges in the cyber network, whose failure may significantly affect the communication efficiency. The failure of the cyber nodes or edges will cause possible larger latency or erroneous data collection, which greatly affects real-time data collection and protection system. Four different topologies of the cyber network will be modeled in order to study the effect of topologies on cascading failure.

5.2.3 CPPS Interdependency

In a CPPS, cyber devices are used to collect real-time data and receive control signals, in order to assist normal operation of electrical network. Electrical network provides necessary power for cyber devices to operate normally. Therefore, a bidirectional interdependency is established in the proposed model as shown in Fig. 5.2. The figure illustrates a sample CPPS with two areas (zones), which are defined in application-specific ways or by individual utility company decisions, and are normally controlled by a corresponding area dispatch center. In addition to the area dispatch center, there are higher level control centers such as state control center and regional control center. For simplicity, only regional control center is considered in the proposed model. It is used to monitor and control the entire power system in the region.

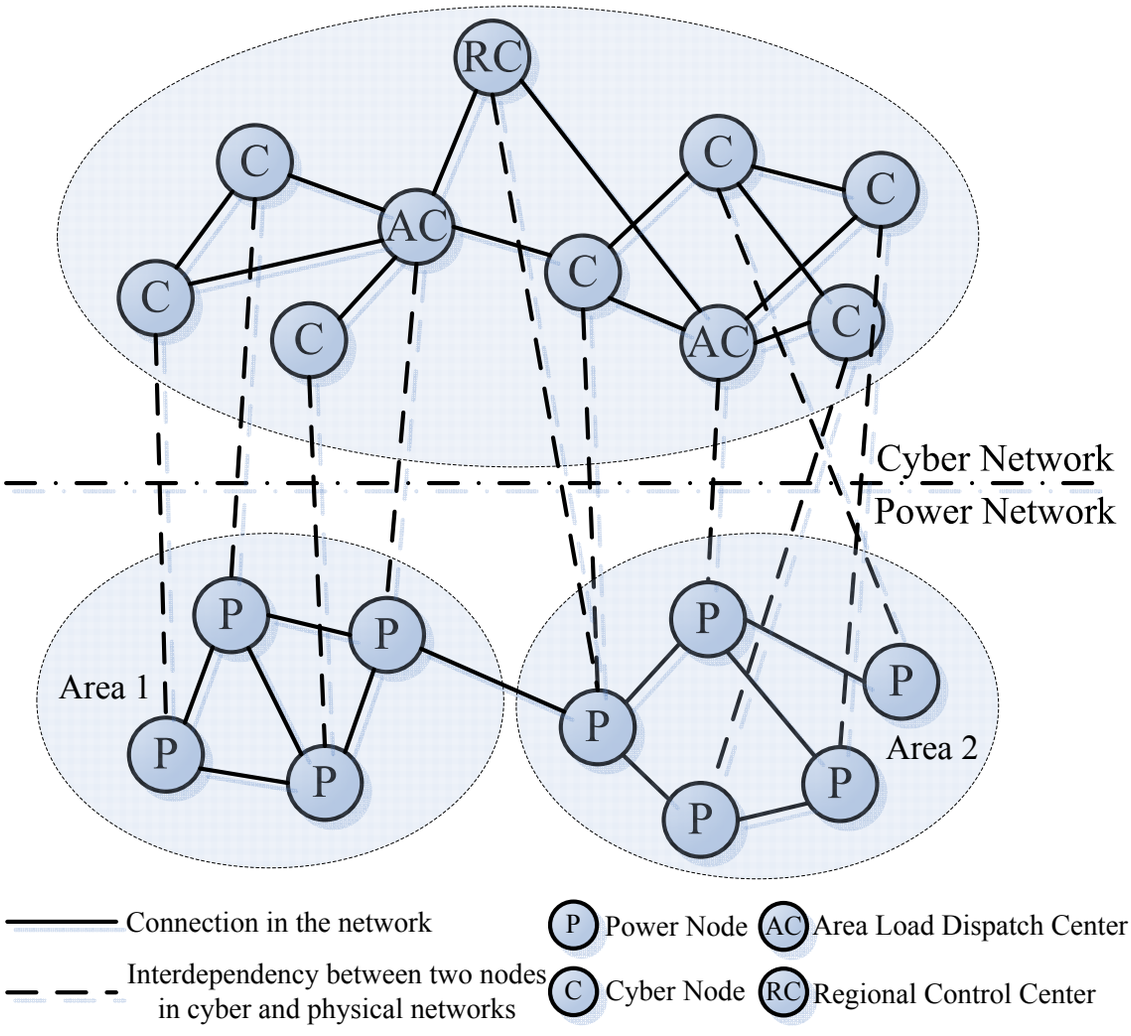


Figure 5.2: Example of cyber-physical interdependency in a two-area CPPS.

5.2.4 Cascading Failure Mechanism in the Proposed Model

Cascading failure mechanism is a complex procedure and is extremely demanding on computations. The proposed cascading failure model uses stochastic methods to model the failure propagation in CPPS. Furthermore, without losing generality and accuracy, basic circuit theories, i.e., Kirchhoff's

circuit laws, are taken into consideration for the purpose of power flow calculation. The cascading failure mechanism in the proposed model can be generally described as follows: (i) One or several nodes or edges in CPPS will be selected as the initial failure. (ii) When a node fails, all edges connected to it will lose their functions and are regarded as fail. (iii) The failure in the power network will cause possible islanding, i.e., the network is divided into several small islands, and the power flow of each island will be recalculated. Particularly, when a power node fails, its interdependent cyber node will be failed as well due to the power outage. (iv) The failure in the cyber network will not directly trigger any other component failures in CPPS; however, the failed component will bring larger latency in communication and control system. To be specific, when a cyber node fails, its dependent power node will have larger probability to fail due to the unobservability and uncontrollability. (v) When control centers fail, the area or region they are responsible for will be greatly affected, as the information process and control actions can not be performed. (vi) Cyber nodes with connections to their control centers have larger probability to lead a failure on its interdependent power node [188].

On the other hand, the cascading failure mechanism is modeled using both stochastic and deterministic methods. The Markov process is adopted to describe the cascading failure propagation within the stochastic methods. Several critical factors during the cascading failure in CPPS using stochastic methods are considered in the proposed model: (i) Hidden failure is the second frequent cause that aggravates the cascading failure according to Department of Energy [2]. In the proposed model, hidden failures are potential risks to all the elements in the power network, where the failure probability of power nodes is depicted as P^n_{hf}, and that of edges in power network as P^e_{hf}. (ii) The connecting nodes in the cyber network can exchange and transfer data in a short time. Therefore, when a cyber node fails, it is likely to cause larger latency and possible false data exchange, which may affect its neighbors and create potential risks. This situation has been carefully considered in the study, where the cyber node i_n connecting to a failed one has an extra failure probability $P_{cn}(i_n)$. (iii) Failure of area load dispatch centers and regional control centers may increase the failure probability of their controlled area, where the element i_n in the area has an extra failure probability $P^a_{cc}(i_n)$ and $P^r_{cc}(i_n)$ for area control center and regional control center respectively. In addition, if the cyber network breaks into several sub-islands, the cyber nodes that have no connection to their control centers are regarded having the extra failure probability $P^a_{cc}(i_n)$ and $P^r_{cc}(i_n)$ respectively, to their interdependent power node. (iv) Different from many previous models [7, 192, 193], overloading components in the power network will not be tripped instantaneously. In fact, the protection relays in the power system are designed to be able to work for a period of time before it is tripped [176]. In the proposed model, the tripping time and probability are determined by the extent of overloading. In other words, a more severely overloaded element will have higher chance and less time to be tripped. Furthermore, in order to better mimic a more realistic transmission line protection system, a novel two-end-relay protection system for transmission line is incorporated into the model. By considering the cyber control network, it is possible to model the two-end-relay protection system, where protection relays on both ends of the transmission line are controlled by the two ending cyber nodes respectively. Either relay that gets tripped will cause the transmission line failure. Therefore, the function of overloading stress and failure

probability of transmission line i_e at time t can be expressed as,

$$P_L(i_e,t) = \begin{cases} a_{i_e}\left(\frac{L_{i_e}(t)-C_{i_e}}{C_{i_e}}\right), & \textbf{if } L_{i_e}(t) > C_{i_e} \\ 0, & \textbf{if } L_{i_e}(t) \leqslant C_{i_e} \end{cases} \tag{5.1}$$

$$\lambda_{i_e}^S(t) = \overline{\overline{P_L(i_e,t)} \cdot \overline{\lambda_{i_e}^A(t)} \cdot \overline{\lambda_{i_e}^B(t)}} \tag{5.2}$$

where $L_{i_e}(t)$ is the power loading of element i_e obtained by power flow calculation at time t, a_{i_e} is the basic unit rate (trippings per second), and C_{i_e} is the capacity of element i_e that is proportional to its initial load, i.e., $C_{i_e} = (1+\alpha)\cdot L_{i_e}(0)$. Term $L_{i_e}(0)$ is the initial power loading of the element i_e, $\alpha(\alpha > 0)$ is the tolerance parameter, and $\lambda_{i_e}^A(t)$ and $\lambda_{i_e}^B(t)$ are failure probabilities of two nodes A and B of a transmission line. The bar of a probability means its complement set.

The failure probability of node i_n at time t is the combination of all failure possibilities and can be expressed as,

$$\lambda_{i_n}^S(t) = \overline{\overline{S_{hf}^n P_{hf}^n} \cdot \overline{S_{hf}^e P_{hf}^e} \cdot \overline{S_{cn}(i_n,t)P_{cn}(i_n)} \cdot \overline{S_{cc}^a(i_n,t)P_{cc}^a(i_n)} \cdot \overline{S_{cc}^r(i_n,t)P_{cc}^r(i_n)}} \tag{5.3}$$

where S_{hf}^n, S_{hf}^e, $S_{cn}(i_n,t)$, $S_{cc}^a(i_n,t)$, $S_{cc}^r(i_n,t) \in \{0,1\}$ represent the corresponding status of the element i_n at time t. The status is 1 when the component is experiencing one or multiple designated failures, with its extra failure probability being multiplied. The status is 0 when the component is not in any abnormal conditions.

Deterministic methods in the proposed model include power flow calculation, islanding detection, and CPPS interdependency inspection. Power flow and islanding detection will be processed each time a failure occurs in the power network. Power flow calculation is used to determine the loading condition of each component. The existence of generators will be checked for each island during the islanding detection. Those islands with no generators will be regarded as a failed sub-system. CPPS interdependency will be checked each time a power node fails, so that its interdependent cyber node will also be set as fail.

5.2.5 State Transition During the Cascading Failure

Nodes and links are regarded as basic elements of the CPPS, where the tripping event of basic elements is referred to as Element State Transition (EST). In this case, the cascading failure propagation in the power network can be regarded as a sequence of ESTs. Let $s_i(t)$ be the state of element i of a given network, where $s_i(t) = 0$ represents the normal condition of an element i at time t, and $s_i(t) = 1$ corresponds to a failed element i at time t, as shown in Fig. 5.3. In the state transition, $\lambda_i(t)$ stands for the transition rate of node i from "0" to "1", and $\mu_i(t)$ represents the transition rate from "1" to "0". The future state of an element is only determined by its present state and the transition rule, which can thus be described by a Markov process. The initial state of the CPPS for Markov process is given according to power grid information, and the transition rule is described as follows. The transition rate $\lambda_i(t)$ is affected by power networks, cyber networks, and the coupling effect between them. The transition rate in the power network can be divided into stochastic transition $\lambda_i^S(t)$ or deterministic

transition $\lambda_i^D(t)$. $\lambda_i^S(t)$ can be achieved from equations (5.2) and (5.3), and $\lambda_i^D(t)$ can be determined according to Section 5.2.4. For the cyber network, the transition rate is only determined by $\lambda_i^D(t)$. The transition rate $\mu_i(t)$ stands for the recovery process of a failed component, which normally takes half an hour to several hours to complete. In this study, the focus is on the fast cascading failure process and no components could be repaired within a short time, $\mu_i(t)$ can thus be treated as 0. A time interval limit T_{lim} between the two consecutive failures will be set, meaning that beyond T_{lim} the mitigation and control strategies could be fully implemented and the power system can get back to its normal operation state without incurring any further cascading failures.

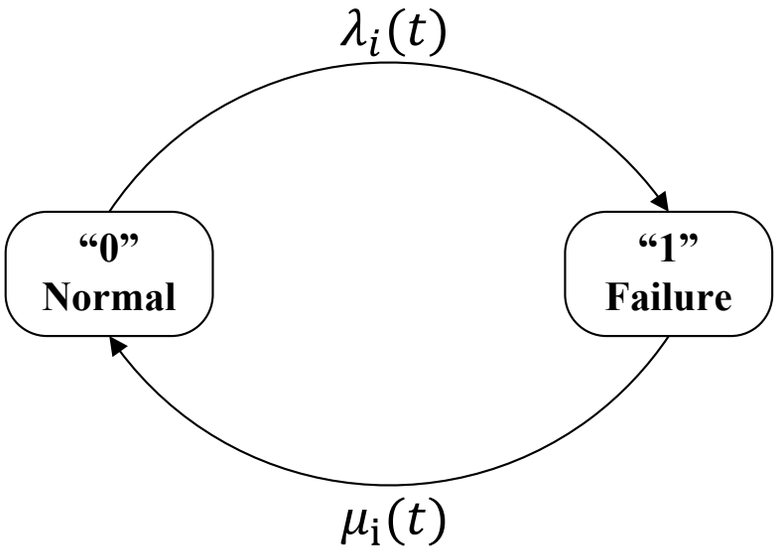


Figure 5.3: Dynamic description of failure in terms of state transition. State "0" is the normal operation state; state "1" represents that the component fails. The transition rate of two states are $\lambda_i(t)$ and $\mu_i(t)$, where $\lambda_i(t)$ consists of stochastic transition rate $\lambda_i^S(t)$ and deterministic transition rate $\lambda_i^D(t)$.

The power network can be represented by an undirected graph P with p elements, and cyber network represented by C with q elements. The state of $P-C$ coupling network can be described as $S(t) = \{s_{P_1}, s_{P_2}, \cdots, s_{P_p} s_{C_1}, s_{C_2}, \cdots, s_{C_q}\}$. Therefore, the cascading failure propagation can be expressed by the evolution of $S(t)$, which can be divided into two procedures: (i) determine the time τ of the next state transition; (ii) identify the element i_t that will transit (be tripped).

Consider the current state in a Markov process. Suppose $S(t) = N_S$, which is a specific network state at time t. Only those elements in state "0" may undergo a state transition. Let τ denote the time interval between two adjacent state transitions, $F(\tau)$ denote the accumulative Probability Density Function (PDF) that the next stochastic transition occurs before $t+\tau$, i.e., $F(\tau) = 1-P[S(t+\tau) = N_S|S(t) = N_S]$. In this study, the transition rule of Markov process is demonstrated as follows. The state transition during the stochastic stage can be derived through an extended Gillespie method [47], which was originally used for analyzing coupled chemical reactions [194, 195]. Therefore, $F(\tau)$ can be derived as follows, where detailed mathematical derivations and explanations can be found in [46],

$$F(\tau) = 1 - e^{-\lambda^{S*}(t)\tau} \tag{5.4}$$

where $\lambda^{S*}(t)$ is the sum of the element state transition rates of all "0" state elements in the power network during the stochastic stage. Equation (5.4) shows that $F(\tau)$ follows an exponential distribution, which has been in proven and adopted in relevant studies [153]. The physical meaning of $\lambda^{S*}(t)$ can be interpreted as the stress of CPPS. In order to determine the element that undergoes the next transition and the time when it happens, without losing the consideration of the property of exponential distribution of $F(\tau)$ and the characteristics where the element with a higher transition rate has a larger chance to be selected for the next transition, the following equations are used:

$$\tau = F^{-1}(z_1) = \frac{ln(1-z_1)}{-\lambda^{S*}(t)}, \tag{5.5}$$

$$\sum_{j=0}^{i_t-1} \frac{\lambda_j^S}{\lambda^{S*}} \leqslant z_2 < \sum_{j=0}^{i_t} \frac{\lambda_j^S}{\lambda^{S*}} \tag{5.6}$$

where z_1 and z_2 are two uniformly and independently generated random numbers in $(0, 1)$, λ_j is the jth component among all the "0" state elements. Therefore, from equations (5.5) and (5.6), the time τ and the transit element i_t can be determined from the stochastic method. By solving the equation (5.6), i_t could be determined. For the deterministic stage during the cascading failure, all ESTs are instantaneous, with the transitional elements being selected according to Section 5.2.4.

Some assumptions are made in the proposed model: (i) Power system transient states during the cascading failure are ignored due to the limited effects, complexity and computational burdens; (ii) For simplicity, all cyber devices connected to a single bus are abstracted as a lumped device. (iii) Real-time mitigation strategies are not considered during the cascading failure, since this chapter mainly focuses on investigating the different cyber network structures and interdependent effects in CPPS.

5.2.6 Flow Chart of the Proposed Model

The cascading failure mechanism in the proposed model is described in the flow chart shown in Fig. 5.4.

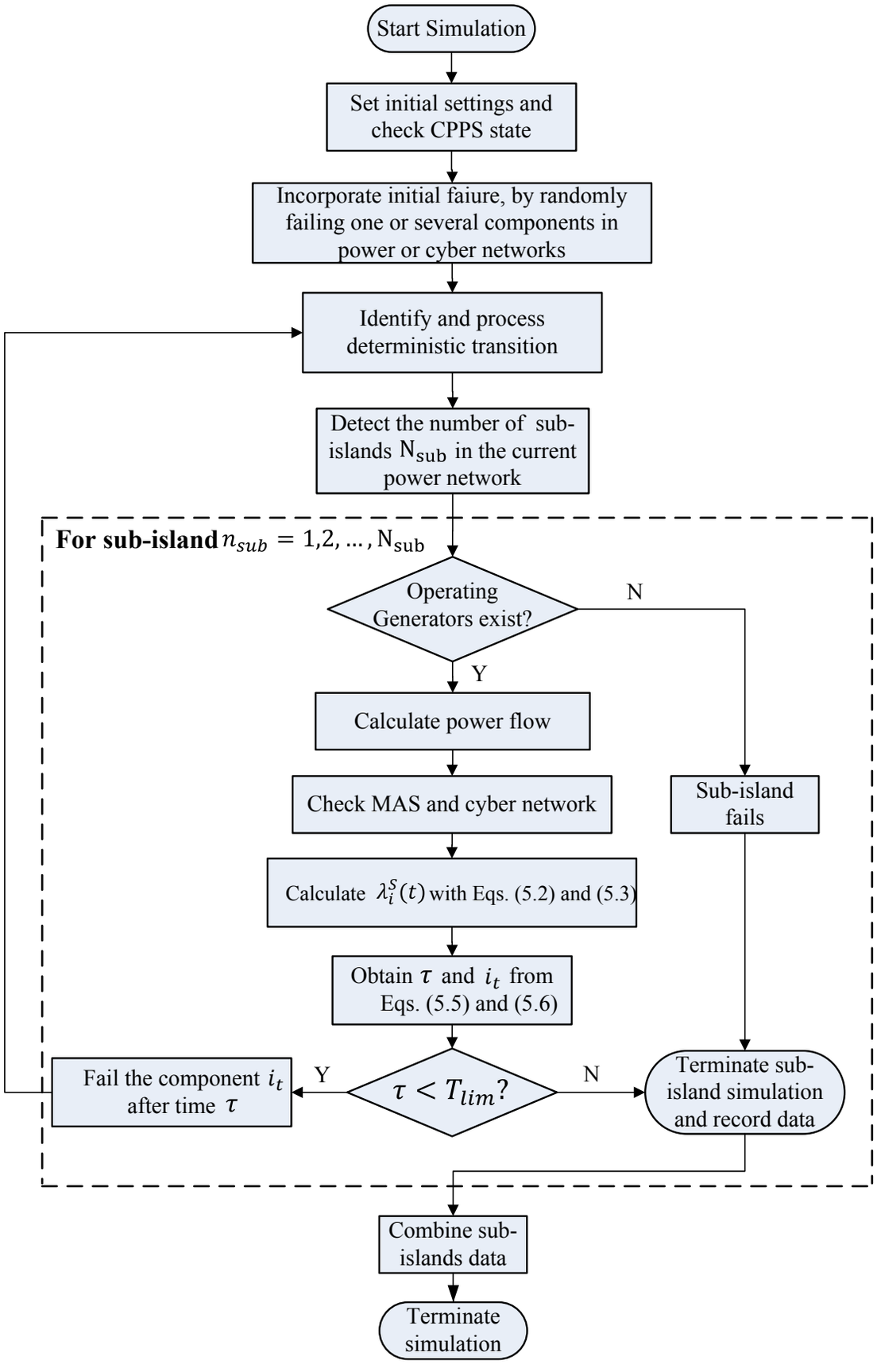


Figure 5.4: Flowchart of the proposed CPPS cascading failure model

Proposed CPPS Cascading Failure Algorithm

Step 1: Initialization

- Initialize CPPS with given information.
- Run CPPS simulation for steady-state operation.

Step 2: Initial failure implementation

- Implement an initial failure, i.e., failure of one or several cyber nodes, power nodes or edges.

Step 3: Deterministic transition

- Detect and process deterministic transitions during the failure step. Firstly, fail all physical edges connected to a failed node. Secondly, fail all physical nodes with no edges. Finally, fail all cyber nodes whose interdependent power nodes have failed.

Step 4: Detect the number of sub-islands

- After the initial failure and deterministic transition, the topology of CPPS may have changed, which can possibly cause the entire system to be divided into several sub-islands. The number of all sub-islands in power network is identified and denoted by N_{sub}. Simulations for each sub-island is proceeded individually and simultaneously.
- Run from ***Step 5*** to ***Step 9*** for sub-island $n_{sub} = 1, 2, \cdots, N_{\text{sub}}$.

Step 5: Check operating generators in the sub-island

- If there are operating generators, go to ***Step 6***.
- Otherwise, the sub-island is regarded "fail". Go to ***Step 9***.

Step 6: Calculate power flow and check MAS and cyber network

- Calculate power flow for the sub-island.
- Check MAS and cyber network. Record all failed area load dispatch centers and regional control centers. Identify all the states of cyber nodes, i.e., $S_{cn}(i_n, t)$, $S^a_{cc}(i_n, t)$, and $S^r_{cc}(i_n, t)$.

Step 7: Identify failed element and time

- Transition rate $\lambda^S_i(t)$ can be achieved from equations (5.2) and (5.3).
- According to equations (5.5) and (5.6), the transited element i_t and the time τ can be obtained.

Step 8: Failure determination

- Compare τ with time interval threshold T_{lim}.
- If $\tau < T_{lim}$, the failure occurs. Trip the component i_t after time τ and go back to ***Step 3***
- Otherwise, no more cascading failure occurs in the sub-island. Continue to ***Step 9***

Step 9: Terminate sub-island simulation

- Terminate sub-island simulation and record data.
- If no more sub-islands to proceed, go to ***Step 10***; otherwise, go back to ***Step 5***.

Step 10: Data combination and simulation termination

- After simulations for all sub-islands, combine the results and terminate simulation.

5.3 Case Study

5.3.1 Test System Introduction

The power network used in the study is modified from IEEE Standard New England 68-bus test system, as shown in Fig. 5.5, which is a simplified equivalent power network of the inter-connected New England Test System (NETS) and New York Power System (NYPS) [179]. The data of the cyber network and the interdependency of CPPS are synthesized and modified with similar characteristics according to [196, 197]. Furthermore, by considering both geographical factors and topological features, the area load dispatch center are selected at nodes 68, 33, 18, 42, 41 for each area, which are colored in blue in Fig. 5.5, and the regional control center is chosen to be located at node 30, with the node colored red.

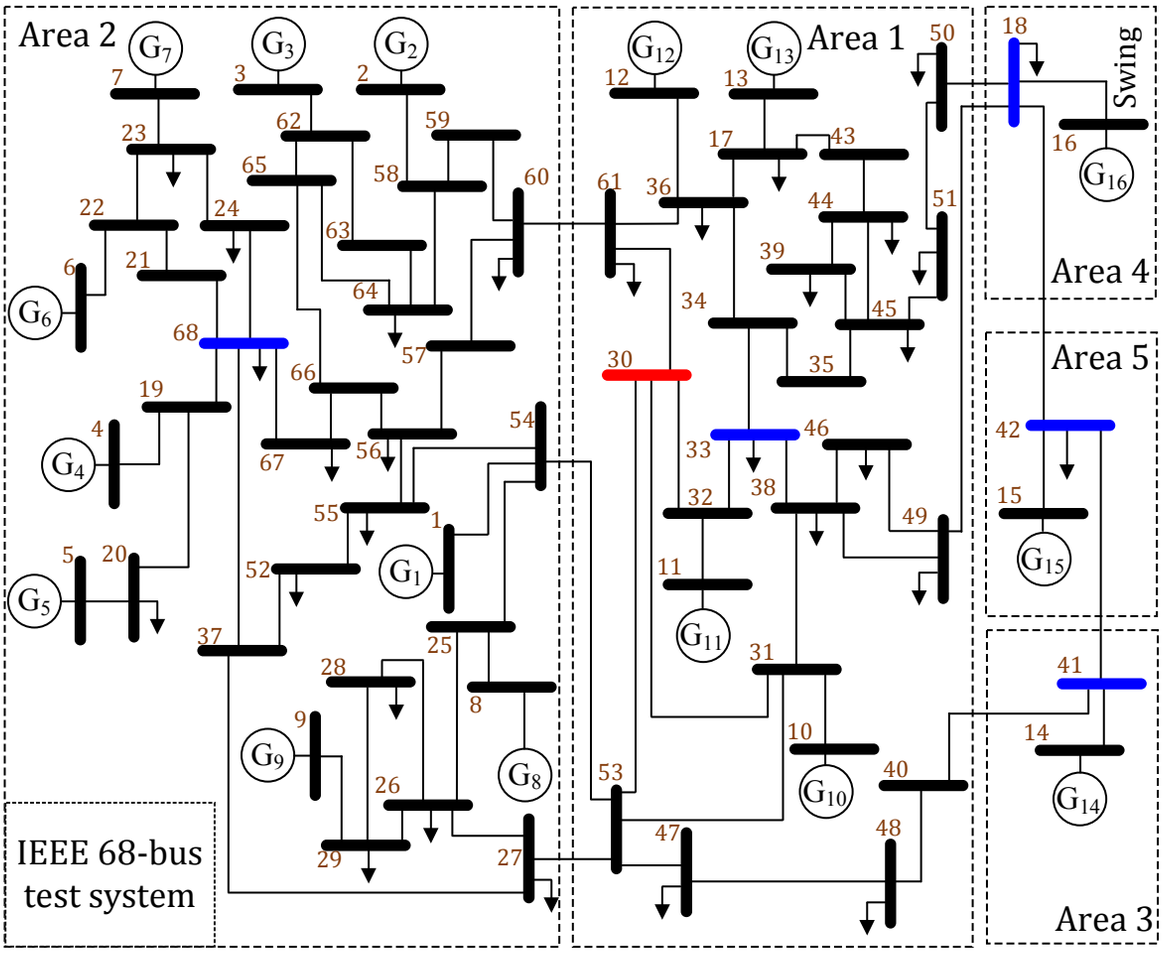


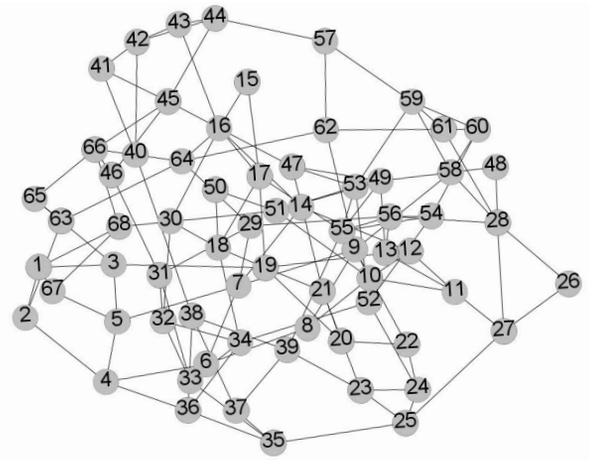
Figure 5.5: IEEE 68-bus test system diagram, where area load dispatch center are colored in blue, and regional control center in red.

The synthesized cyber networks used in this study include: 1) random network, 2) scale-free network, 3) small-world network, 4) same topology with power network except that 15% of edges are randomly selected and rewired [198]. By investigating random cyber networks, it is able to provide a more general statistical result on the cascading failure in comparison to other results. Scale-free and small-world networks are two typical networks in many complex systems [132, 133], which have been proved to be widely used as the cyber network in power systems in China, known as "double-star

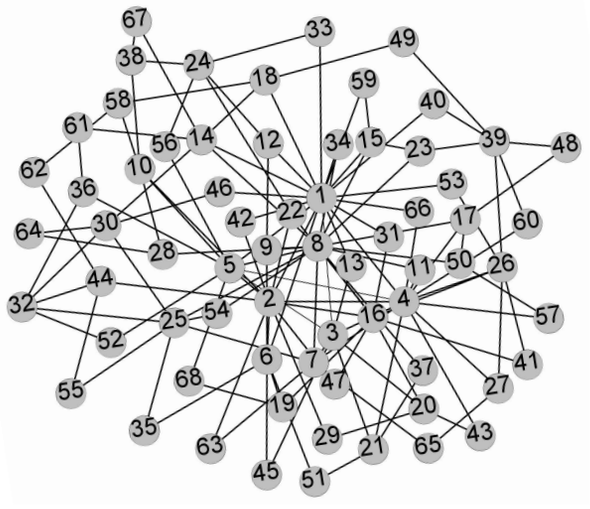
structure" and "mesh structure" respectively [196, 197]. In some cases, part of the cyber networks are constructing along the power networks, which result in a similar topology between cyber and power networks [198]. Therefore, it is reasonable to firstly build a cyber network that has the same topology with the power network, and then randomly choose 15% of edges to rewire. These four cyber topologies shown in Fig. 5.6 provide useful information on investigating the cascading failure in CPPS. The statistical parameters of the networks used in the study are listed in Table 5.1, where the statistical indexes are number of nodes and edges, the average degree $\overline{k}$, the average path length L, the clustering coefficient cc, and the maximum degree k_{max}. Small-world network has a balanced degree distribution, while the scale-free network has only a few hub nodes, i.e., nodes with higher degrees. Table 5.1 also demonstrates that $L_{\textbf{small world}} > L_{\textbf{random}}$ and $C_{\textbf{small world}} > C_{\textbf{random}}$, which are typical characteristics of small-world networks. Term k_{max} in scale-free network is much larger than that of other networks, which displays the key feature of the scale-free network. These hub nodes have been proved to make the network significantly vulnerable to intentional attacks [23, 199]. The average shortest path length L shows the efficiency of data transmission, where a smaller L indicates a more efficient network information transmission ability.

Table 5.1: Statistical parameters of CPPS networks

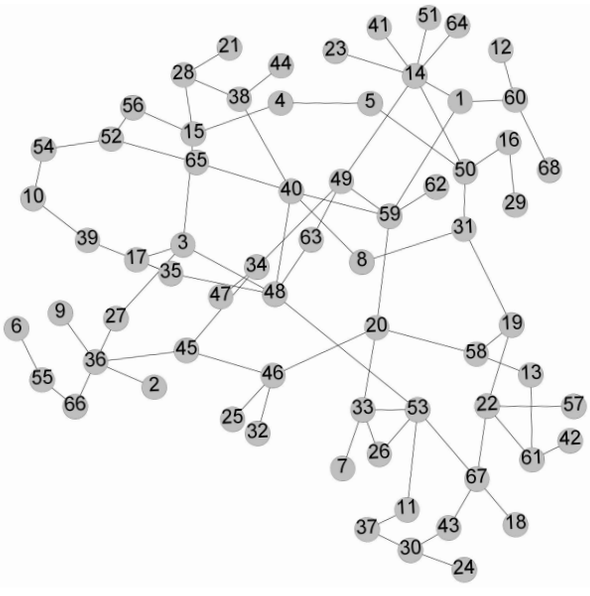
		Nodes	Edges	$\overline{k}$	L	D	cc	k_{max}
IEEE 68-bus Power Network		68	83	2.441	5.9175	14	0.0505	5
Cyber Network	Rewired Network	68	83	2.441	5.8648	15	0.0113	5
	Random Network	68	115	3.382	3.5540	9	0.0730	7
	Small-World Network	68	124	3.647	3.6100	7	0.1250	6
	Scale-Free Network	68	120	3.529	3.0729	8	0.1078	10



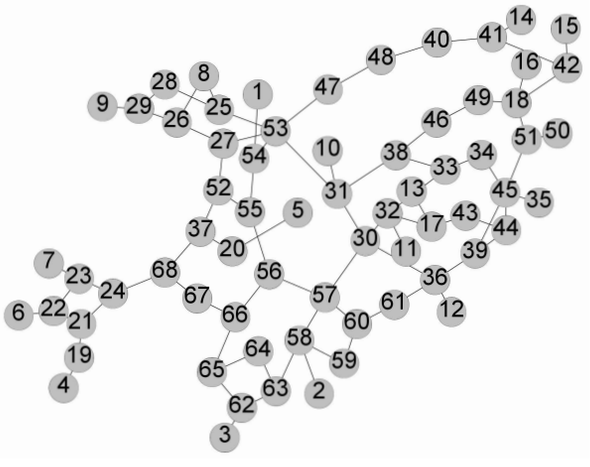
(a) Small-World Network



(b) Scale-Free Network



(c) Random Network



(d) Rewired Network

Figure 5.6: Examples of 4 network structures. Rewired network in (d) is constructed from IEEE 68-bus power network and randomly choose 15% of edges to rewire.

Some important parameters of the proposed CPPS cascading failure model are listed in Table 5.2. Term a_i is set by fitting historical cascading failure data, where the time duration and load demands increments are considered in the simulation [46]. Failure probability under different situations are set and modified according to [200, 201].

Table 5.2: Useful parameters of the proposed CPPS cascading failure model

a_i	$0.035s^{-1}$
T_{lim}	$1800s$
α	0.4
P_{cn}	0.001
P_{hf}^{e}	0.001
P_{hf}^{n}	0.0006
P_{cc}^{a}	0.02
P_{cc}^{r}	0.02

Different interdependencies in CPPS have also been investigated in this study. In practice, interdependencies in CPPS are affected by various factors, such as connections between buses and cyber devices and geographical reasons. However, in general, it follows the rule that important nodes in both networks tend to couple together [133]. Three commonly used centralities, betweenness, degree, and loads (in physical network), are used as parameters for nodes evaluation in this study. By combining these centralities, as well as random interdependencies, seven different cyber-physical interdependencies are generated, which are Betweenness to Betweenness (B-B), Betweenness to Degree (B-D), Betweenness to Load (B-L), Degree to Degree (D-D), Degree to Betweenness (D-B), Betweenness to Load (D-L), and Random (R). Take B-L as an example, nodes in the cyber network are ranked according to betweenness centrality, and the power nodes are ranked based on their initial loads. Afterwards, the interdependency is established by connecting the same ranking cyber and power nodes.

5.3.2 Simulation Results

The study mainly focuses on the robustness and vulnerability of CPPS against the cascading failure, by considering different interdependencies between CPPSs, and the impacts of four typical cyber network structures. Since the proposed model is a stochastic model, 100 simulations with the same network topology for each case are conducted in order to achieve a more reliable result.

Robustness of CPPS

The Blackout Size (BS), depict as η, of CPPS is used in this study in order to quantitatively investigate the scale of cascading failure on CPPS. It counts the final surviving components in the CPPS, thus quantitatively measuring the damage of the cascading failure to the system. A smaller η indicates a more robust system, and η can be expressed as follows,

$$\eta = \frac{p_s + q_s}{p + q} \tag{5.7}$$

where p_s and q_s are the number of surviving components in power network and cyber network respectively, which are obtained by the proposed CPPS cascading failure model in Section 5.2.

Transmission line tolerance parameter α is a very important index, which reflects the transmission line capacity given the normal operation state, whose value is not only critical to the system safety operation, but also an economic indicator when building or upgrading the transmission lines. A higher α represents a higher over-current capacity of the transmission line, improving the safety of power

transmission, but at the same time increasing the infrastructure cost. Therefore, to find a suitable value for α that fits both safety and economic requirements is of great importance. By gradually increasing the tolerance parameter α from 0.1 to 1, the robustness of CPPS can be observed. The simulation is conducted by failing each component in CPPS as the initial failure, i.e., failing each transmission line and bus in power network, and each node in cyber network. Thereafter, η is calculated by averaging the 100 simulation results for each case. The Average Blackout Size (ABS) η_{avg} for the whole CPPS is obtained by averaging all η.

Fig. 5.7 shows the trend of change of η_{avg} against α under different interdependencies, with the random cyber network structure. Fig. 5.8 demonstrates the robustness of CPPS against α with different cyber network structures and D-D interdependencies, as well as the robustness of non-coupling power network. It can be found that an "elbow" occurs at $\alpha = 0.5$, where the η_{avg} changes sharply when α is less than 0.5, and η_{avg} changes slowly beyond that. This indicates a relatively economic threshold setting of α when considering the overall robustness of a CPPS, since a lower α value may dramatically decrease the overall robustness, while a higher α value does not increase the robustness too much from an economic perspective. In Fig. 5.7, simulation results of D-D, D-L and B-L interdependencies perform slightly better than other cases. Results of different interdependencies result in a similar η_{avg} when α changes, meaning that the interdependencies in CPPS have less impact on the robustness of the system than the different cyber network structures do. From Fig. 5.8, it can be found that the robustness of CPPS under small-world and rewired cyber network structures perform better, and scale-free network structure has the largest average blackout size. Furthermore, the result of no-coupling power system also shows the "elbow" characteristic and performs much better without considering the cyber network impact. This phenomenon proves the necessity of investigation on the cyber-physical coupling effect when studying power system cascading failure.

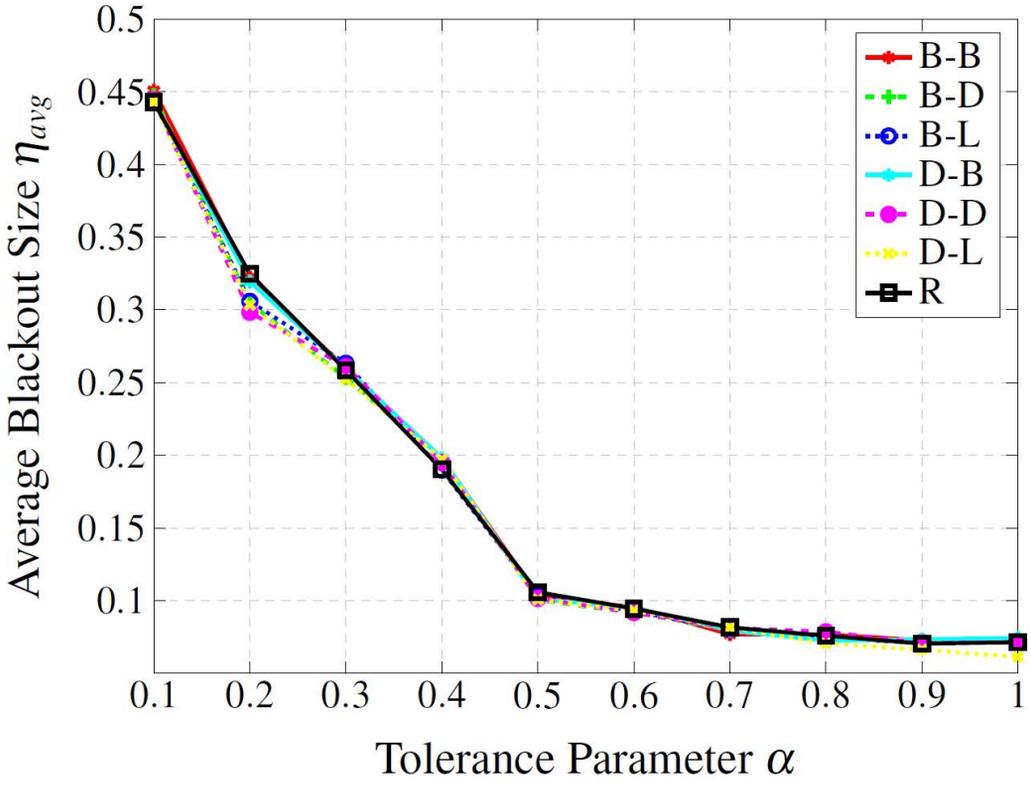


Figure 5.7: Robustness evaluation with various interdependencies using average blackout size η_{avg} with different tolerance paramter α. The simulation is using the random cyber network structure.

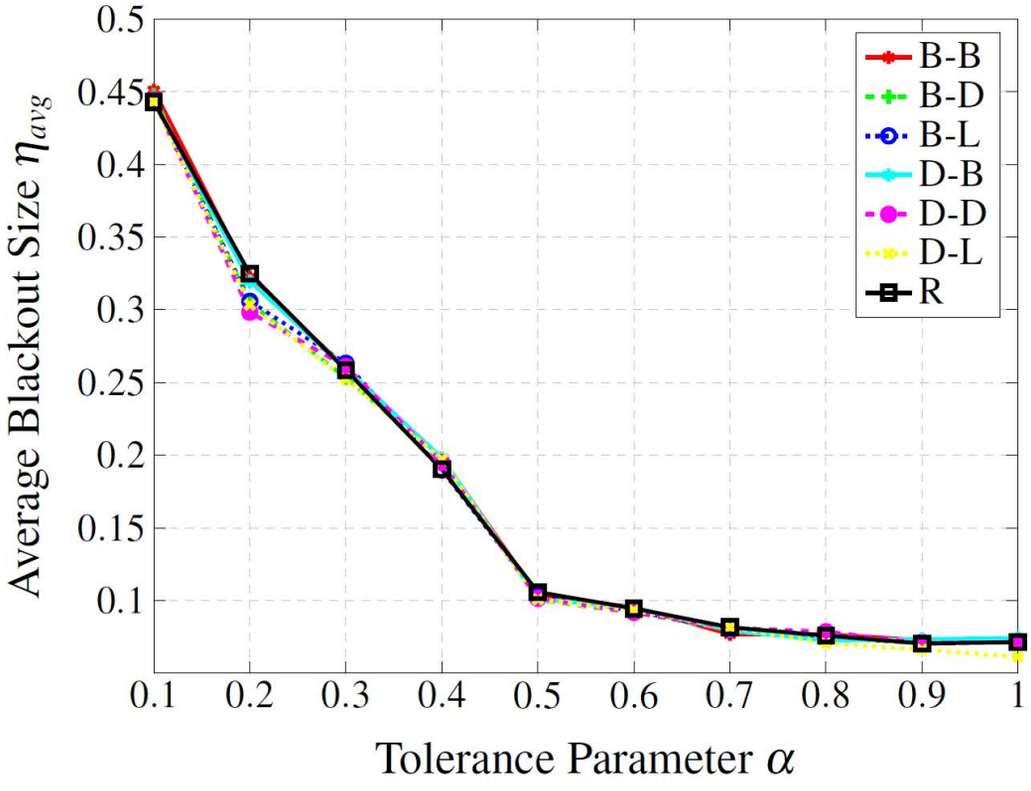


Figure 5.8: Robustness evaluation with various network structures using average blackout size η_{avg} with different tolerance paramter α. The simulation is using the D-D interdependency.

Large Cascading Failure in CPPS

In addition to the overall robustness of CPPS, the probability of Large Cascading Outages (LCOs) is also an important evaluation criterion. Common outages that affect only a small area can be restored within tens of minutes to several hours, while the restoration of a LCO may take several hours to several days. There is not a precise definition for the scale of a LCO. In this study, LCO is defined as cases that $\eta_{avg} \geq 20\%$, which indicates a large area outage and a severe impact on the normal operation of CPPS.

The simulation result is shown in Fig. 5.9. It can be found that CPPS with scale-free cyber network has the largest probability of LCO while small-world topology has the lowest. The rewired cyber network structure performs slightly better than the random one. The simulation result also shows that the interdependencies have limited effect on the probability of LCO.

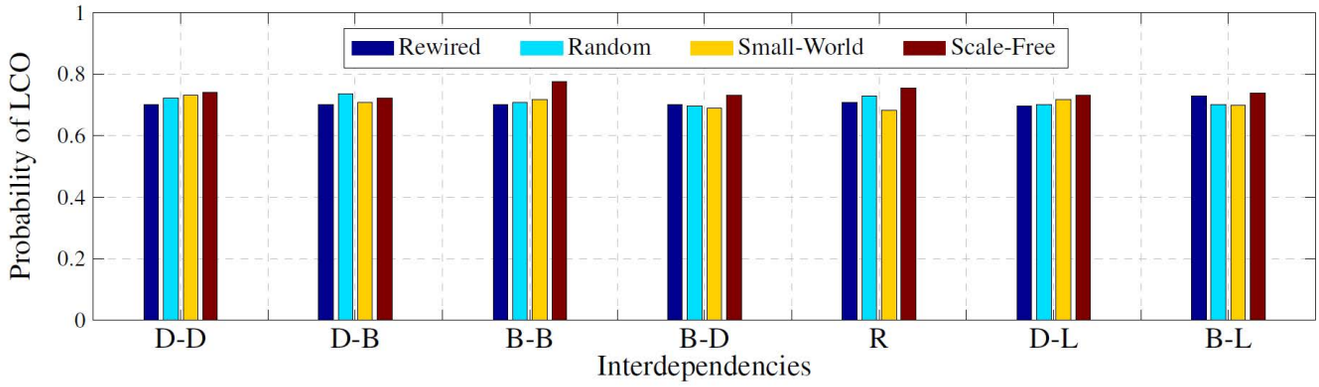


Figure 5.9: Probability of large cascading outages under different cyber network structures and interdependencies. Tolerance parameter α is set as 0.2.

Critical and Vulnerable Components in CPPS

In power system operations, there are usually a few critical and vulnerable components with the biggest impact on cascading failures in a CPPS. Critical components are those whose failures may lead to a large blackout. Vulnerable components represent the most frequently failed components among all possible CPPS outages. In this study, the tolerance parameter α is fixed at 0.2. In Tables 5.3 and 5.4, a list of top 15 critical and vulnerable components under different situations are demonstrated. In the table, NC stands for non-coupling power system, RW is rewired network, R represents random network, SW is small-world network, and SF is scale-free network. Simulation results from column 3 to column 6 are comparing different cyber network structures under B-B interdependencies, and the results from column 7 to column 13 are comparing different interdependencies with randomly coupled cyber network.

In Table 5.3, it can be found that the ranking of components varies with or without considering the coupling cyber network, while the difference brought by various cyber network structures and interdependencies are indiscernible. The most critical components when considering the coupling effect are mostly the nodes that support a control center, i.e., node 30 is the location of regional control center. This result indicates the importance of the location of the control center. Other critical components, except these control centers, show certain agreement, such as the most critical components listed in

NC column also appears several times in other columns, where coupling effects are considered. On the other hand, when considering the interdependent cyber network, some new critical components are observed, i.e., node 35, which appears 7 times in the table, but is absent in the NC column. This again shows the necessity to take cyber network into consideration when investigating cascading failure.

The most vulnerable components presented in Table 5.4 show great similarity between the coupling and non-coupling power systems. The result implies that the consideration of coupling cyber network has little effect on the vulnerable components in a power system, and suggests that these frequently failed components deserve extra care and timely upgrade so that the system can stay at a healthy working condition.

Table 5.3: Critical components of CPPS ($\alpha = 0.2$). Indexes 1-68 are power nodes, indexes over 69 represent power edges.

		B-B				Random						
Rank	NC	RW	R	SW	SF	B-B	B-D	B-L	D-B	D-D	D-L	R
1	54	30	30	30	30	30	30	30	30	30	30	30
2	70	12	33	68	68	33	68	33	33	13	32	68
3	2	68	68	133	2	68	33	127	80	35	126	105
4	80	33	2	54	53	2	80	70	70	36	33	80
5	12	127	36	78	141	36	12	126	68	133	36	109
6	36	133	69	36	11	69	2	68	12	106	11	2
7	133	70	32	70	12	32	54	31	78	81	2	70
8	1	54	11	135	80	11	78	48	54	12	68	136
9	53	2	35	12	35	35	1	80	133	47	70	114
10	81	1	105	80	46	105	69	122	31	127	78	81
11	69	78	80	105	32	80	82	105	111	51	12	84
12	31	53	54	31	36	54	32	2	13	54	80	12
13	10	46	12	44	79	12	36	90	57	136	54	53
14	105	47	56	32	126	56	57	79	136	55	46	61
15	11	35	79	34	70	79	47	35	35	53	56	35

Table 5.4: Vulnerable components of CPPS ($\alpha = 0.2$). Indexes 1-68 are power nodes, indexes over 69 represent power edges.

		B-B				Random						
Rank	NC	RW	R	SW	SF	B-B	B-D	B-L	D-B	D-D	D-L	R
1	97	97	97	97	97	97	97	97	97	97	97	97
2	143	143	143	143	100	143	143	143	100	143	143	143
3	100	100	100	100	143	100	100	100	143	100	100	100
4	28	28	28	28	28	28	28	28	28	147	28	28
5	99	99	99	99	99	99	99	99	99	118	99	99
6	101	101	101	101	101	101	101	101	101	99	101	101
7	118	118	118	118	118	118	118	118	118	28	118	118
8	86	147	147	147	147	147	147	147	147	101	147	147
9	117	117	117	117	117	117	117	117	117	86	117	117
10	82	84	84	84	84	84	84	86	86	50	84	84
11	14	86	86	86	82	86	86	84	84	84	86	86
12	41	18	16	18	86	16	16	18	82	18	16	16
13	84	16	18	16	14	18	18	16	14	122	18	18
14	16	82	82	82	18	82	82	82	41	16	50	82
15	115	41	14	14	41	14	14	14	18	115	82	41

Blackout Distribution

Probabilistic Risk Assessment (PRA) is a commonly used systematic and comprehensive methodology to evaluate risks associated with the power system against cascading failure outages. The PRA is mainly characterized by two quantities, 1) the blackout size η; 2) the probability of occurrence of each η. The PRA is carefully investigated in this study, where the results of both coupling and non-coupling power systems are demonstrated in Fig. 5.10. The result for coupling CPPS shown in the figure uses B-B interdependency and the random cyber network structure. The original data and fitted curve of the results are illustrated, which follows the power-law distribution. The simulation result for the non-coupling power system also shows the power-law distribution.

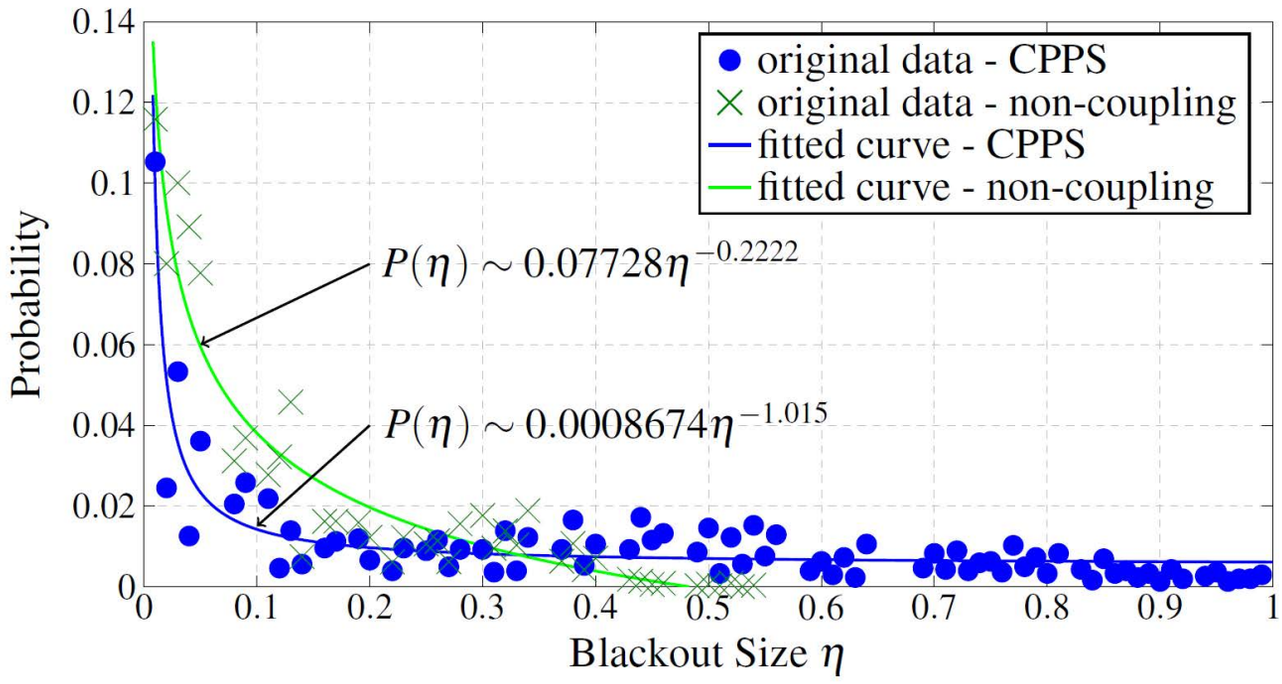


Figure 5.10: Blackout size and probability distribution, where the probability is fitted as a function of blackout size η

The fitted curve in Fig. 5.10 for CPPS is $P(\eta) \sim 0.0008674\eta^{-1.015}$, and for non-coupling one is $P(\eta) \sim 0.07728\eta^{-0.2222}$. It can be found that, compared to the non-coupling power system, η in CPPS has a lower probability when $\eta \leq 0.34$, while it has much higher probability when $\eta > 0.34$. The blackout distribution with other interdependencies and network structures are shown in Tables 5.5 and 5.6. In Table 5.5, it can be seen that the D-B interdependency is prone to having a larger blackout scale, while D-D and D-L interdependencies have higher probability to cause small blackouts. From Table 5.6, scale-free network is most likely to cause larger cascading failures, while small-world shows the opposite trend.

Table 5.5: Fitted curve with different interdependencies, where the simulation is conducted using the random cyber network with α set as 0.2.

Interdependency	Fitted Curve
B-B	$P(\eta) \sim 0.0008674\eta^{-1.015}$
B-D	$P(\eta) \sim 0.0006955\eta^{-1.072}$
B-L	$P(\eta) \sim 0.0007759\eta^{-1.048}$
D-B	$P(\eta) \sim 0.0005493\eta^{-1.124}$
D-D	$P(\eta) \sim 0.002793\eta^{-0.9059}$
D-L	$P(\eta) \sim 0.001074\eta^{-0.9679}$

Table 5.6: Fitted curve with different network structures, where the simulation is conducted using the B-B interdependency with α set as 0.2.

Network Structure	Fitted Curve
Random	$P(\eta) \sim 0.0008674\eta^{-1.015}$
Rewired	$P(\eta) \sim 0.0007467\eta^{-1.054}$
Scale-Free	$P(\eta) \sim 0.000457\eta^{-1.156}$
Small-World	$P(\eta) \sim 0.0001118\eta^{-0.953}$

5.3.3 Simulation Results Discussion

The simulation results indicates the significance of consideration of cyber-physical coupling network. With and without cyber-network, the simulation results various largely. Furthermore, different cyber-networks and coupling methods may greatly affect the robustness of the power system.

5.4 Interim Conclusions

In this chapter, a novel stochastic Cyber-Physical Power System cascading failure model has been proposed. Multi-Agent System is integrated into the proposed model to mimic a more realistic cyber system with modern communication and control systems. The stochastic mechanisms during the cascading failure have also been modeled in order to demonstrate a trade-off between the accuracy and the computational burden of the model. Furthermore, the dual transmission line relay model is incorporated. Simulations are conducted using the IEEE 68-bus test system with four typical cyber network structures and diverse interdependencies. The simulation results demonstrate that D-D and D-L interdependencies, and small-world cyber network structure are more robust against CPPS cascading failures.

In conclusion, this study has made an attempt for the understanding of cascading failures in CPPS under various cyber-network structures and cyber-physical network interdependencies. The simulation results have provided useful information for the robustness, vulnerability and economic aspects for CPPS given various network conditions and topological features.

Chapter 6

Conclusions

6.1 Contribution of the Thesis

This thesis investigates the modeling and analysis of cascading failure in the complex power networks.

The state-of-art models and analytical tools have been introduced thoroughly. The cascading failure background and its related issues have been presented in details.

The thesis then focus on dynamic modeling of cascading failures. The mathematical models of both generators, loads, relays and automatic load shedding mechanisms are taken into consideration. The IEEE 68-bus test system is chosen as the testbed, where validation of the proposed model is conducted by comparing the simulation results with the historical failure propagation profile. Extensive simulations of several cases have provided additional and meaningful insights on the impact of the power system dynamics on cascading failure propagation profile. To be specific, the proposed model allows researchers to incorporate easily no matter what model they need. Further, the computational cost is much lower compared with other traditional models. The analysis methodologies provide new views to investigate cascading failures.

This thesis also studies a complex network-based model, where a multi-step attack strategy is implemented in order to assess the vulnerability of power network, and thus providing a novel idea to build a corresponding protection scheme. The simulation results show that multi-step attack strategy causes larger blackouts compare with traditional simultaneously attack strategy.

The thesis finally studies cyber-physical power system interdependent model against cascading failure. Multi-agent system is integrated into the proposed model to mimic a more realistic cyber system with modern communication and control systems. The stochastic mechanisms during the cascading failure have also been modeled in order to demonstrate a trade-off between the accuracy and the computational burden of the model. Furthermore, the dual transmission line relay model is incorporated.

In summary, this thesis has made an attempt for the understanding, modeling, and analyzing the cascading failures in complex power systems from different perspectives.

6.2 Future Work

There is substantial amount and significant research work that can be conducted in the future to further improve the models and the analytical approaches to the cascading failure in complex power networks. From the perspectives of modeling, the power system dynamic-based model can be more accurate in order to thoroughly study the cascading failure mechanisms. Moreover, the cyber physical power system interdependent model could be further modified. More control schemes and models could be incorporated into the proposed model. On the other hand, machine learning is definitely an appropriate and suitable method to be implemented to extremely investigate the cascading failure propagation and assess the robustness and vulnerability of the power network.

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